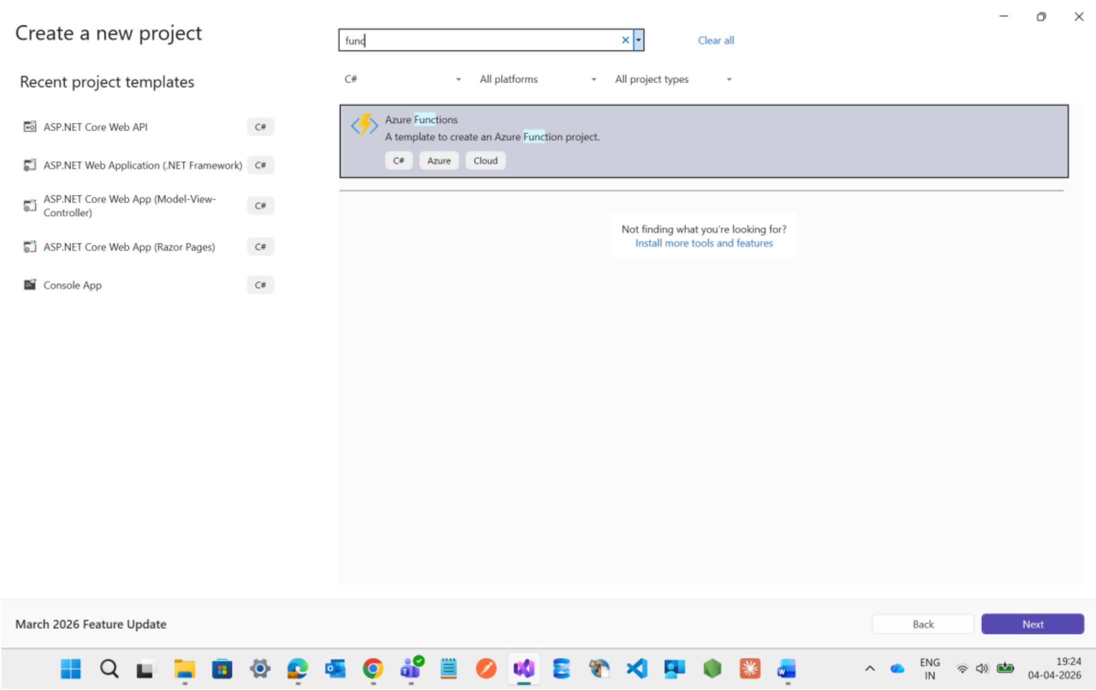
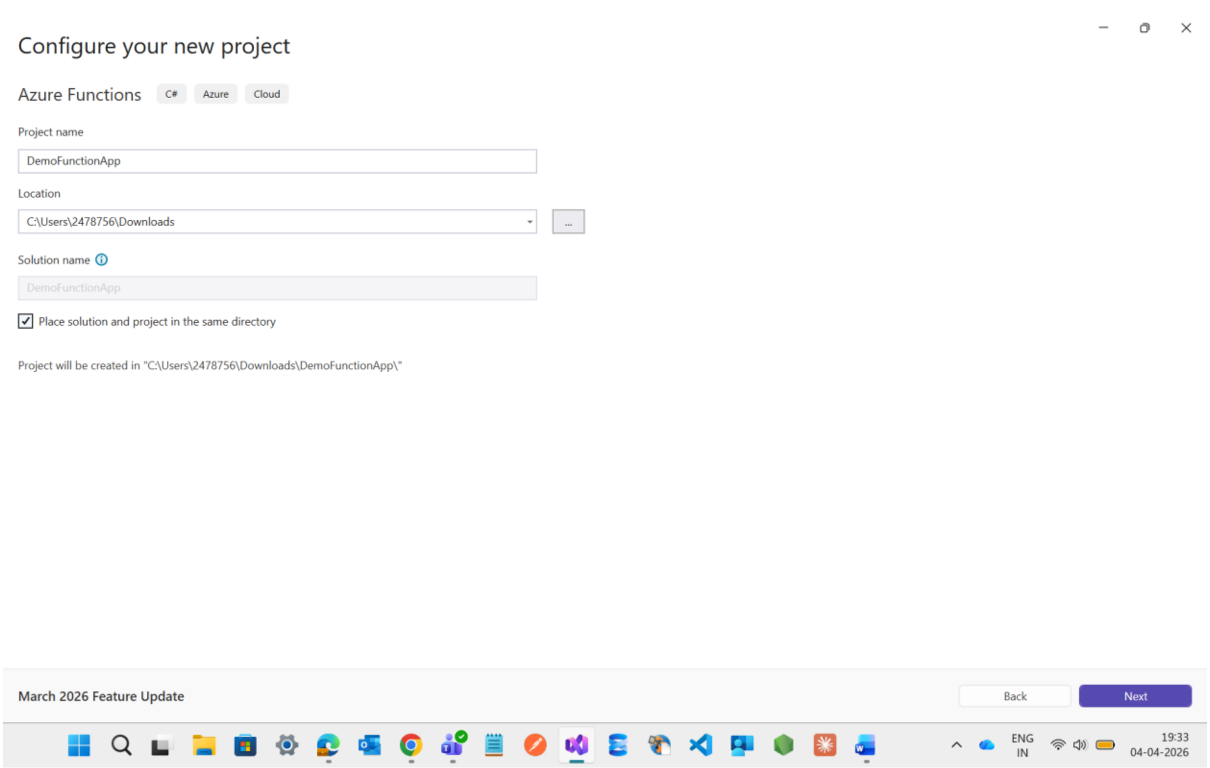
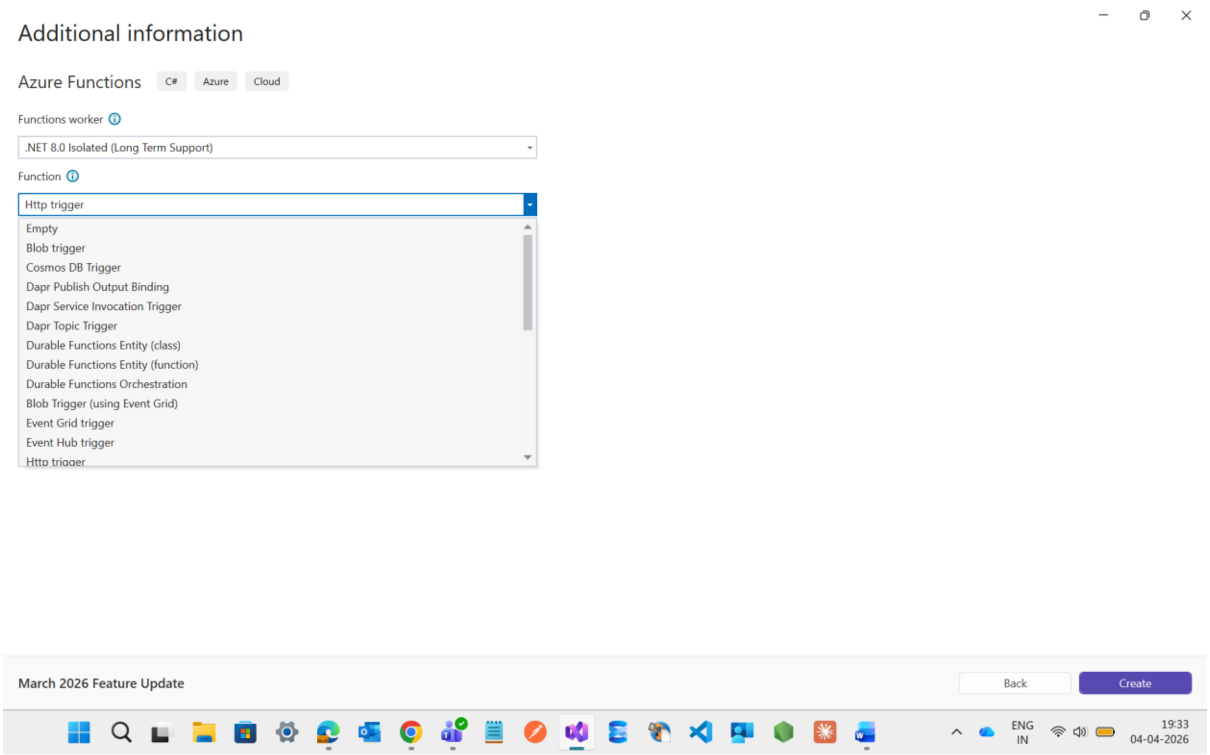
Event Driven Notification system

Now I m going to create one Function App through Visual Studio

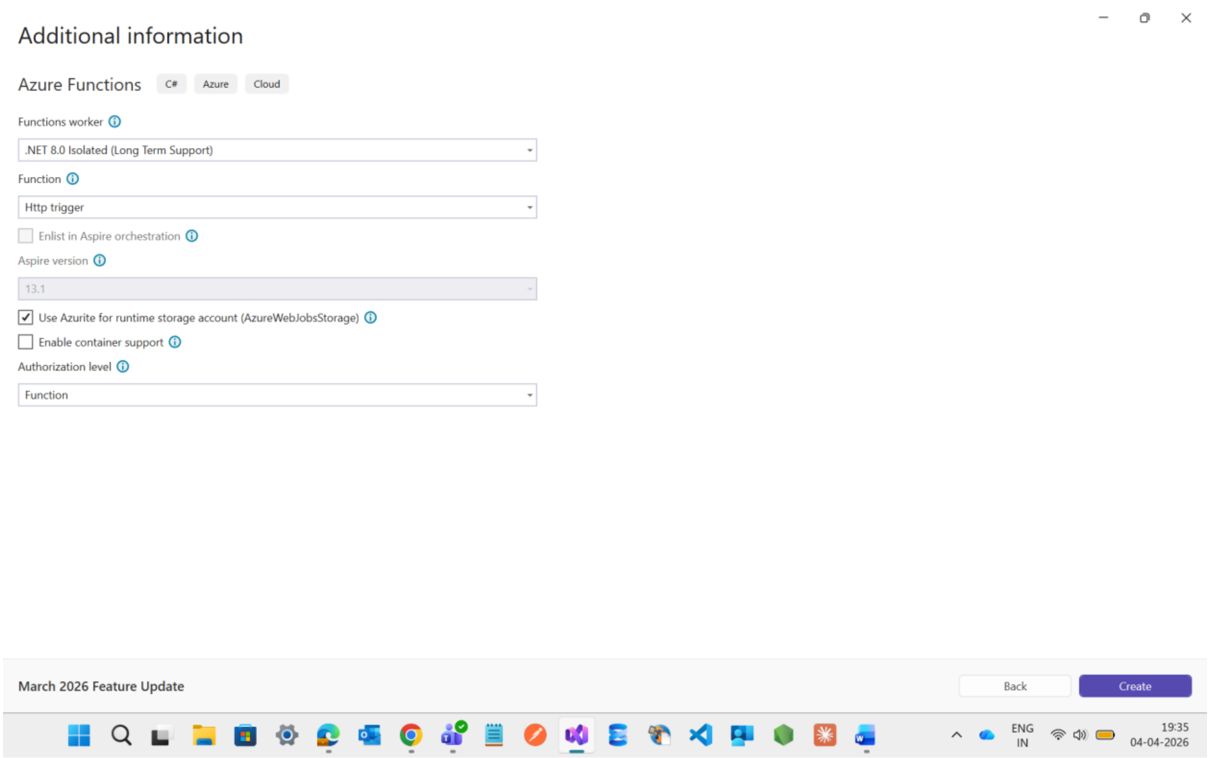




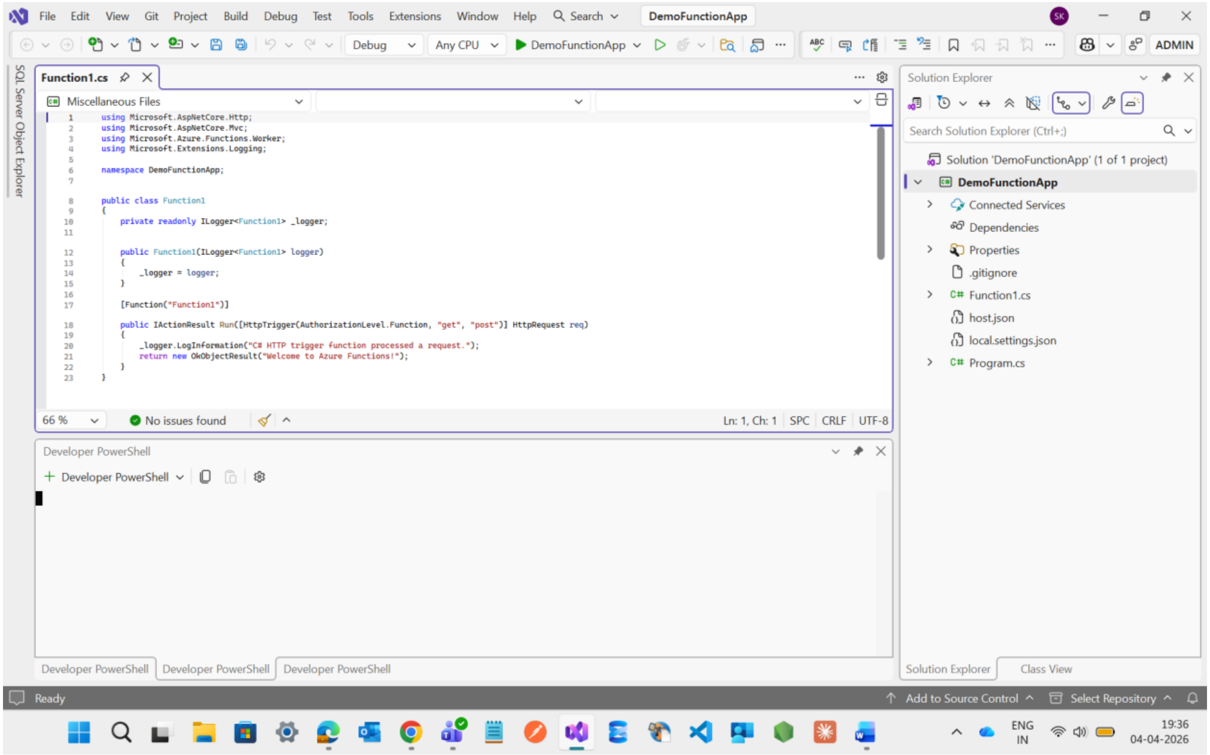
Here In this image we have several types of Trigger but for now only I going with http trigger



See I selected the Http trigger



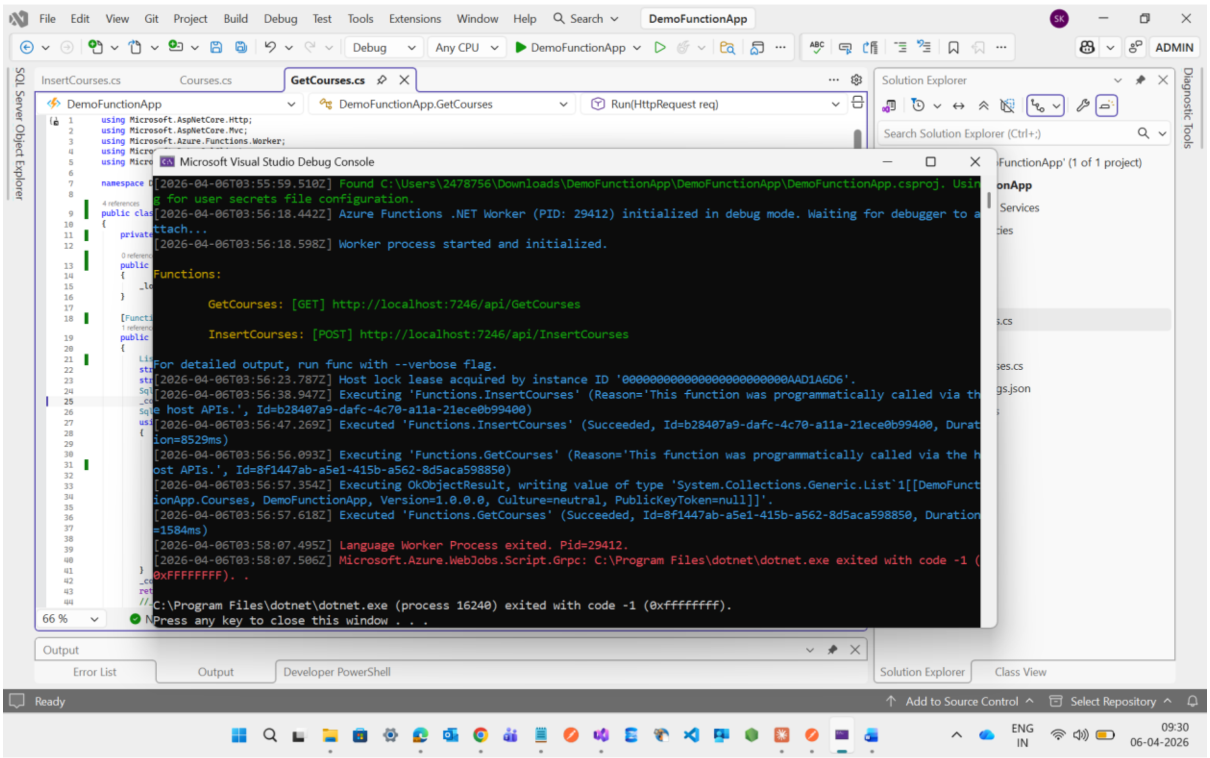
Here it will created a default class with the name of function



Now I m changing the class name of function to GetCourses

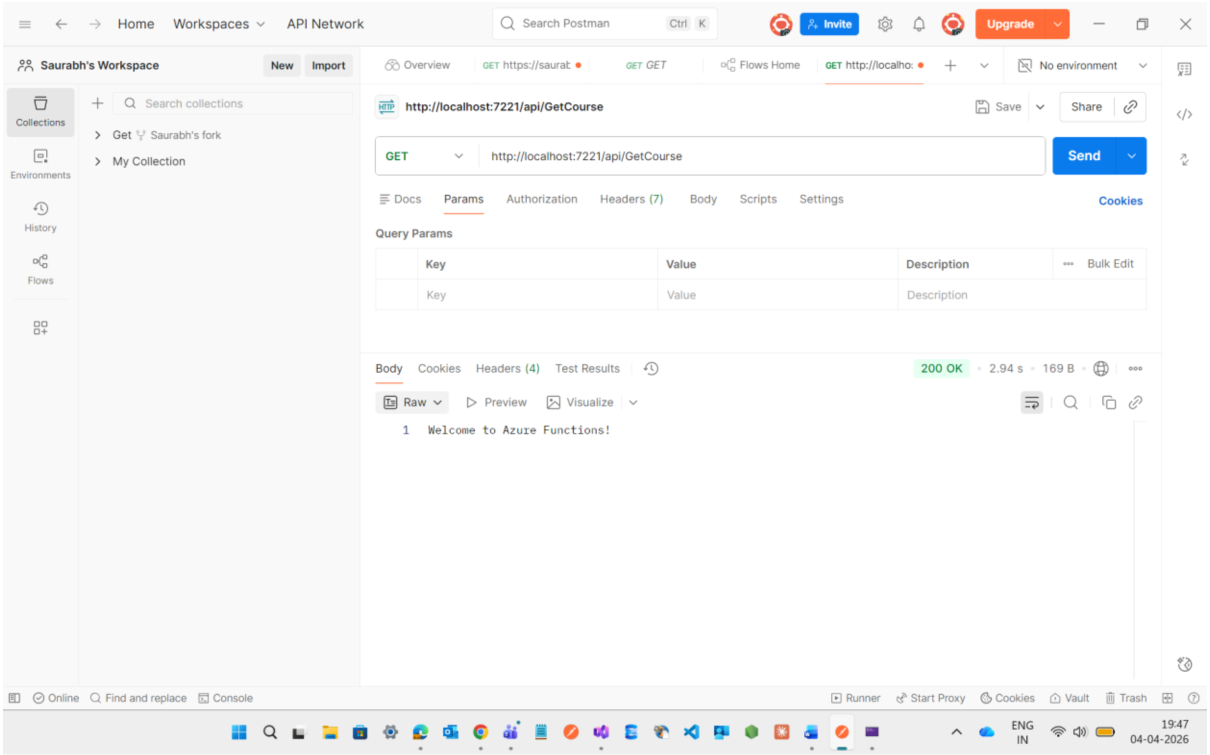
Now I executed the DemoFuctionApp and try to hit the url through Postman

And See After executing it I will get the url of GetCourses

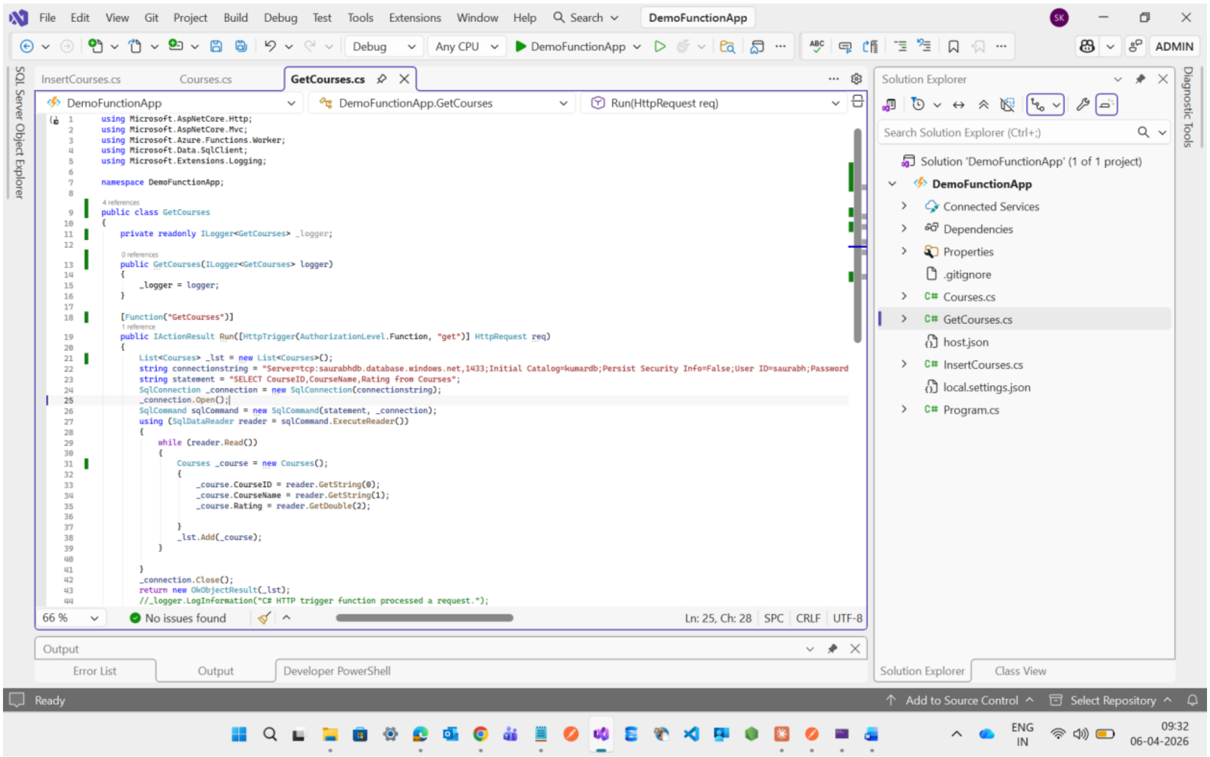


Now I will try to hit this url through Postman to check wheather I will get the output or not

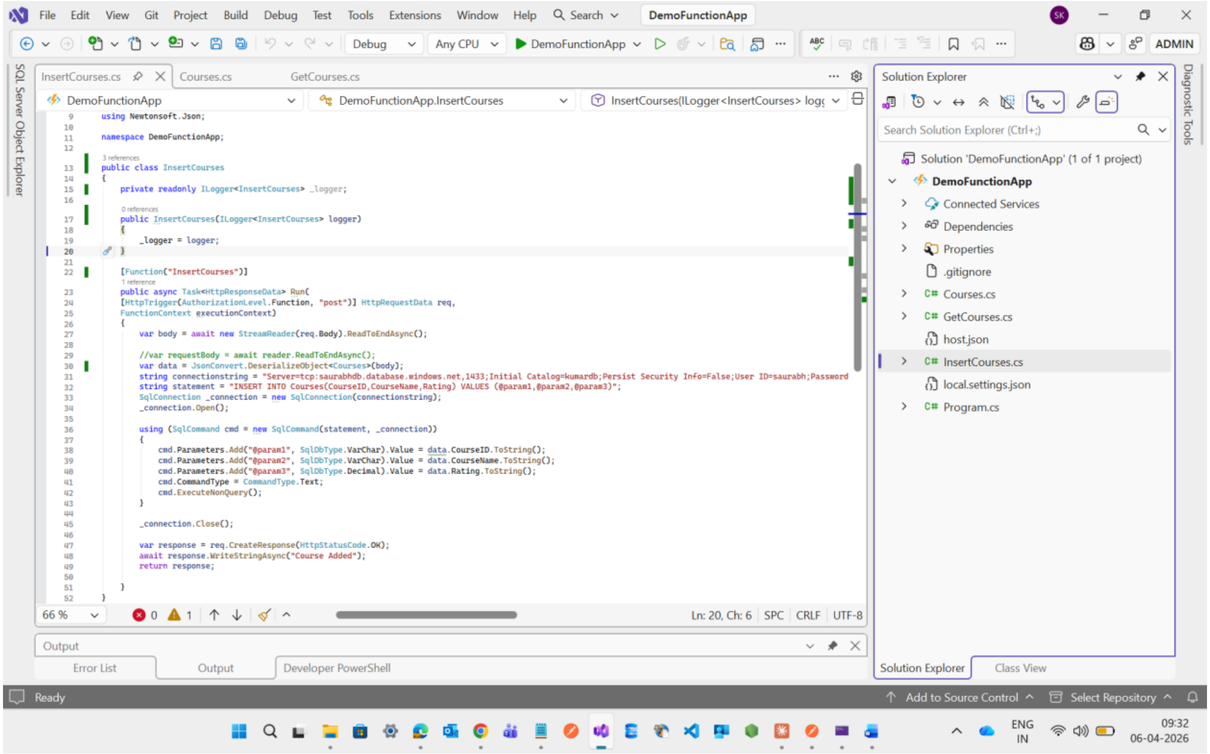
See I will get the output of Welcome to Azure Functions!



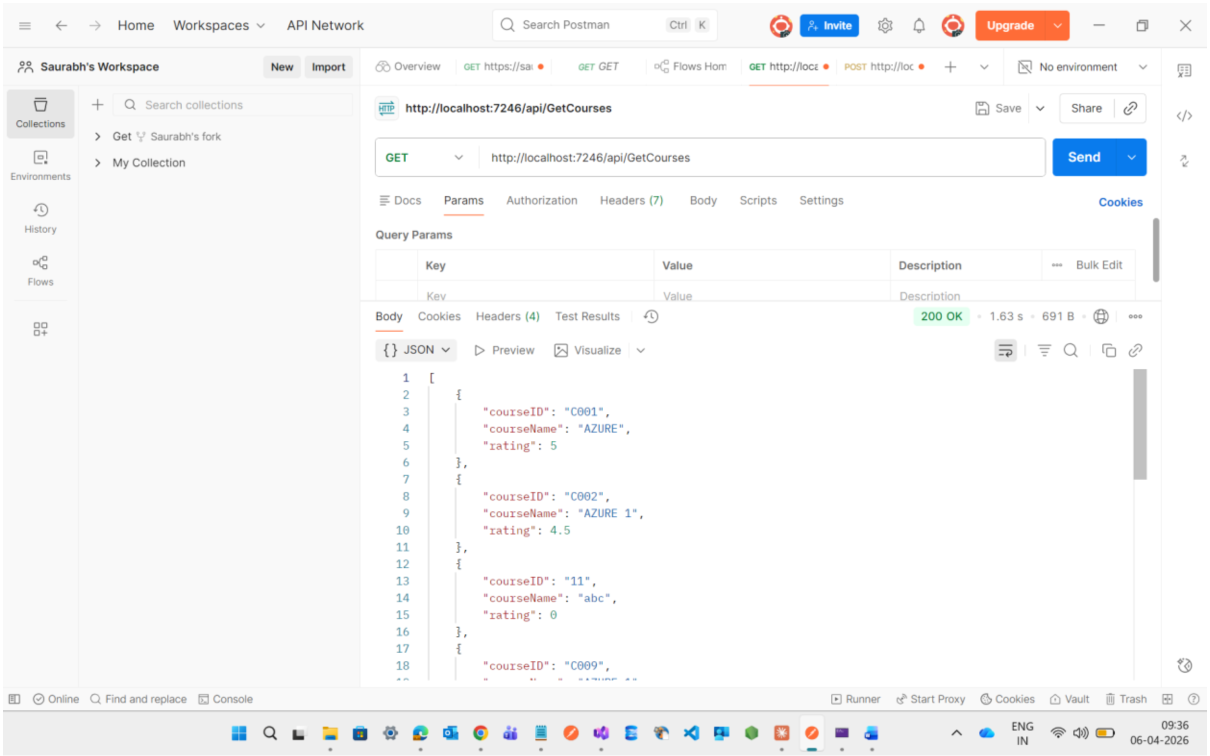
After checking the demo now I will insert the Get Method Code

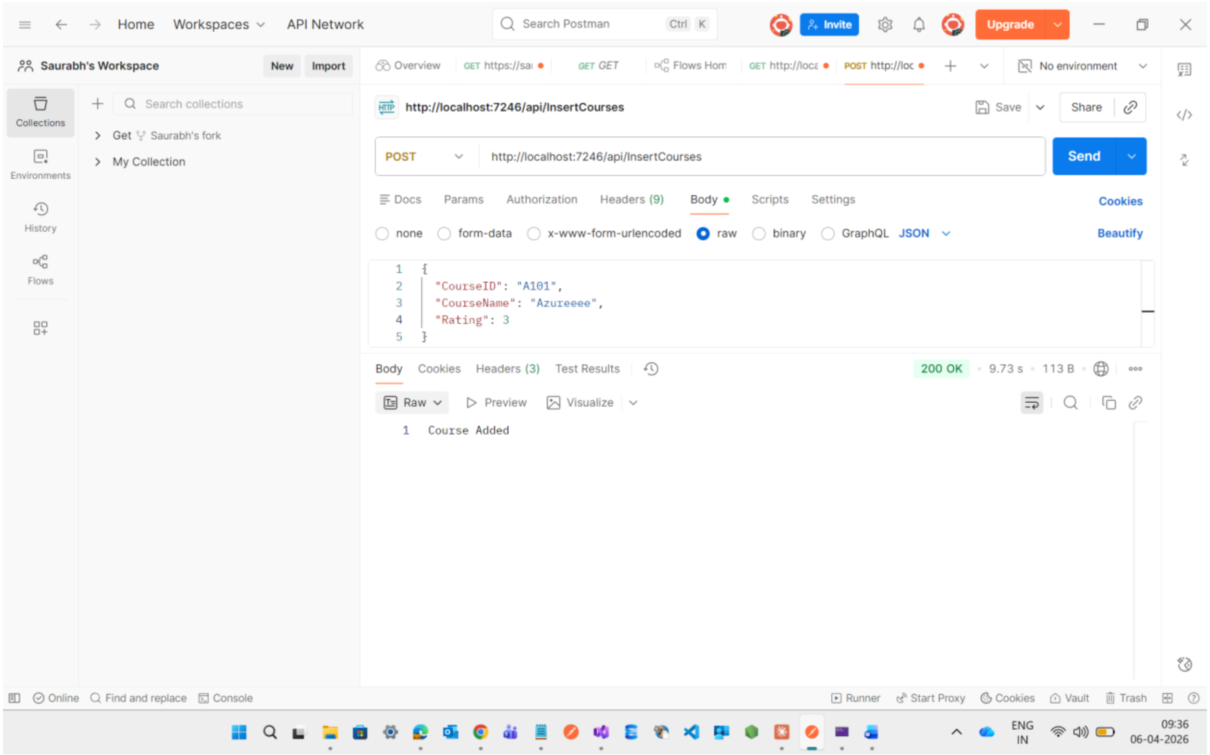


Then I will create one Add Method

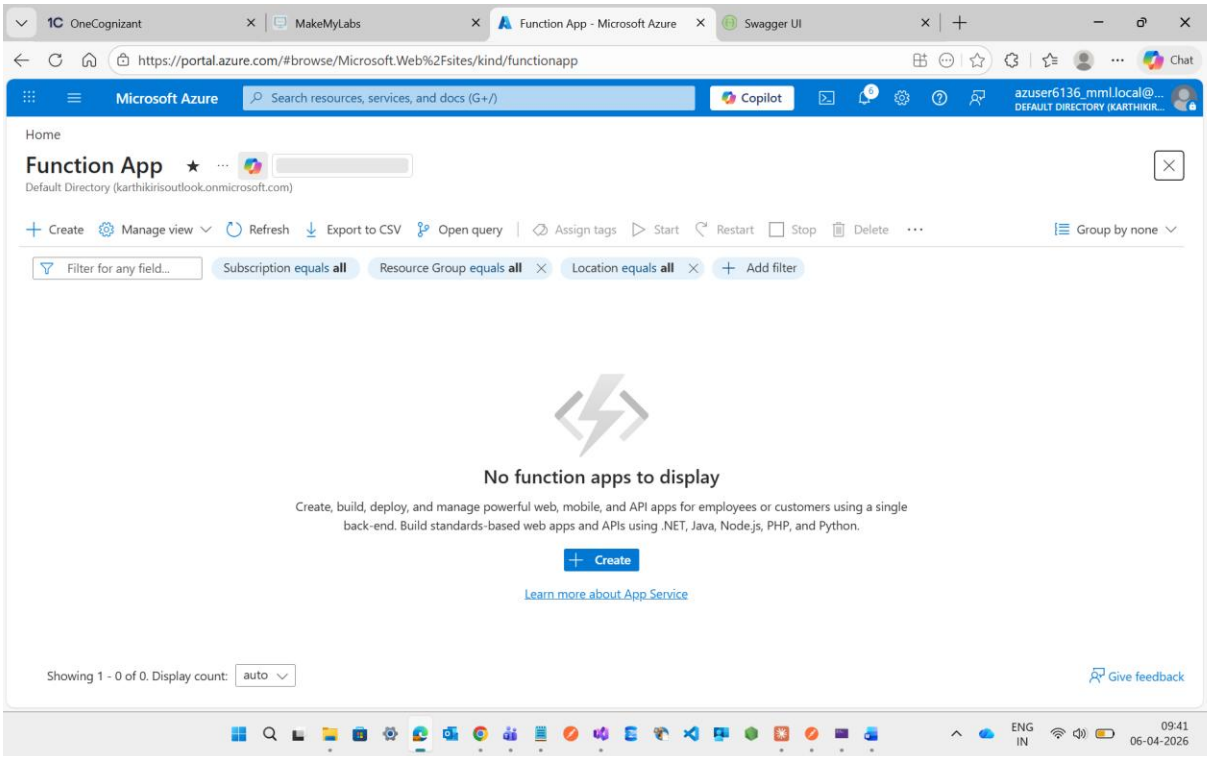


Then I will hit the url through postman it will return the output of both Get and Insert Method

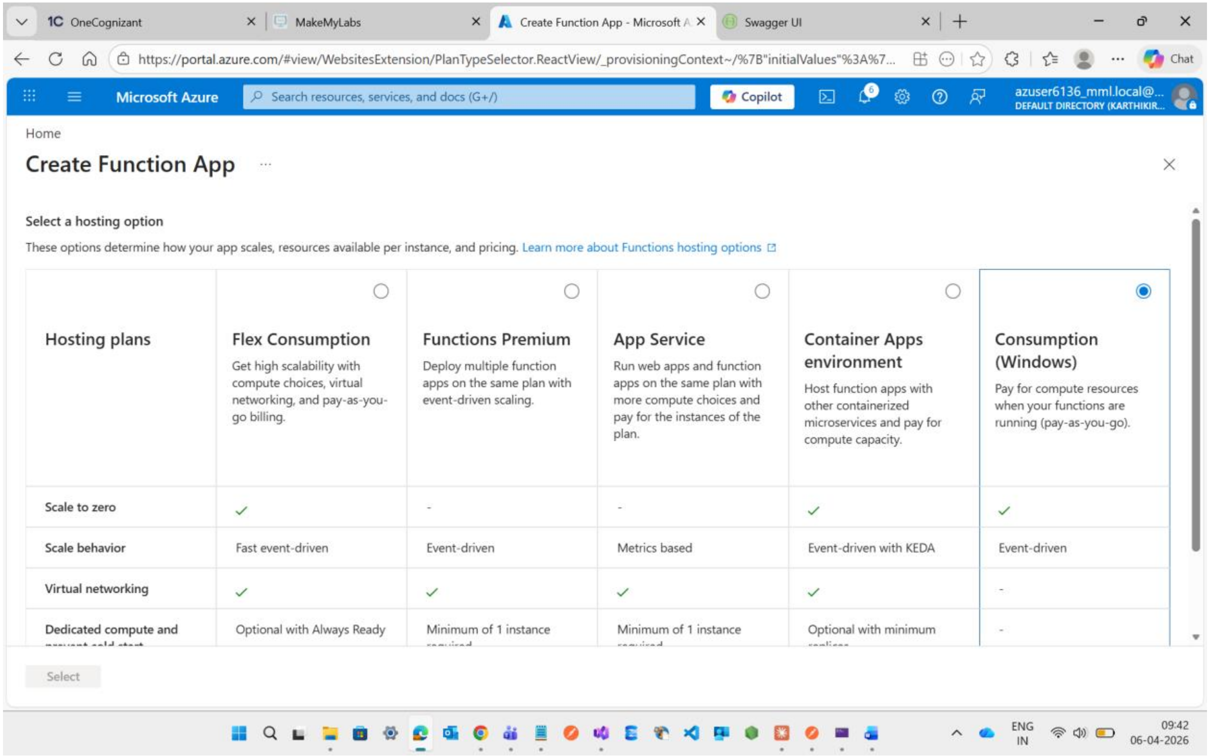


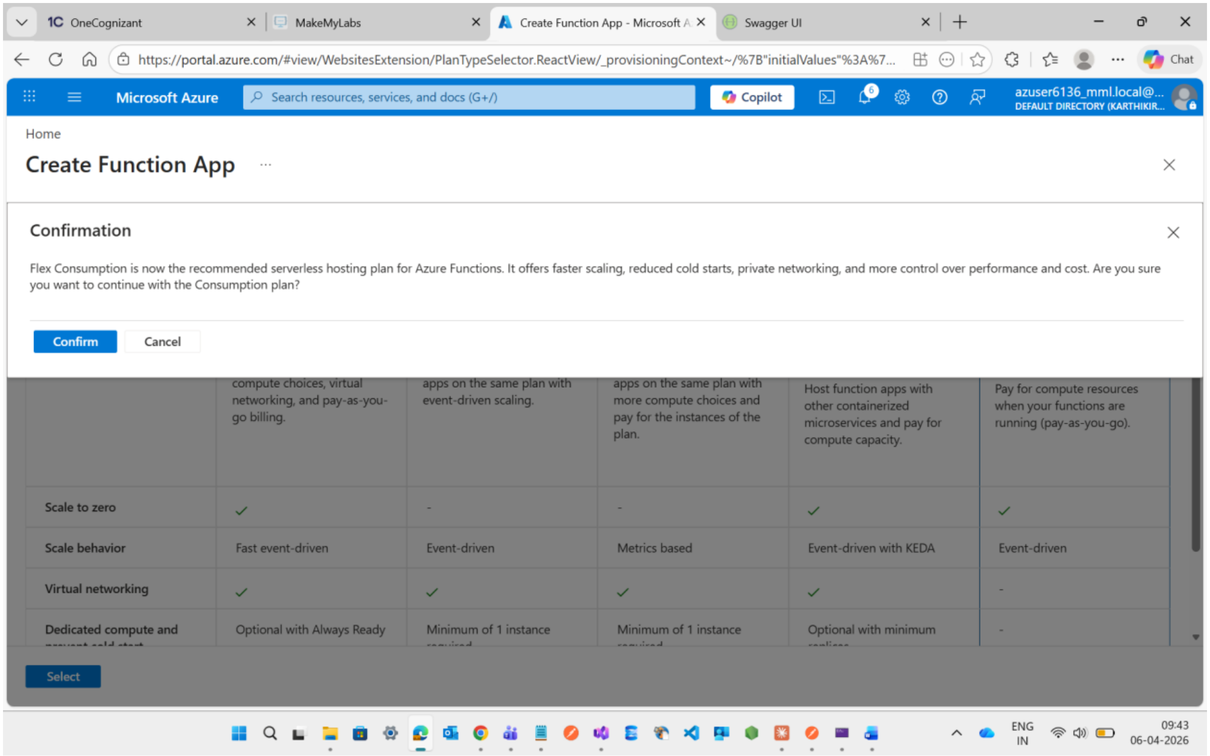


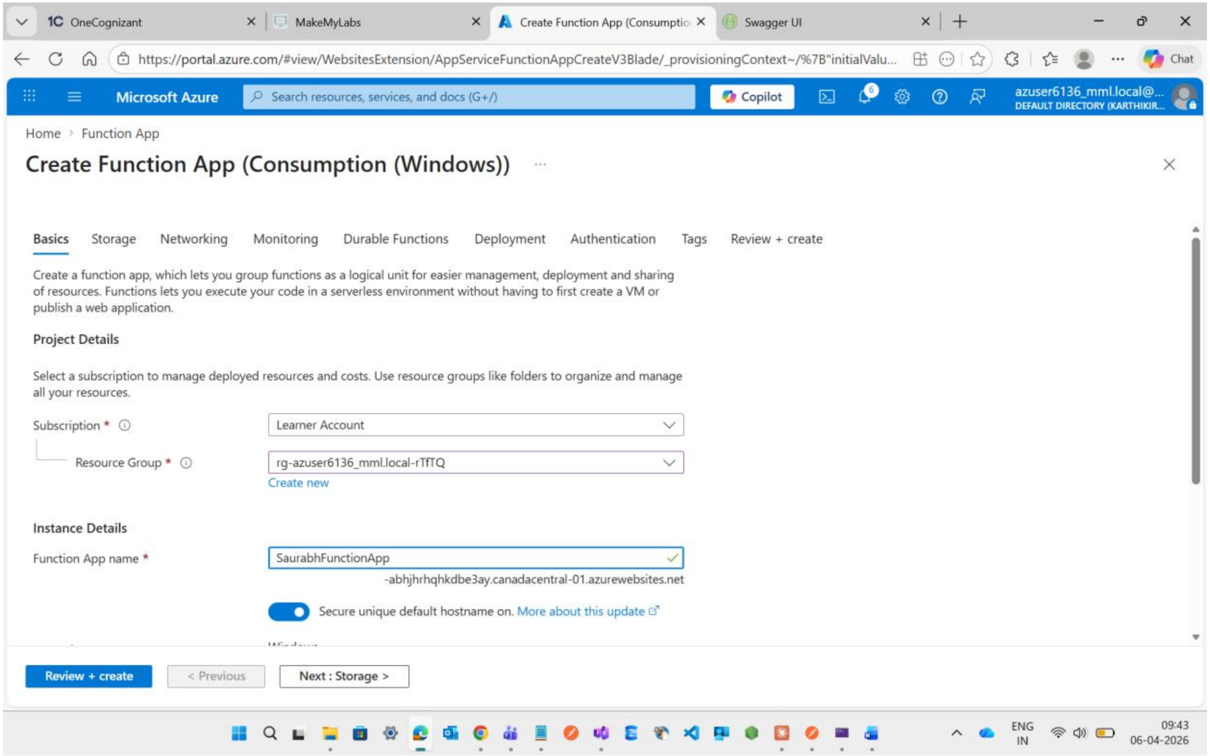
After this I will deploy this Function App on Azure Portal by creating one Function App resource



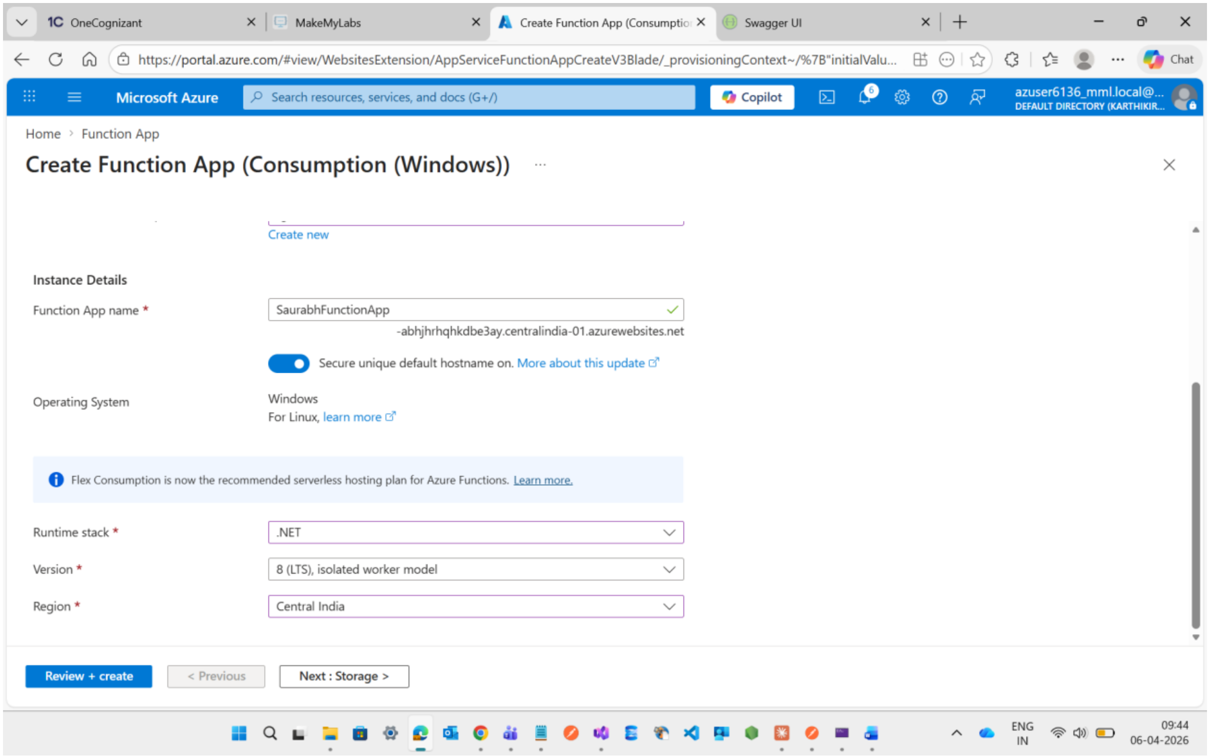
Then I will select the Consumption(windows) for pricing tier


OneCognizant
MakeMyLabs
Create Function App - Microsoft A
Swagger UI
https://portal.azure.com/#view/WebsitesExtension/PlanTypeSelector.ReactView/_provisioningContext~/%7B"initialValues"%3A%7...
Chat
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
azuser6136_mml.local@...
DEFAULT DIRECTORY (KARTHIKR...
Home
Create Function App
Select a hosting option
These options determine how your app scales, resources available per instance, and pricing. Learn more about Functions hosting options
Hosting plans
Flex Consumption
Get high scalability with compute choices, virtual networking, and pay-as-you-go billing.
Functions Premium
Deploy multiple function apps on the same plan with event-driven scaling.
App Service
Run web apps and function apps on the same plan with more compute choices and pay for the instances of the plan.
Container Apps environment
Host function apps with other containerized microservices and pay for compute capacity.
Consumption (Windows)
Pay for compute resources when your functions are running (pay-as-you-go).
Scale to zero
Scale behavior
Fast event-driven
Event-driven
Metrics based
Event-driven with KEDA
Event-driven
Virtual networking
Dedicated compute and
Optional with Always Ready
Minimum of 1 instance
Minimum of 1 instance
Optional with minimum
Select
ENG
IN
09:42
06-04-2026

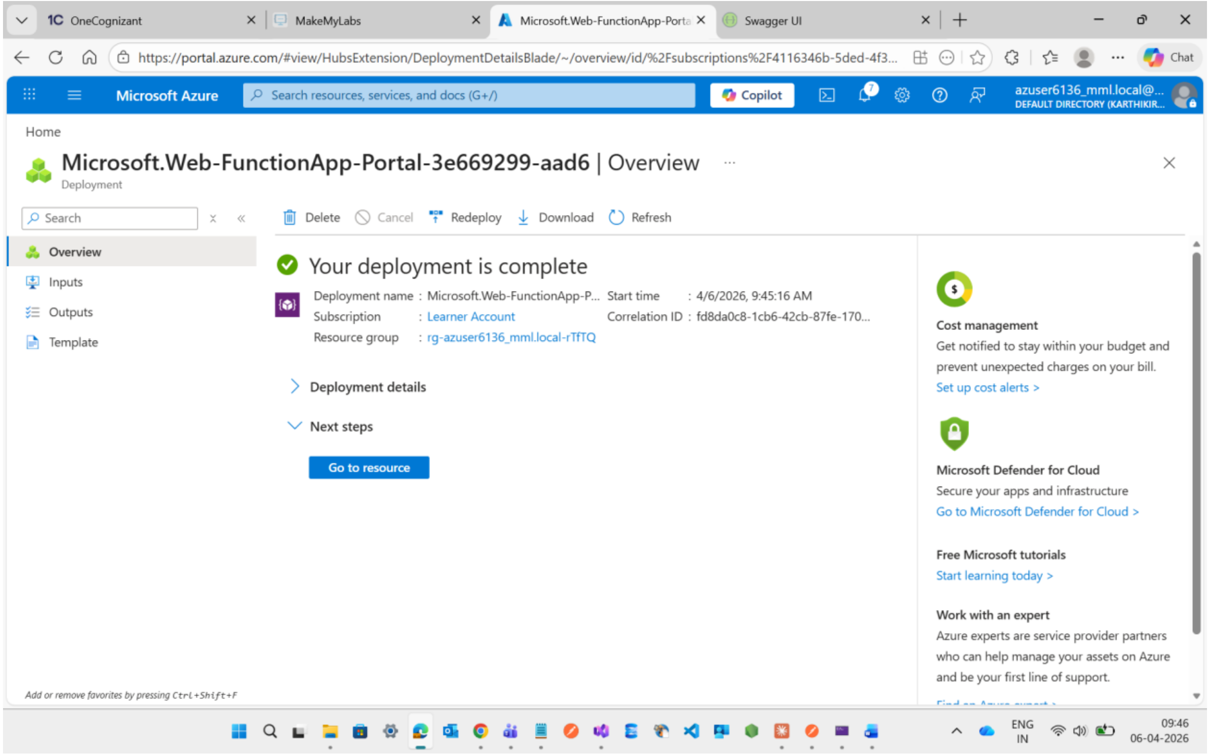

OneCognizant
MakeMyLabs
Create Function App - Microsoft A
Swagger UI
https://portal.azure.com/#view/WebsitesExtension/PlanTypeSelector.ReactView/_provisioningContext~/%7B"initialValues"%3A%7...
Chat
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
azuser6136_mml.local@...
DEFAULT DIRECTORY (KARTHIKR...
Home
Create Function App
Confirmation
Flex Consumption is now the recommended serverless hosting plan for Azure Functions. It offers faster scaling, reduced cold starts, private networking, and more control over performance and cost. Are you sure you want to continue with the Consumption plan?
Confirm
Cancel
Scale to zero
Scale behavior
Fast event-driven
Event-driven
Metrics based
Event-driven with KEDA
Event-driven
Virtual networking
Dedicated compute and
Optional with Always Ready
Minimum of 1 instance
Minimum of 1 instance
Optional with minimum
Select
ENG
IN
09:43
06-04-2026



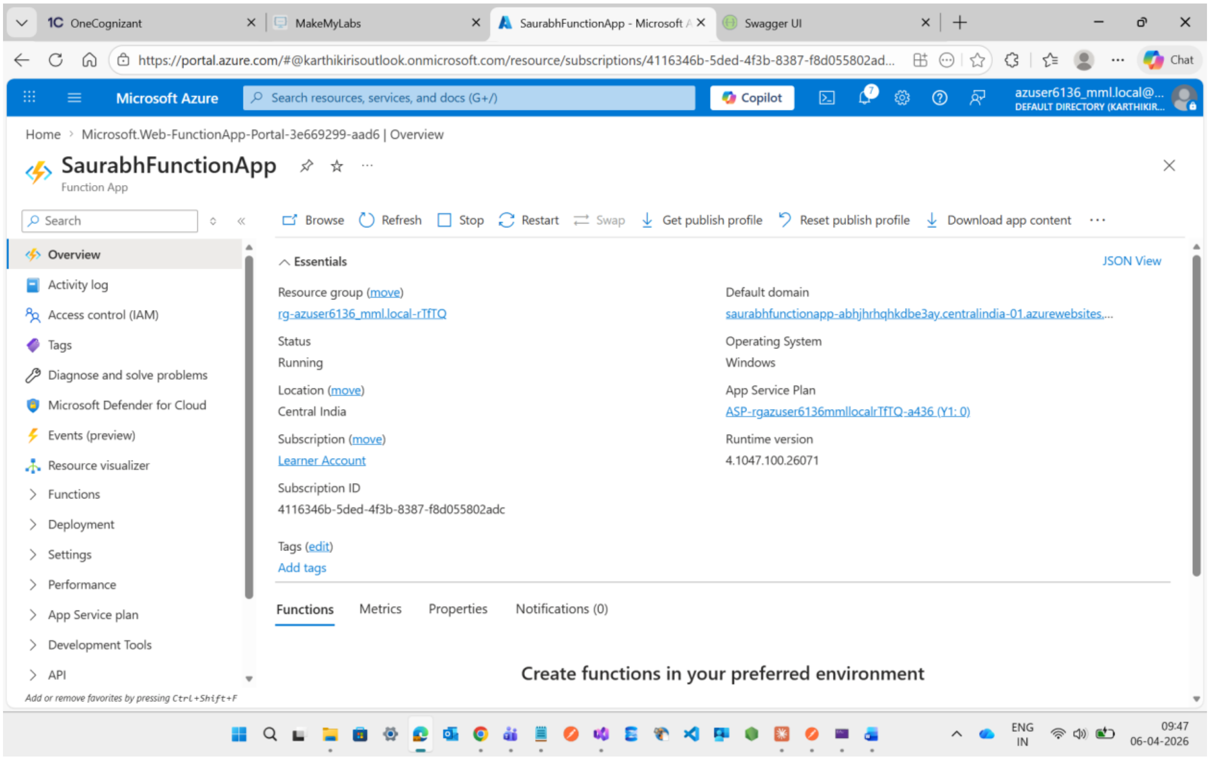
OneCognizant
MakeMyLabs
Create Function App (Consumptio
Swagger UI
https://portal.azure.com/#view/WebsitesExtension/AppServiceFunctionAppCreateV3Blade/_provisioningContext~/%7B"initialValu...
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
azuser6136_mml.local@...
DEFAULT DIRECTORY (KARTHIKR...
Home > Function App
Create Function App (Consumption (Windows))
Basics
Storage
Networking
Monitoring
Durable Functions
Deployment
Authentication
Tags
Review + create
Create a function app, which lets you group functions as a logical unit for easier management, deployment and sharing of resources. Functions lets you execute your code in a serverless environment without having to first create a VM or publish a web application.
Project Details
Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Learner Account
Resource Group *
rg-azuser6136_mml.local-rTfTQ
Create new
Instance Details
Function App name *
SaurabhFunctionApp
-abhjhrhqhkdbe3ay.canadacentral-01.azurewebsites.net
Secure unique default hostname on. More about this update
Review + create
< Previous
Next : Storage >
ENG
IN
09:43
06-04-2026



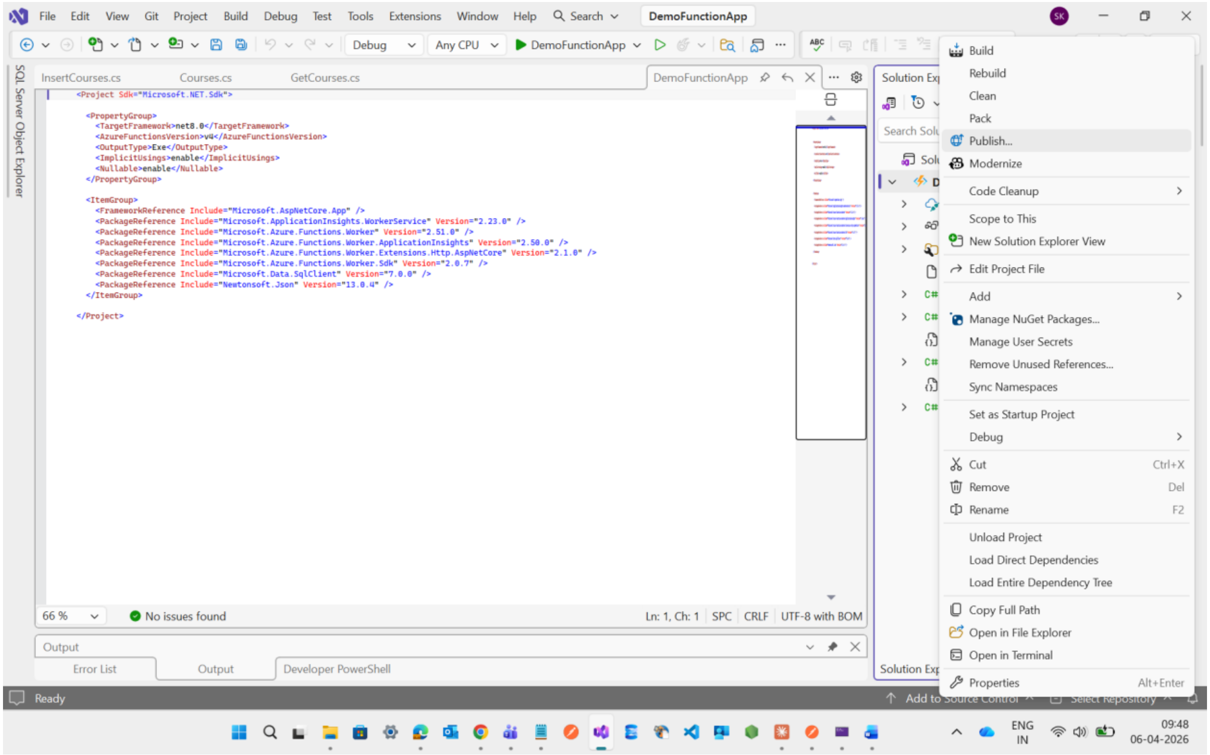
OneCognizant
MakeMyLabs
Create Function App (Consumptio
Swagger UI
https://portal.azure.com/#view/WebsitesExtension/AppServiceFunctionAppCreateV3Blade/_provisioningContext~/%7B"initialValu...
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
azuser6136_mml.local@...
DEFAULT DIRECTORY (KARTHIKR...
Home > Function App
Create Function App (Consumption (Windows))
Create new
Instance Details
Function App name *
SaurabhFunctionApp
-abhjhrhqhkdbe3ay.centralindia-01.azurewebsites.net
Secure unique default hostname on. More about this update
Operating System
Windows
For Linux, learn more
Flex Consumption is now the recommended serverless hosting plan for Azure Functions. Learn more.
Runtime stack *
.NET
Version *
8 (LTS), isolated worker model
Region *
Central India
Review + create
< Previous
Next : Storage >
ENG
IN
09:44
06-04-2026

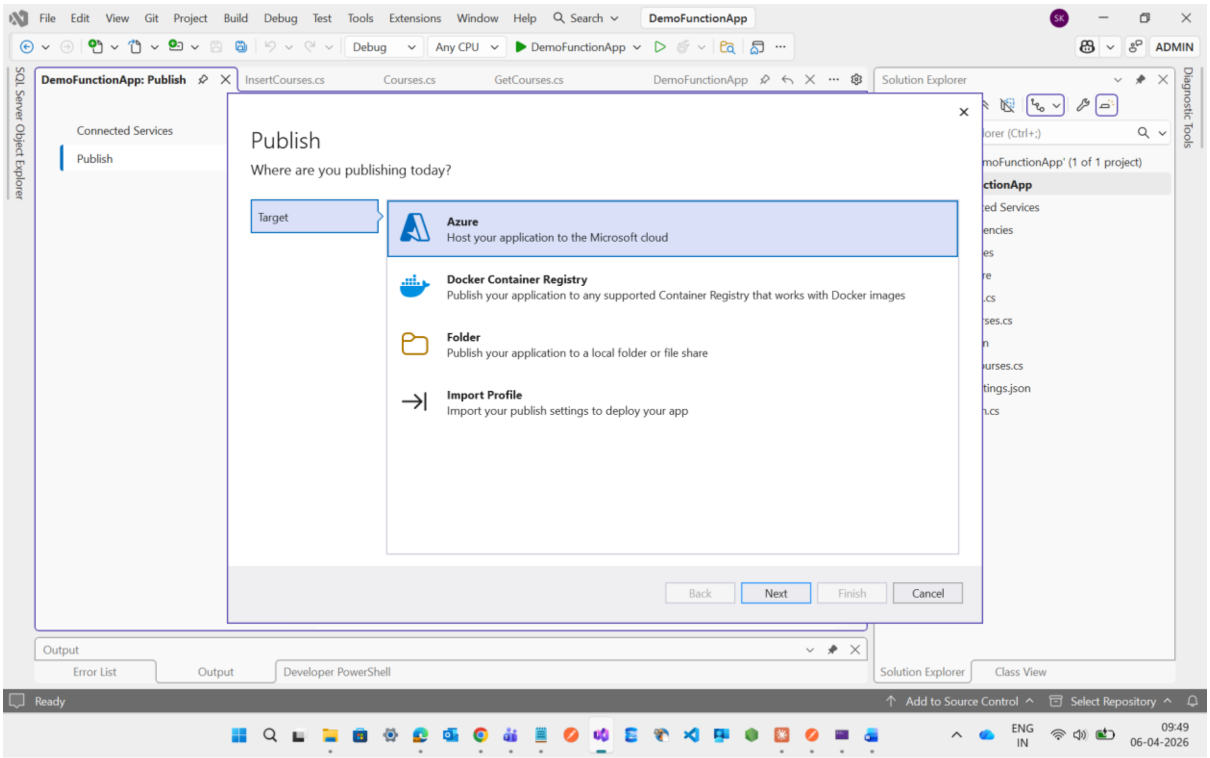


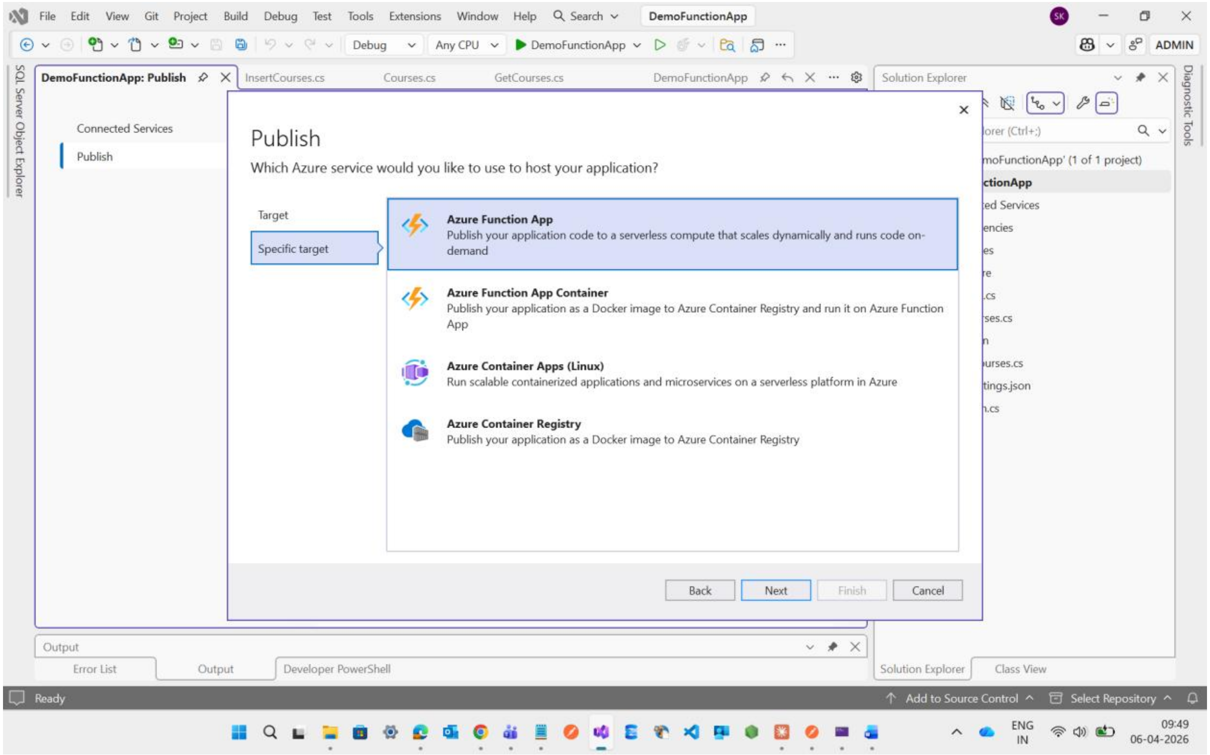
Here Function App is created



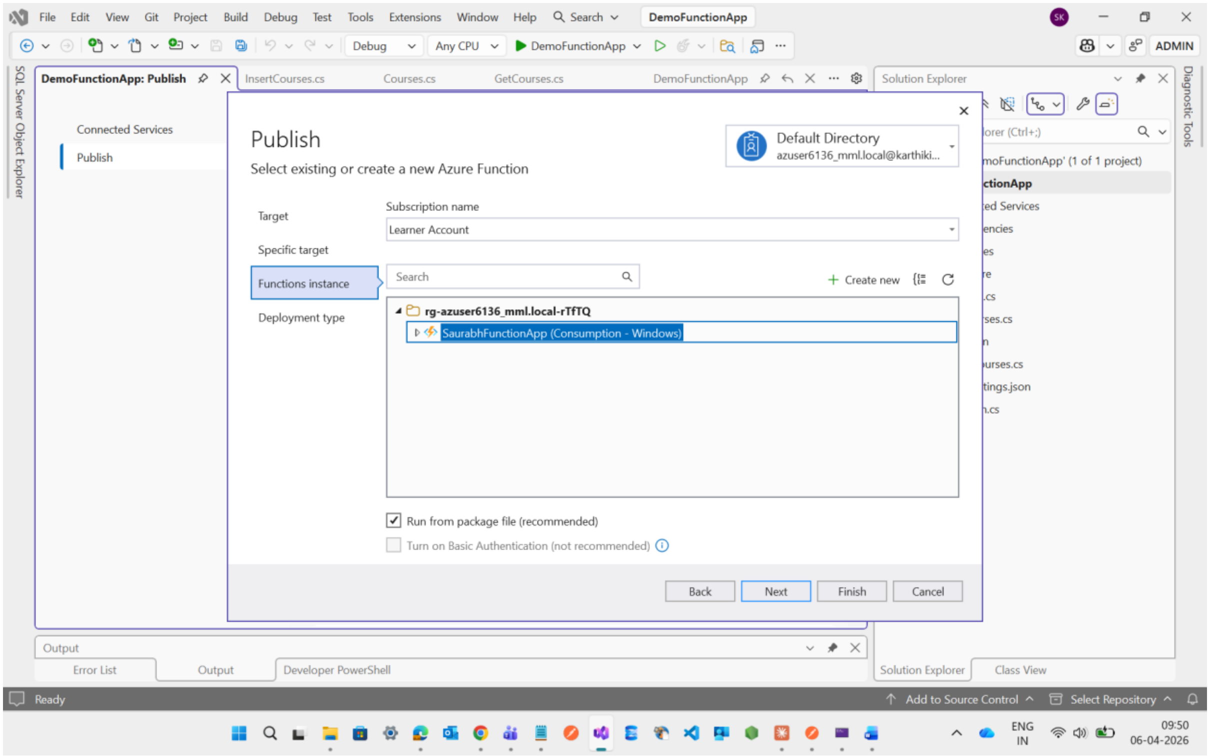
Now I m going to visual studio to publish my function app code


DemoFunctionApp
Build
Rebuild
Clean
Pack
Publish...
Modernize
Code Cleanup
Scope to This
New Solution Explorer View
Edit Project File
Add
Manage NuGet Packages...
Manage User Secrets
Remove Unused References...
Sync Namespaces
Set as Startup Project
Debug
Cut Ctrl+X
Remove Del
Rename F2
Unload Project
Load Direct Dependencies
Load Entire Dependency Tree
Copy Full Path
Open in File Explorer
Open in Terminal
Properties Alt+Enter
No issues found
Ready

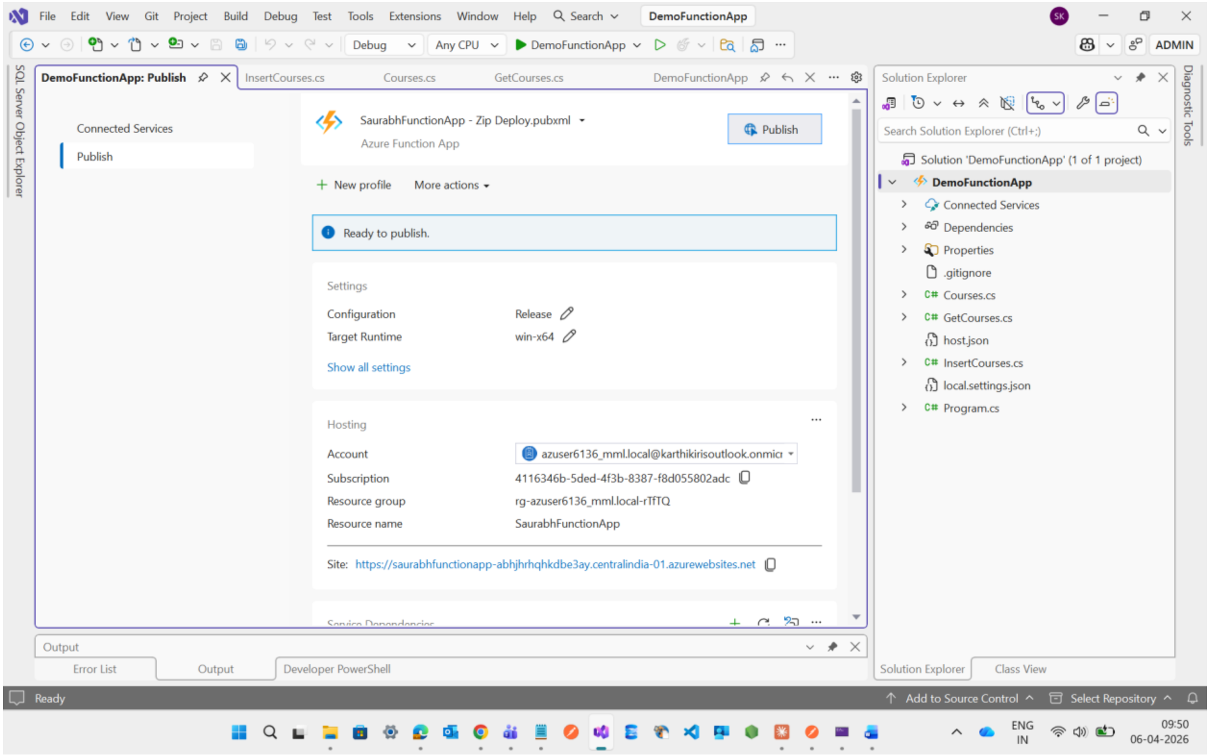

DemoFunctionApp: Publish
Connected Services
Publish
Where are you publishing today?
Target
Azure
Host your application to the Microsoft cloud
Docker Container Registry
Publish your application to any supported Container Registry that works with Docker images
Folder
Publish your application to a local folder or file share
Import Profile
Import your publish settings to deploy your app
Back
Next
Finish
Cancel
Ready

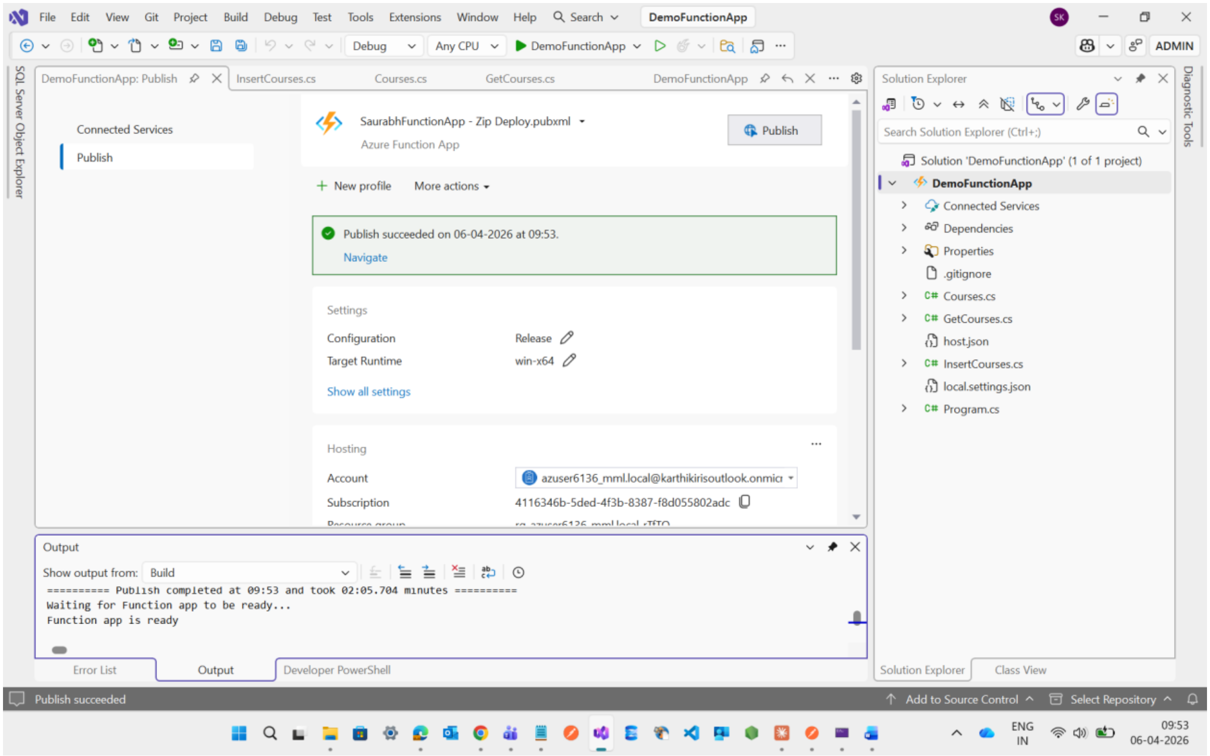


DemoFunctionApp: Publish
Connected Services
Publish
Publish
Which Azure service would you like to use to host your application?
Target
Specific target
Azure Function App
Publish your application code to a serverless compute that scales dynamically and runs code on-demand
Azure Function App Container
Publish your application as a Docker image to Azure Container Registry and run it on Azure Function App
Azure Container Apps (Linux)
Run scalable containerized applications and microservices on a serverless platform in Azure
Azure Container Registry
Publish your application as a Docker image to Azure Container Registry
Back
Next
Finish
Cancel
Solution Explorer
Output
Error List
Developer PowerShell
Ready
Add to Source Control
Select Repository



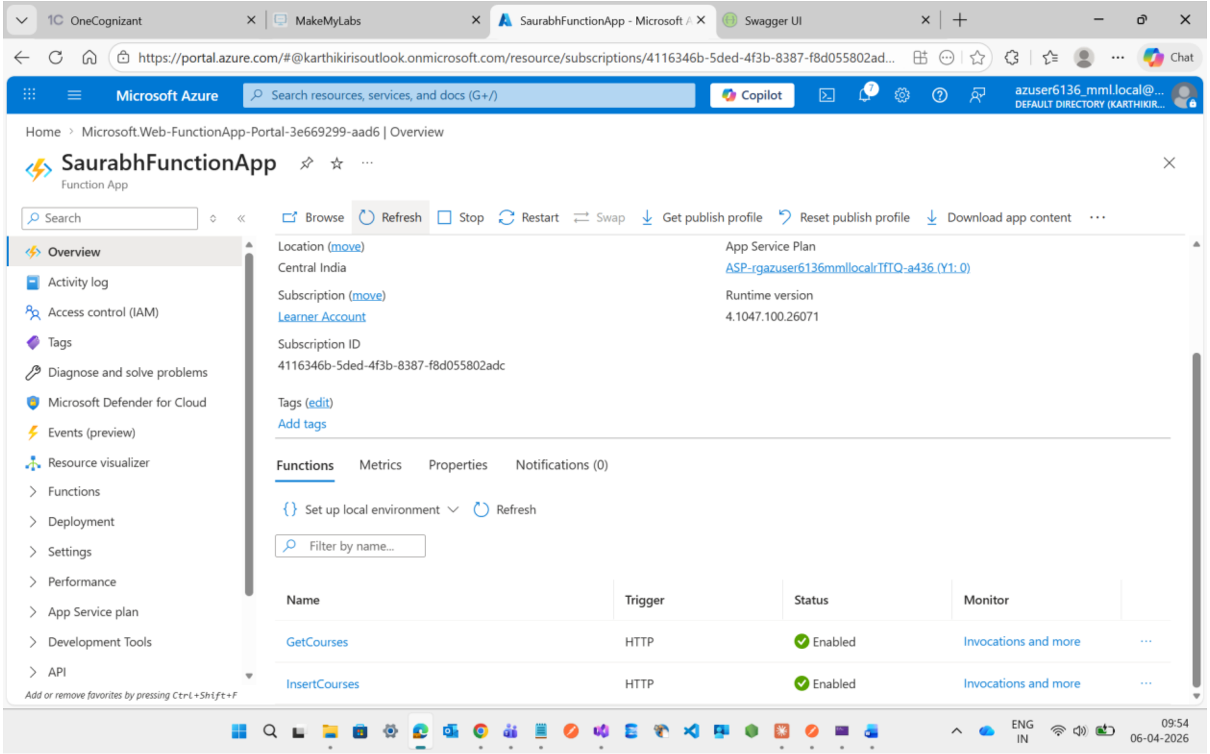
DemoFunctionApp: Publish
Connected Services
Publish
Publish
Select existing or create a new Azure Function
Default Directory
azuser6136_mml.local@karthiki...
Target
Specific target
Functions instance
Deployment type
Subscription name
Learner Account
Search
Create new
rg-azuser6136_mml.local-rTfTQ
SaurabhFunctionApp (Consumption - Windows)
Run from package file (recommended)
Turn on Basic Authentication (not recommended)
Back
Next
Finish
Cancel
Output
Error List
Developer PowerShell
Ready
Add to Source Control
Select Repository





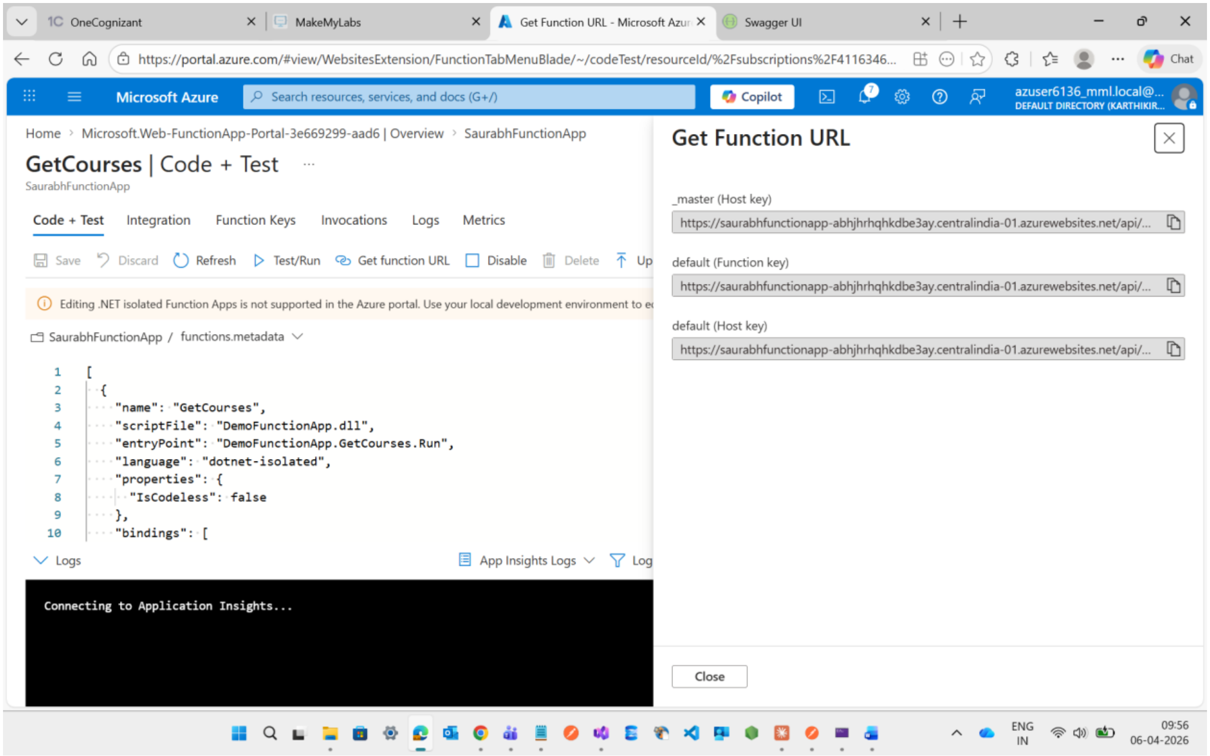
After Published it now I m going to my Function App in Azure with the name of SaurabhFunctionApp

Now see here two method is showing GetCourses, InsertCourses



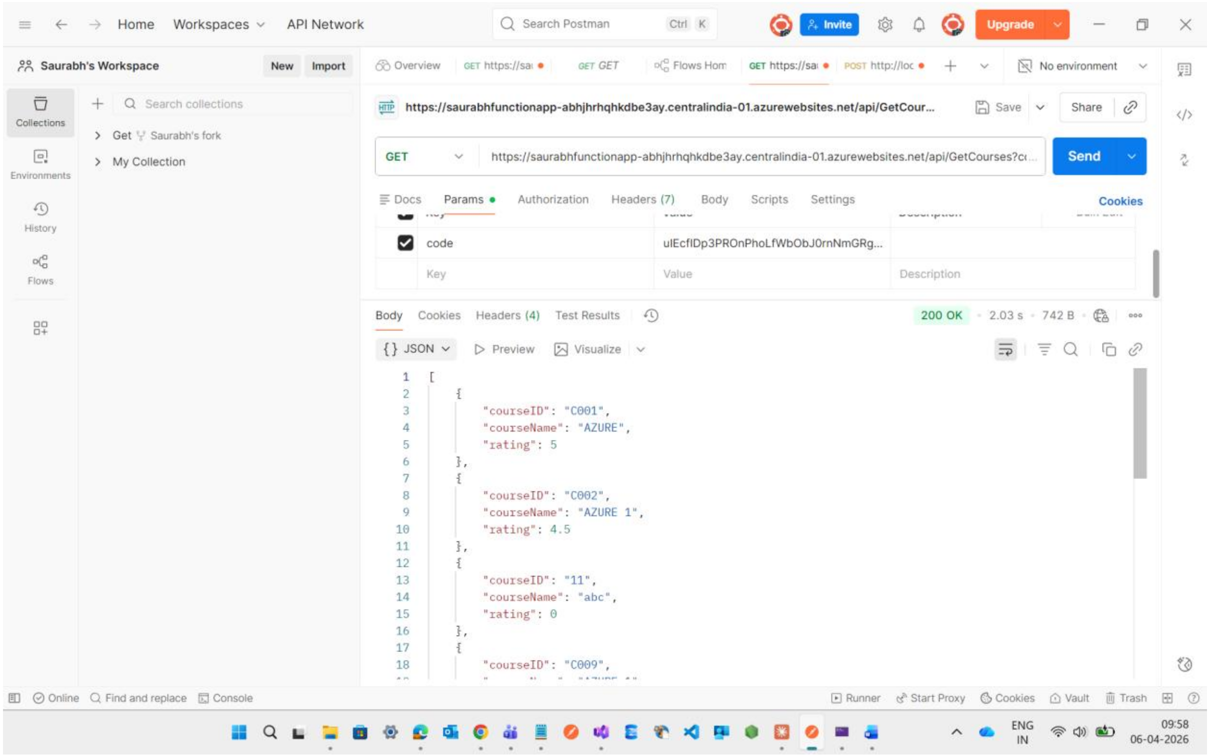
Now I will check the GetCourses Method

Here is the URL of this Get Function

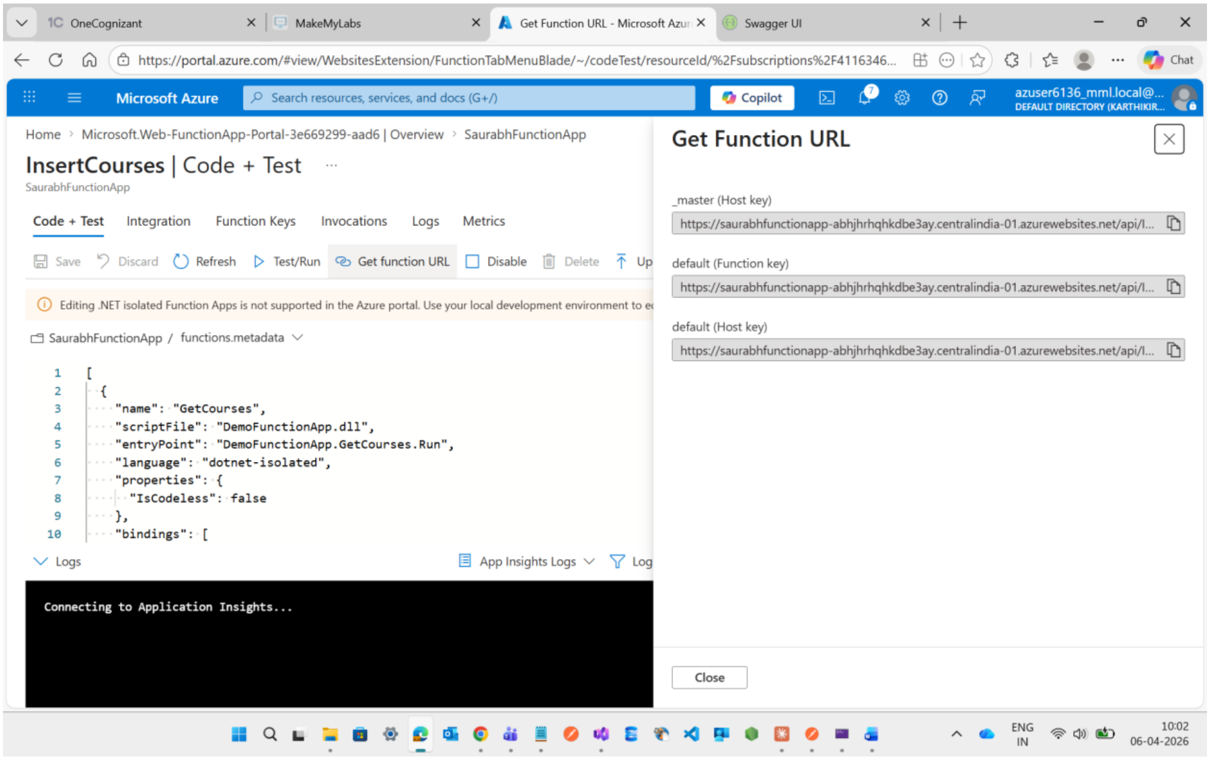


Now I will copy the default link to check it wheather the output is coming or not through Postman

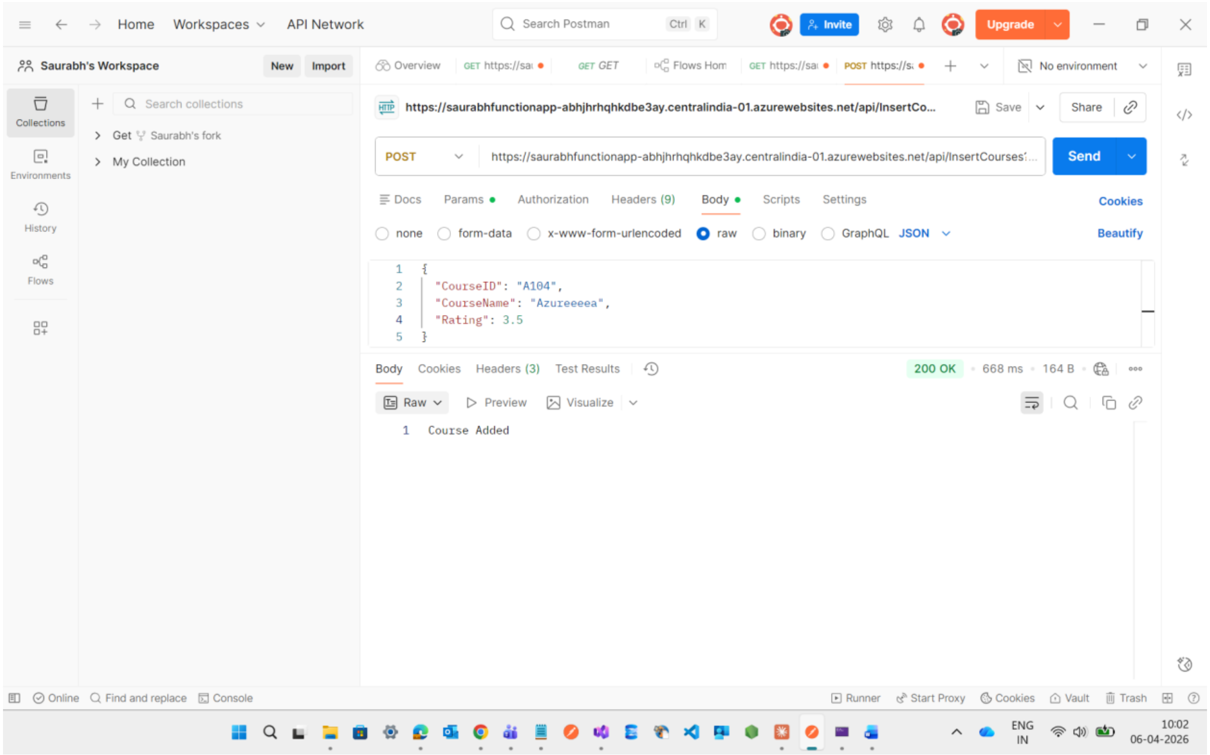
See For Get Function output is clearly visible



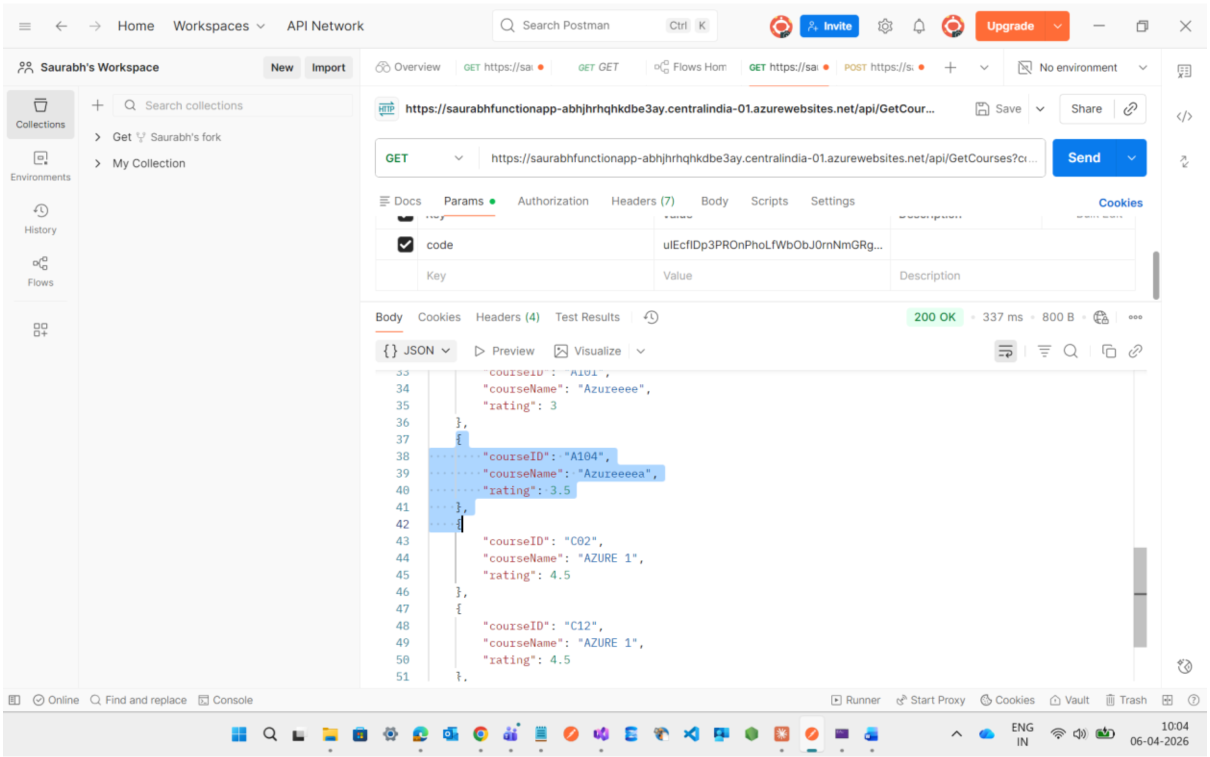
Now I will try with Insert Function through Postman



See this will also work fine

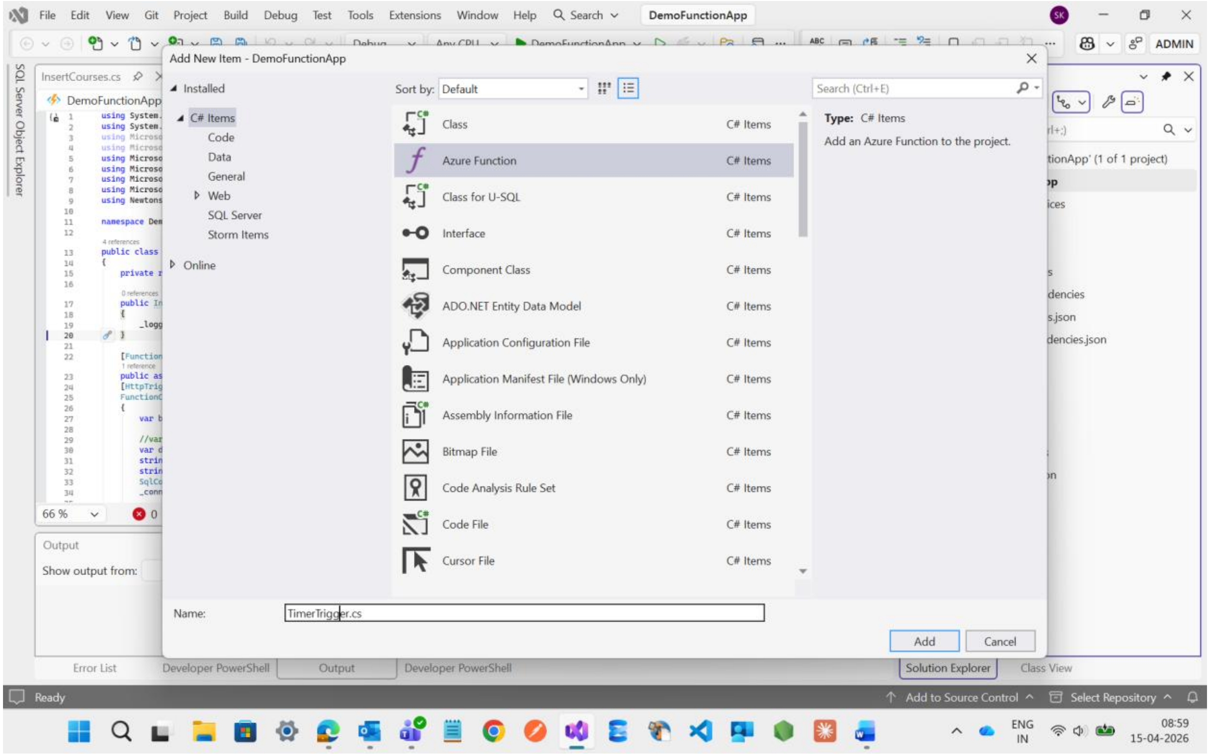


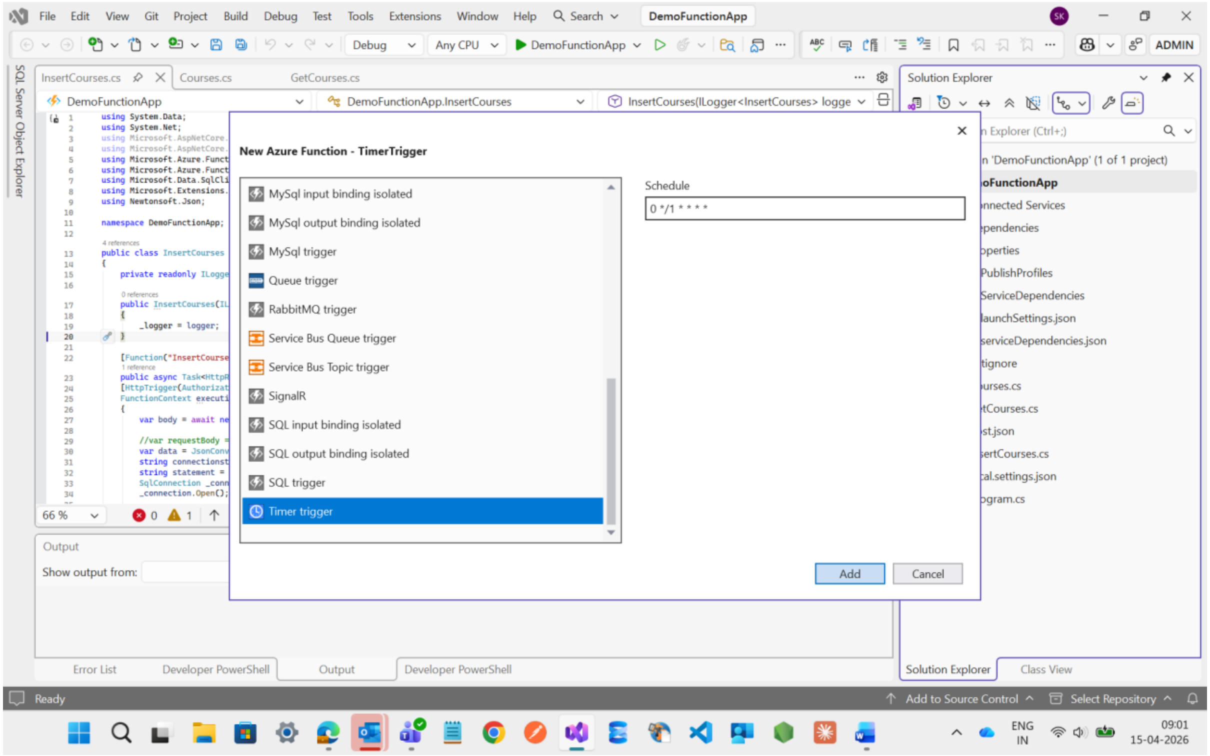
Here the added courses is clearly visible



Timer Trigger

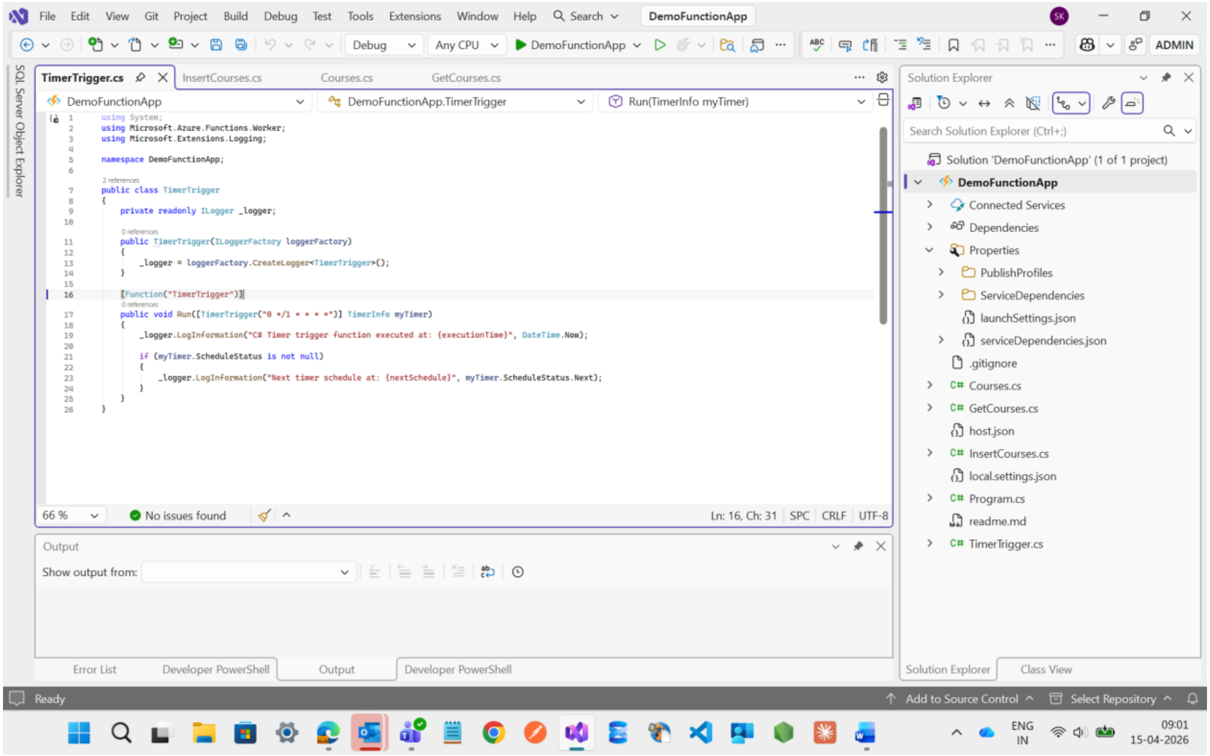
Now I m going to create a one Timer Trigger file in Function App



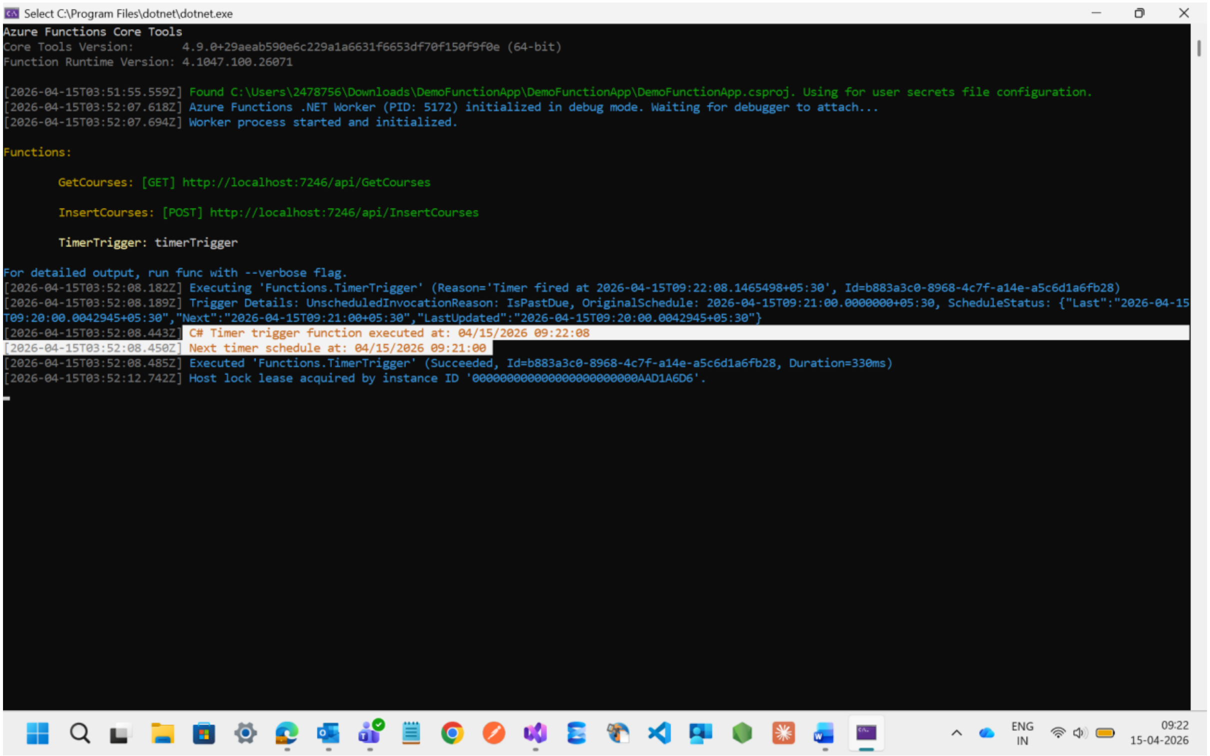


See here I have created a Timer Trigger file.

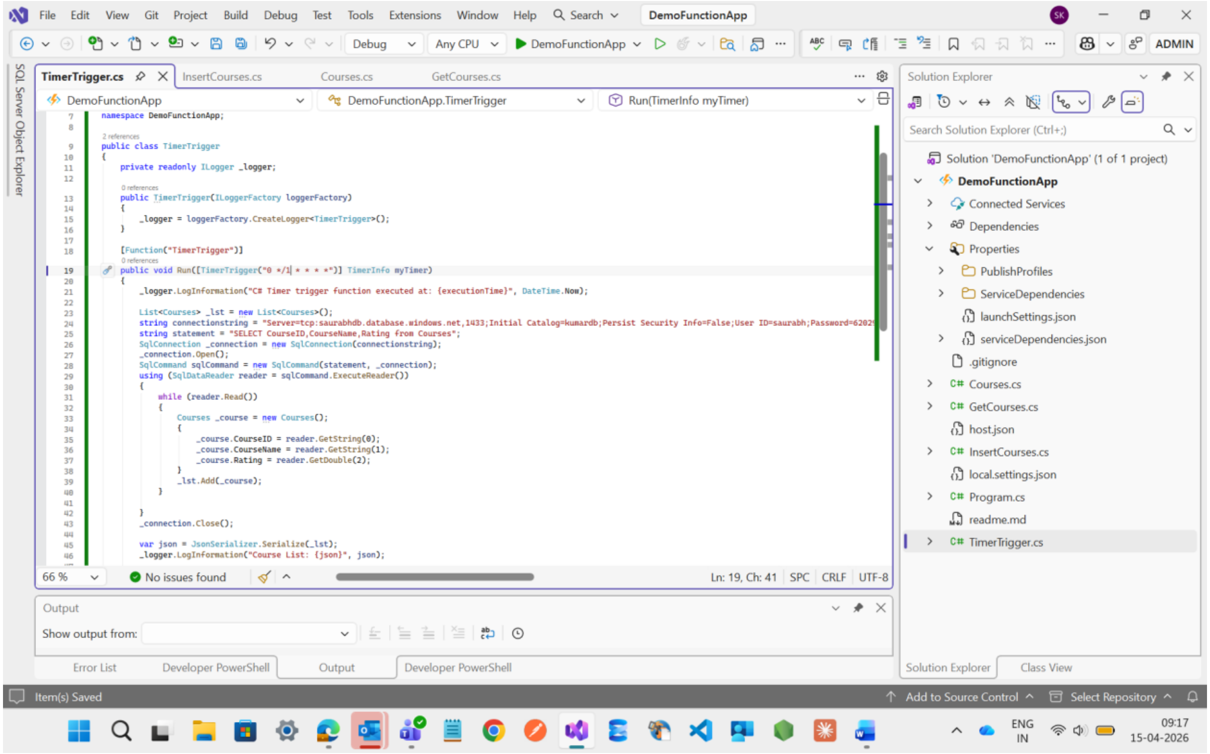
Inside this file I have used 1 minute of timer like every 1 hrs this function will run for 1 min



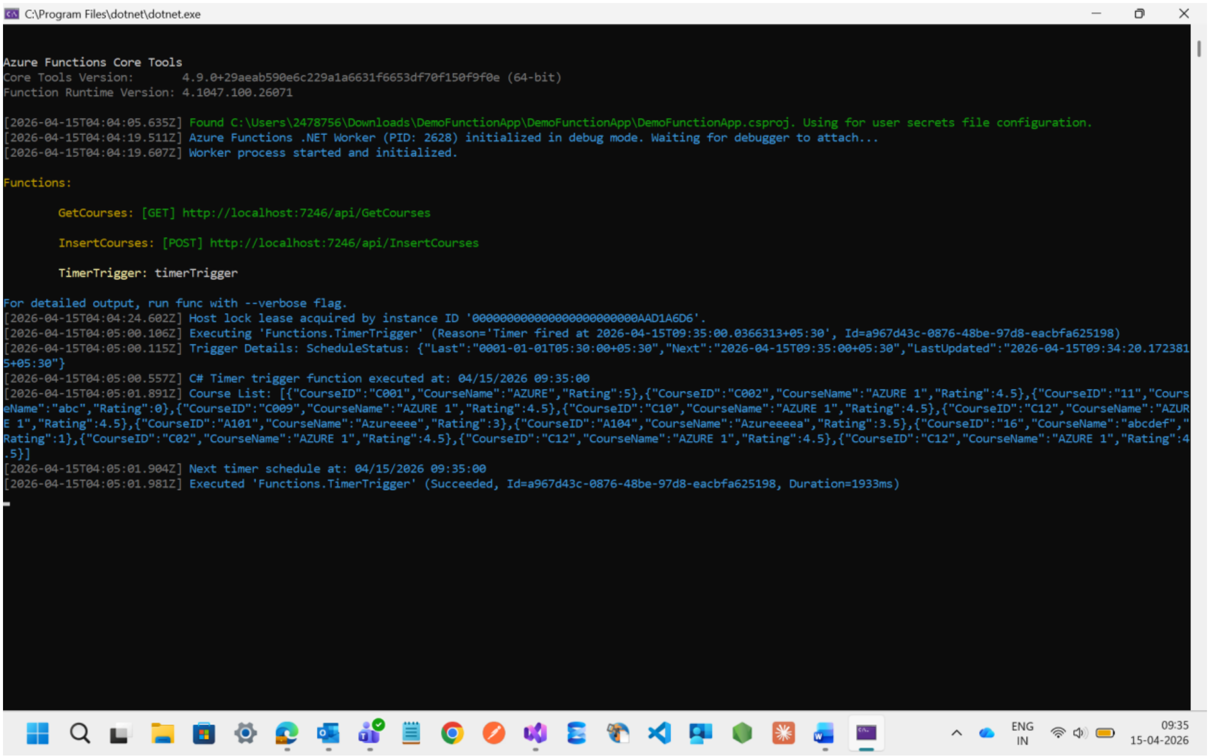
Now I have executed the code it will show the timing when It was executing



Then after I have written the code and provide the connection string inside the timer trigger to fetch the data from database and work as a schedule job

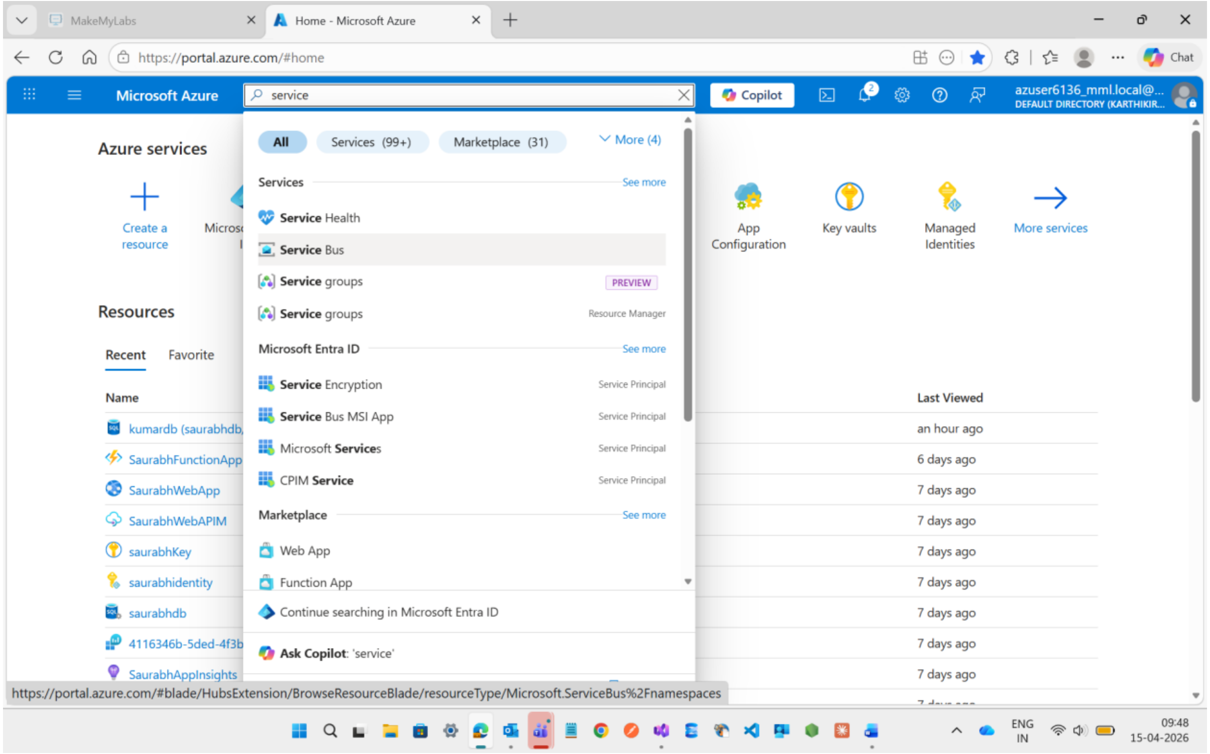


See here it will show the data which is available inside the database based on the time given to run the timer trigger

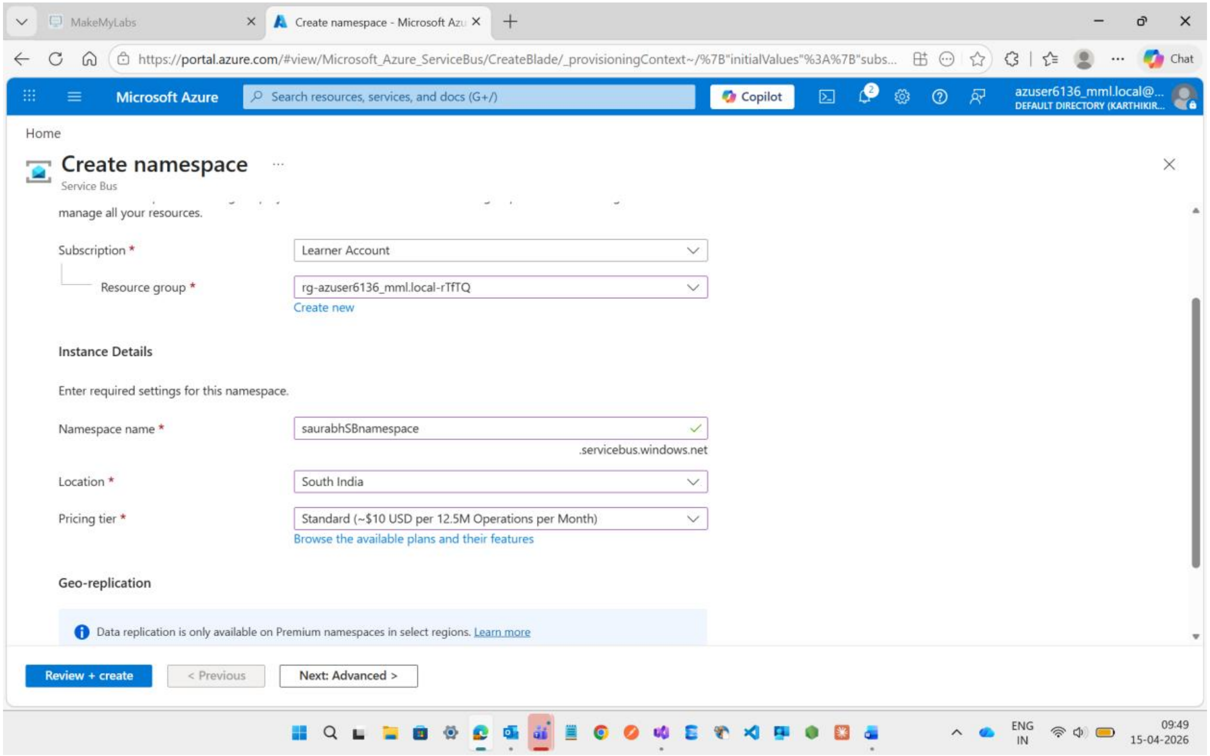


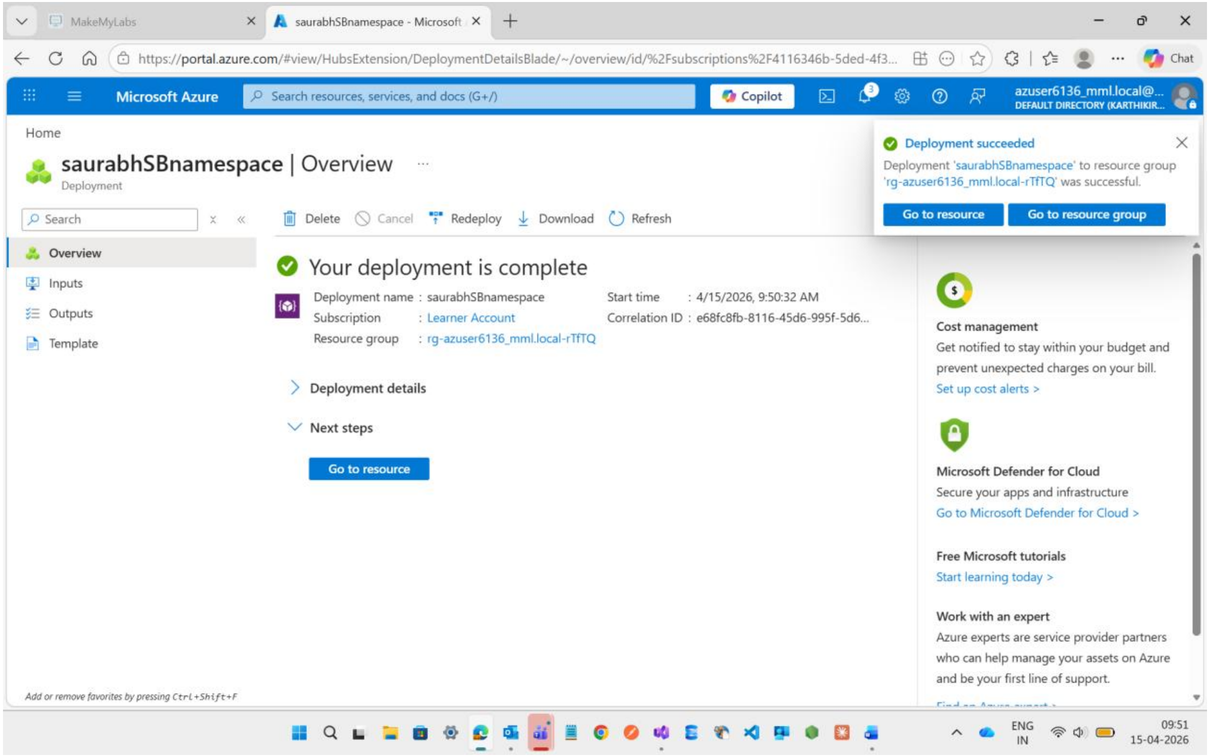
Now I m going to create a Service Bus Trigger

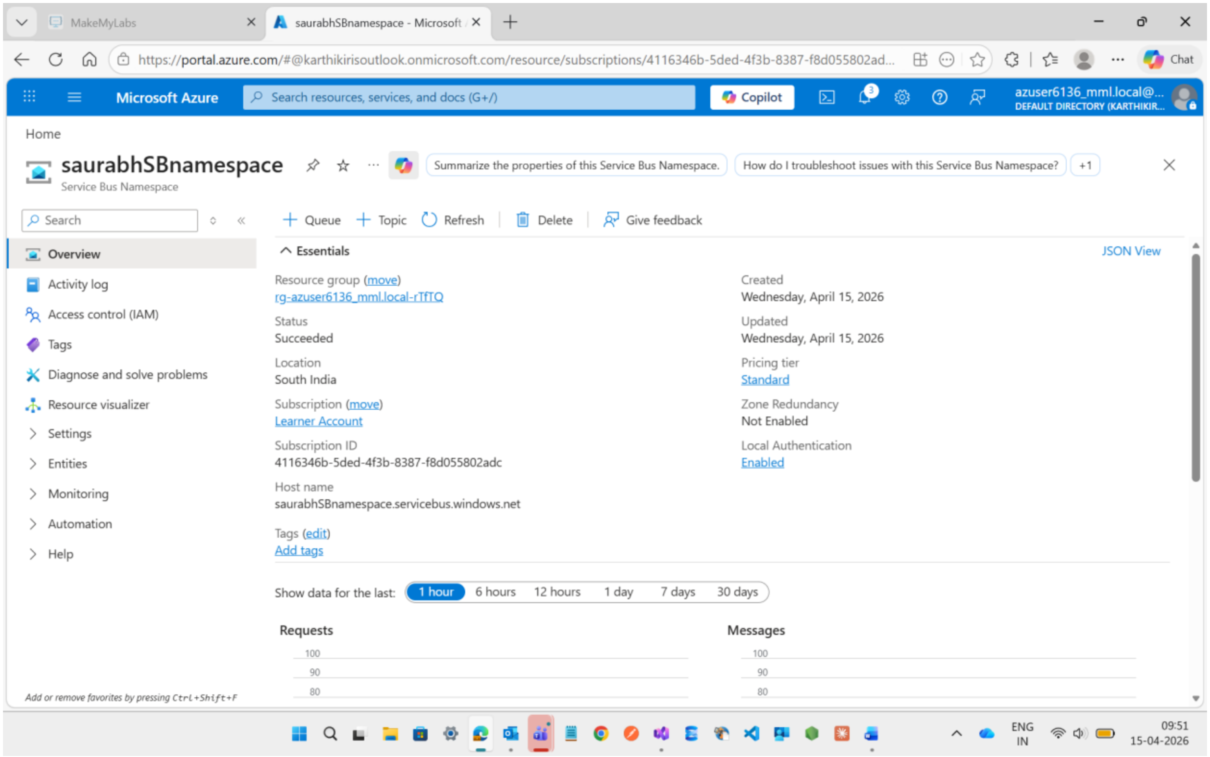
It is a messaging service which is communicating between sender to receiver



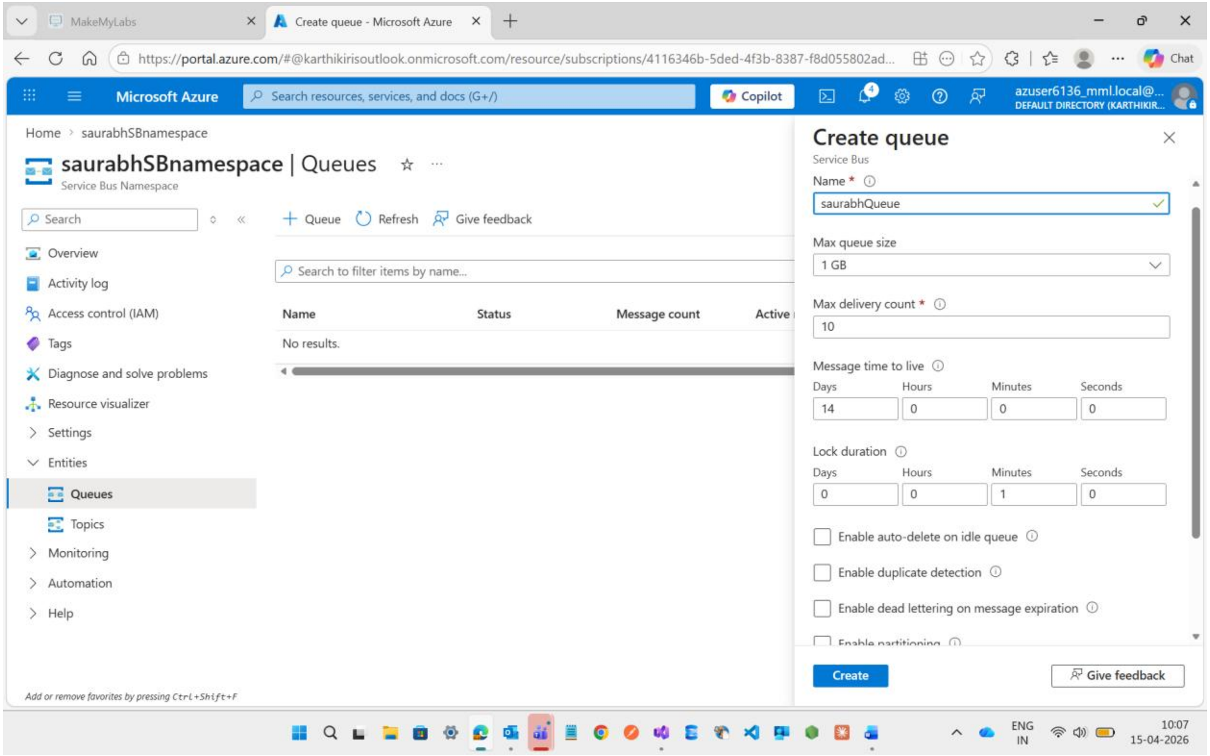
Here I have created a namespace with the name of saurabhSBnamespace



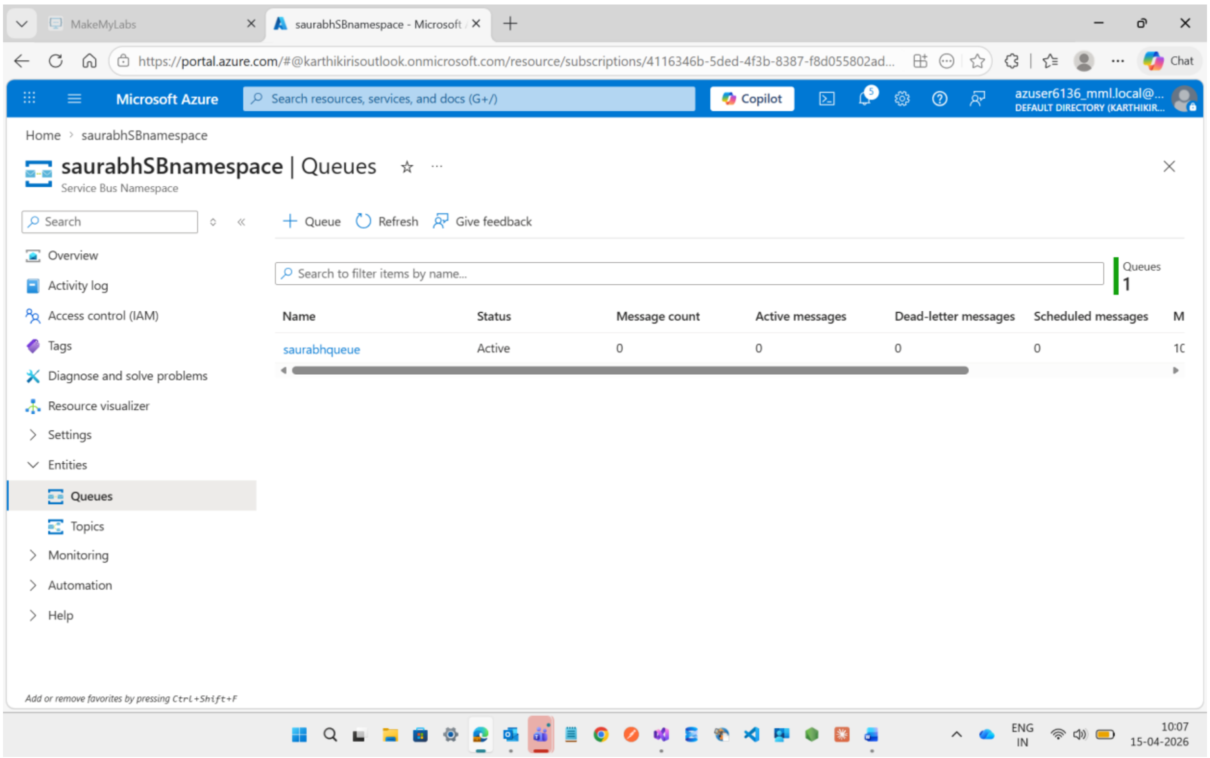




Then after I m going to the Queues and create it

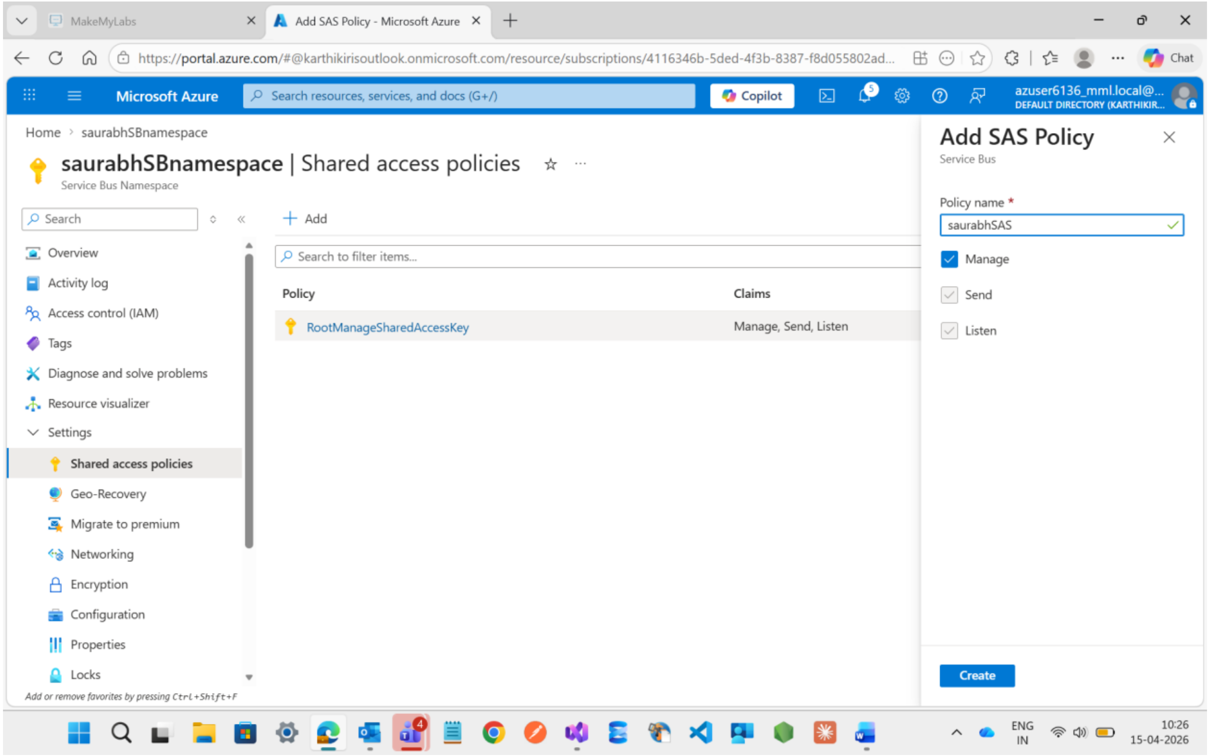


See the queues is created

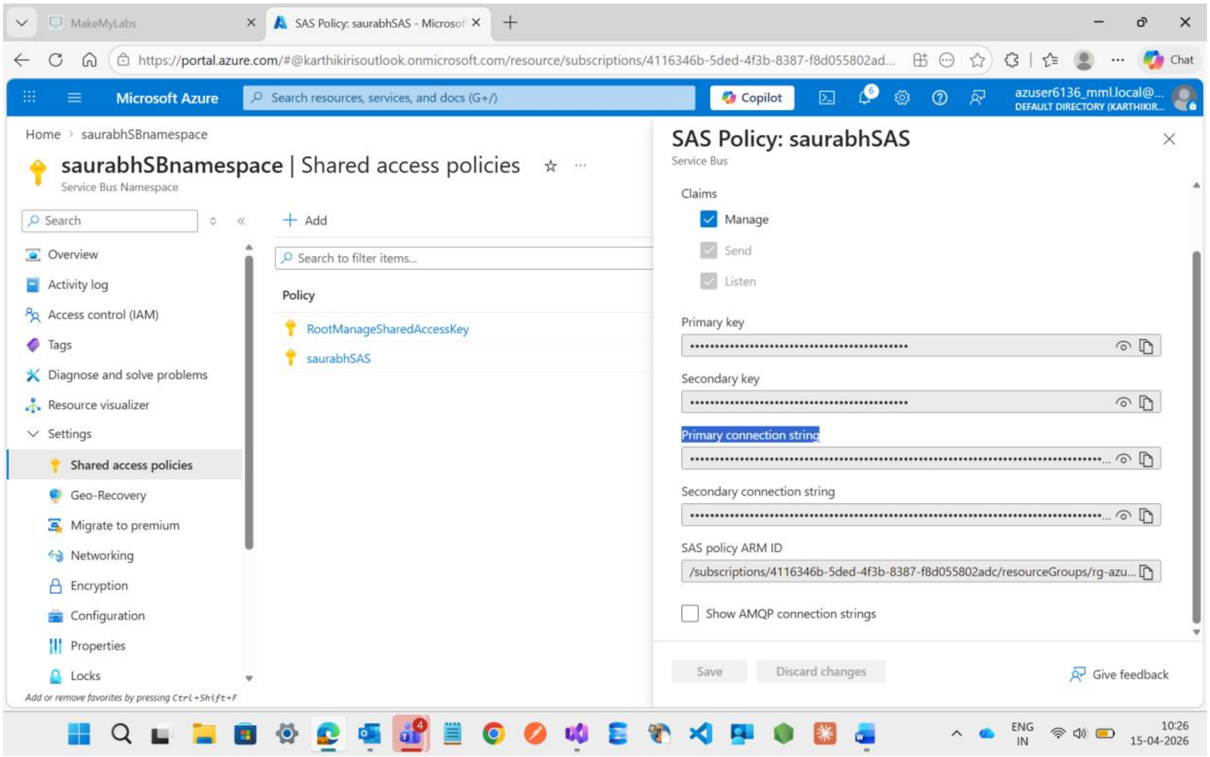


Then after I need to create a connection string to establish the connection for service bus for both entities Queues and Topics to established the connection to our database

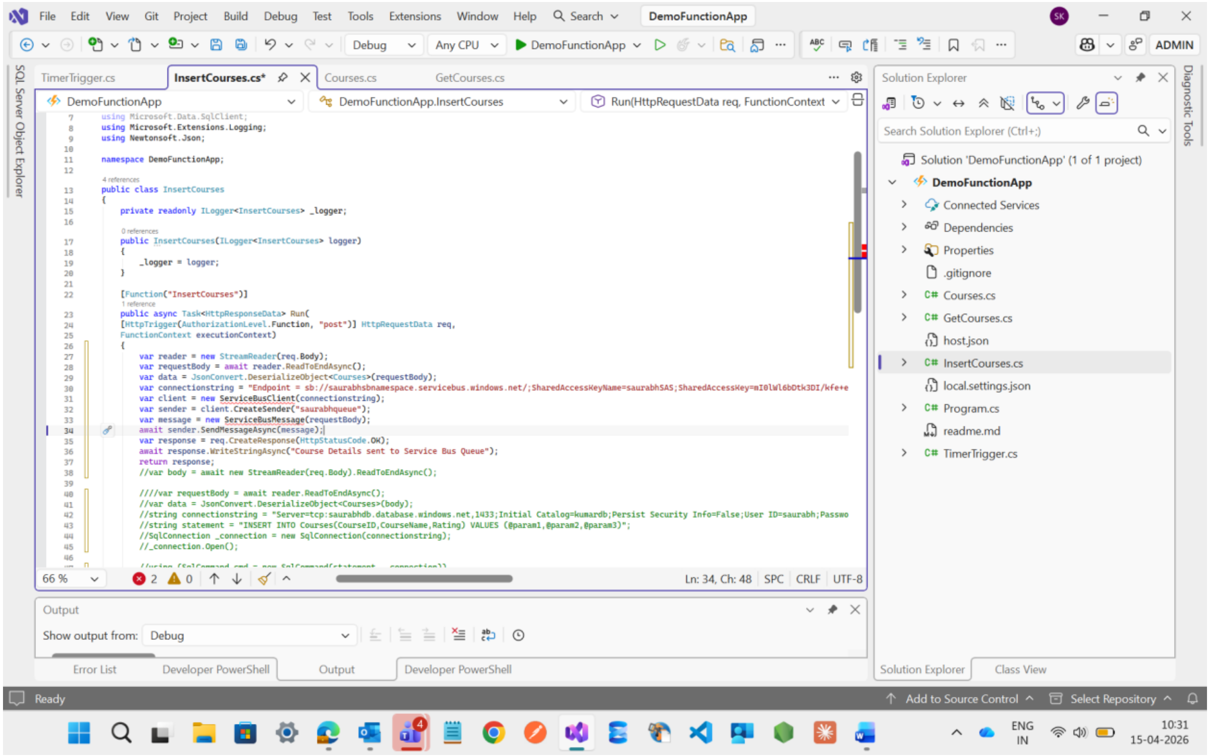
I go to the settings inside the saurabhSbnamespace and open the Shared Access Policy and Add Managed and Provide a name saurabhSAS



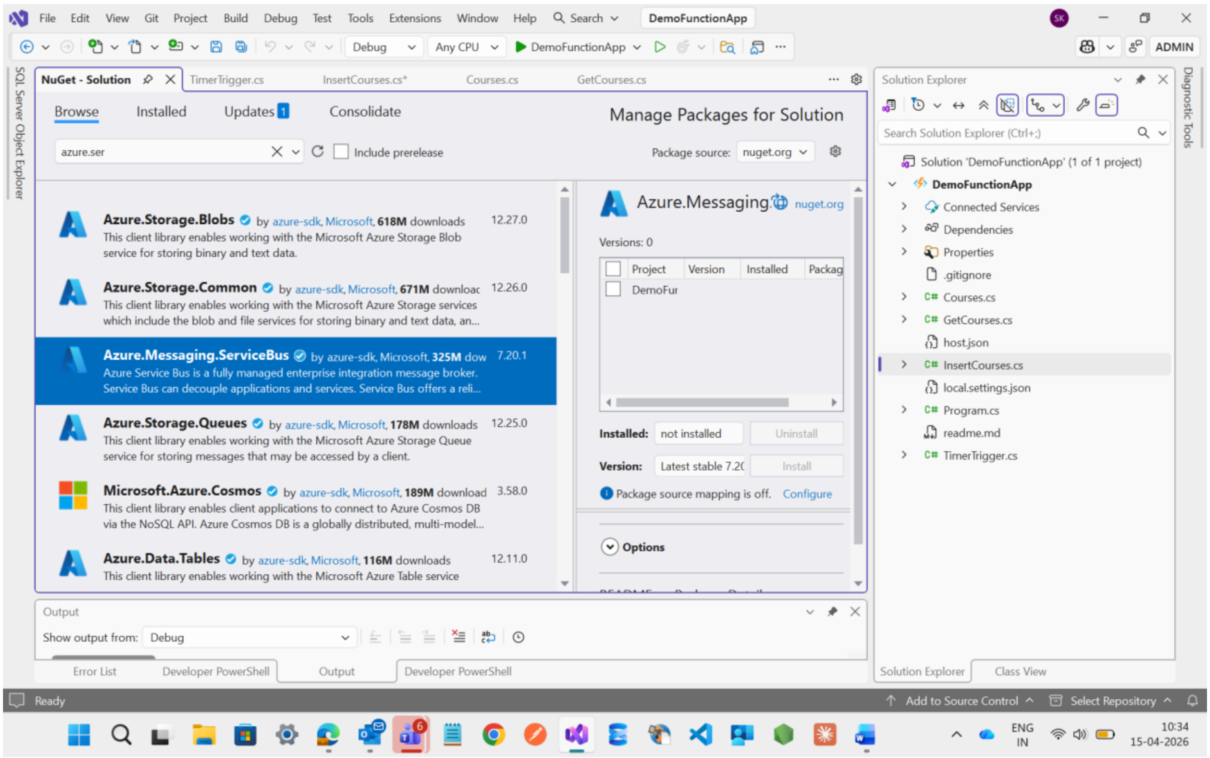
Then open the saurabhSAS and copy the primary connection and paste inside the insert method



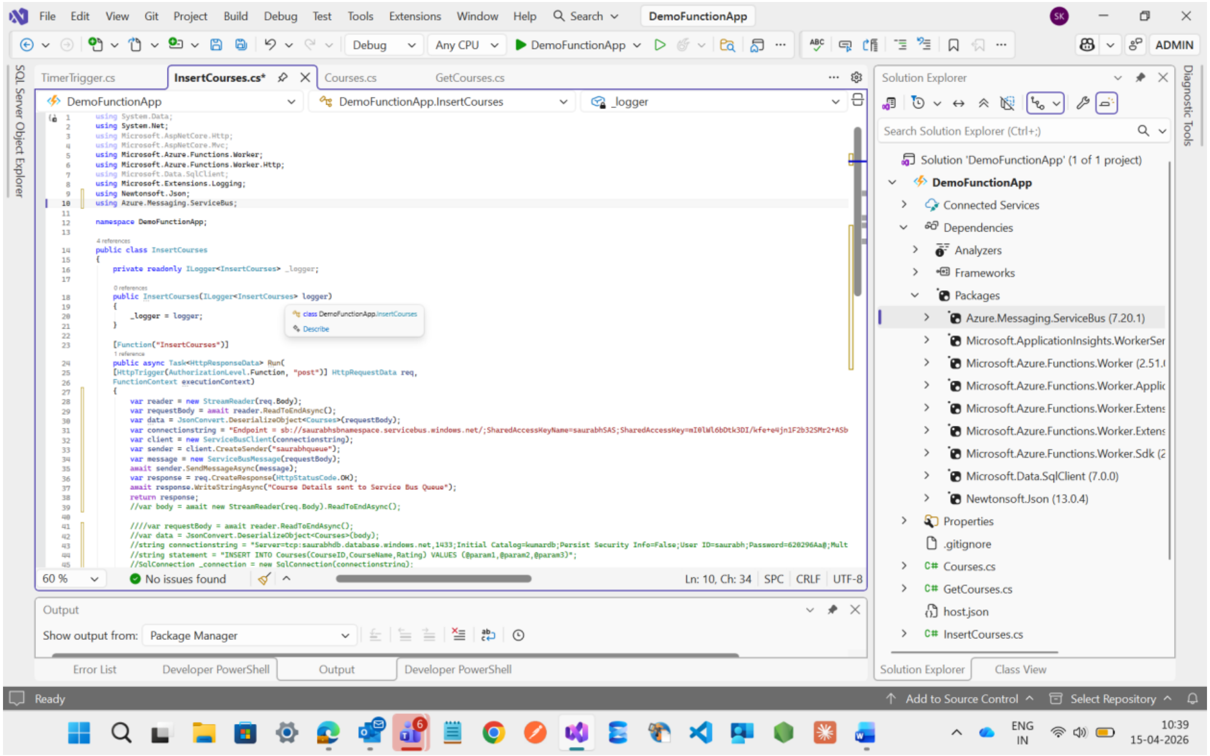
See here I have updated the insert method code to service bus Queues.



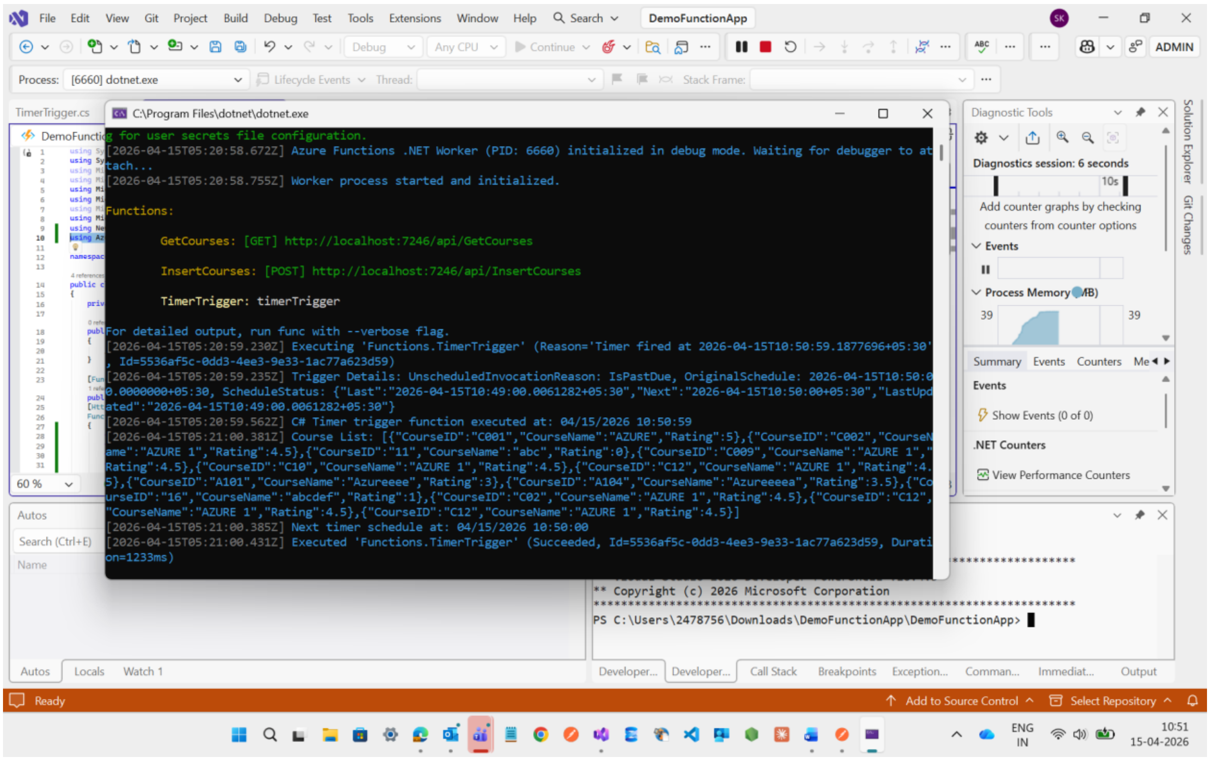
And see inside the above code one error message is appearing to resolve this we need to install one NuGet package with name of Azure Messaging ServiceBus



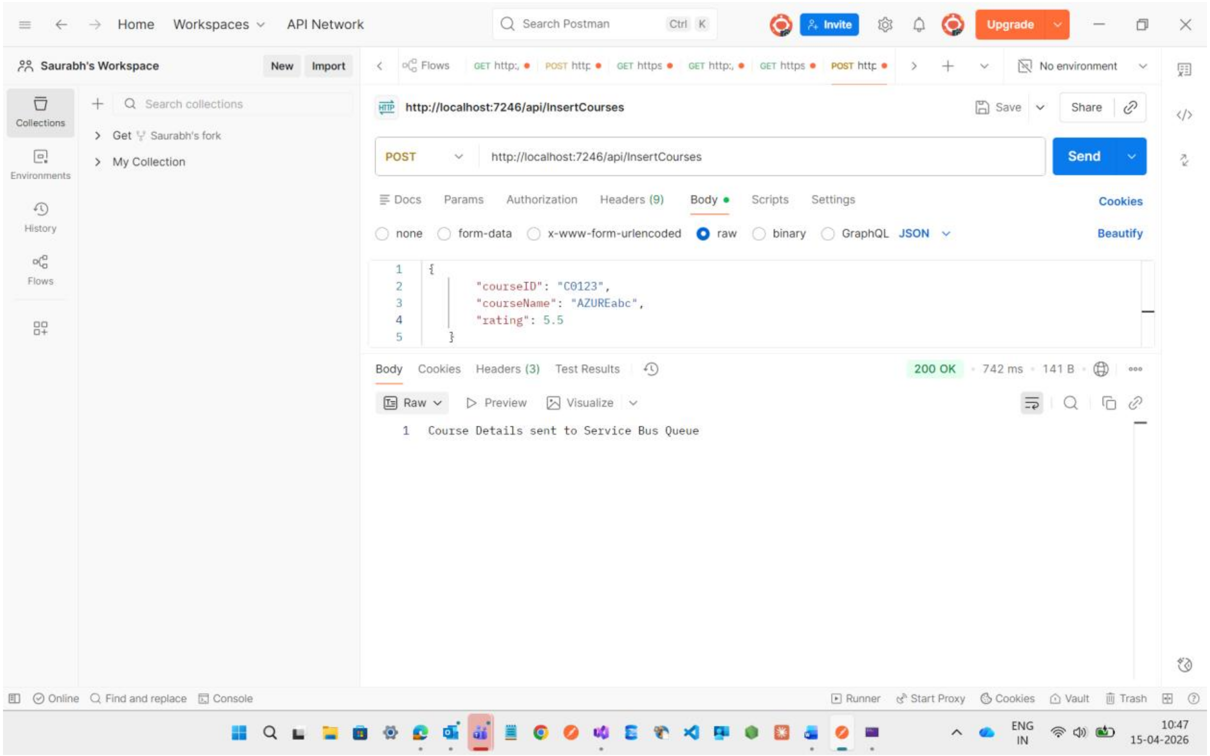
Now see the below code the error will resolved but one thing more after installing the NuGet Package we need to write this header using Azure.Messaging.ServiceBus;



After doing this I will execute the code and try to check with the postman wheather the Queues is working is not



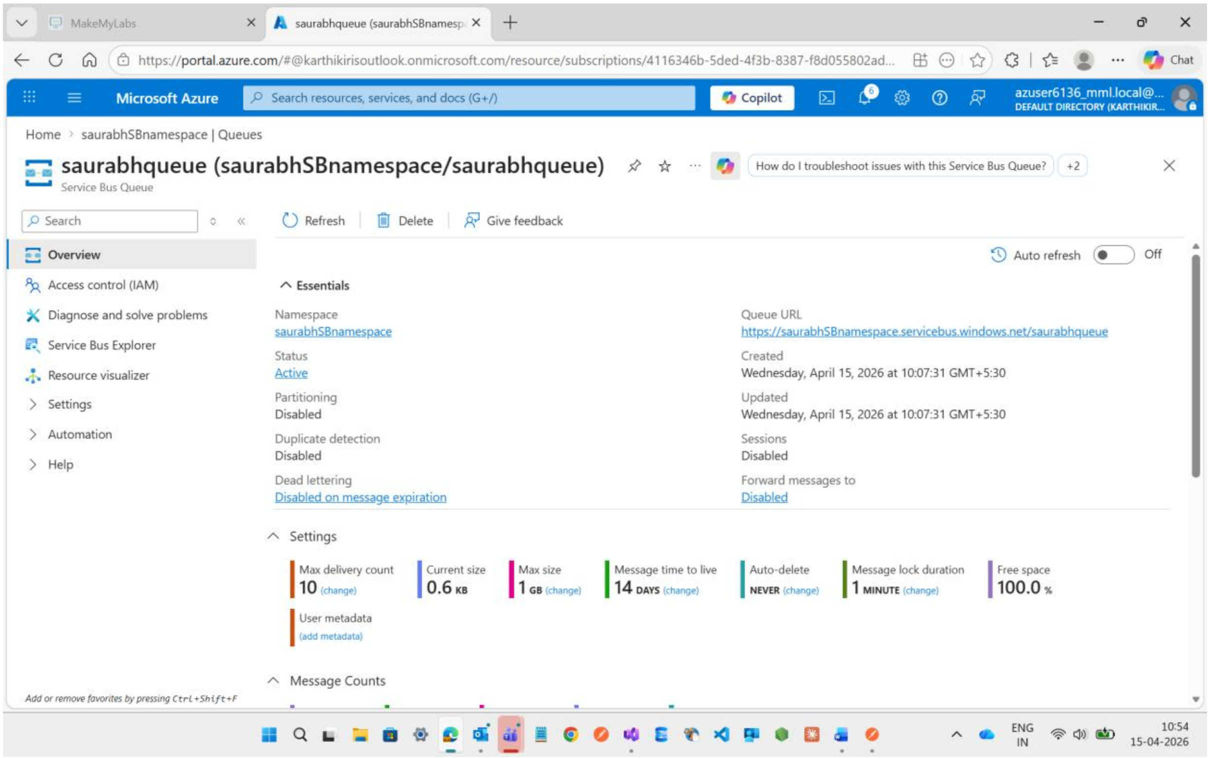
See after testing the url with the POST method inside the postman it will work fine



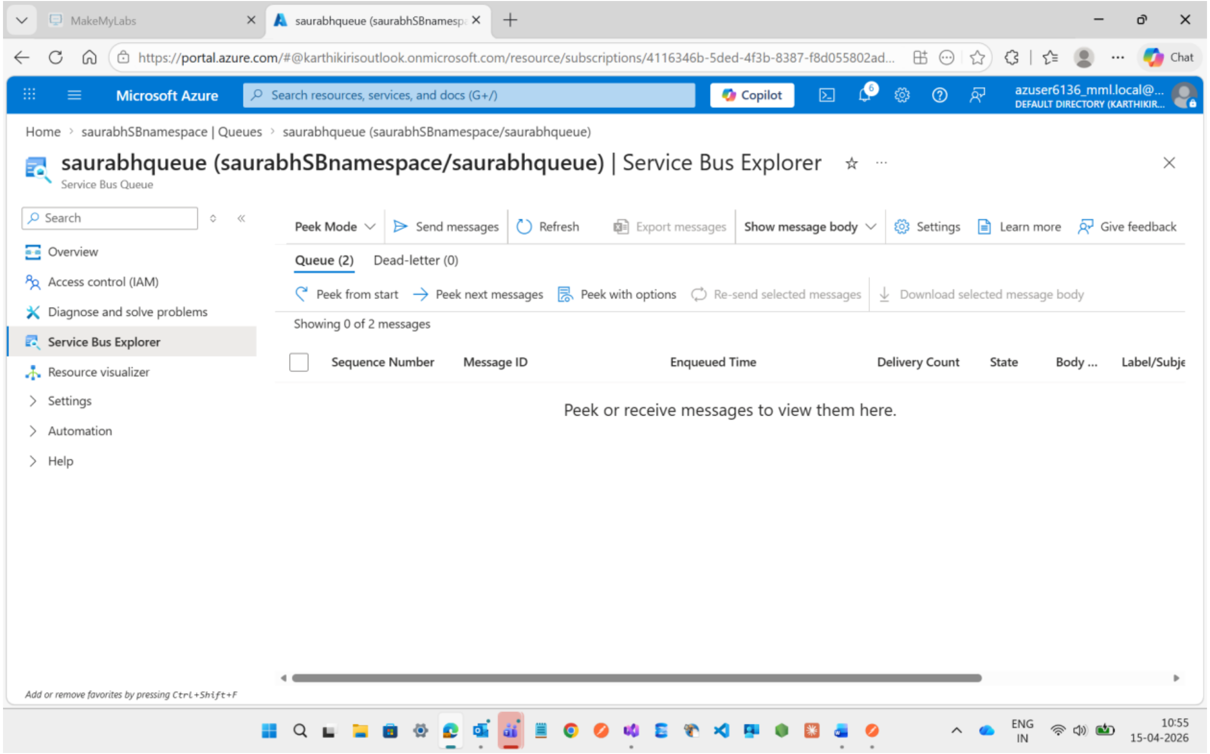
Then After this the Course details will send to the Service Bus Queue.

Now I m going to check inside the Queue wheather the message will come or not

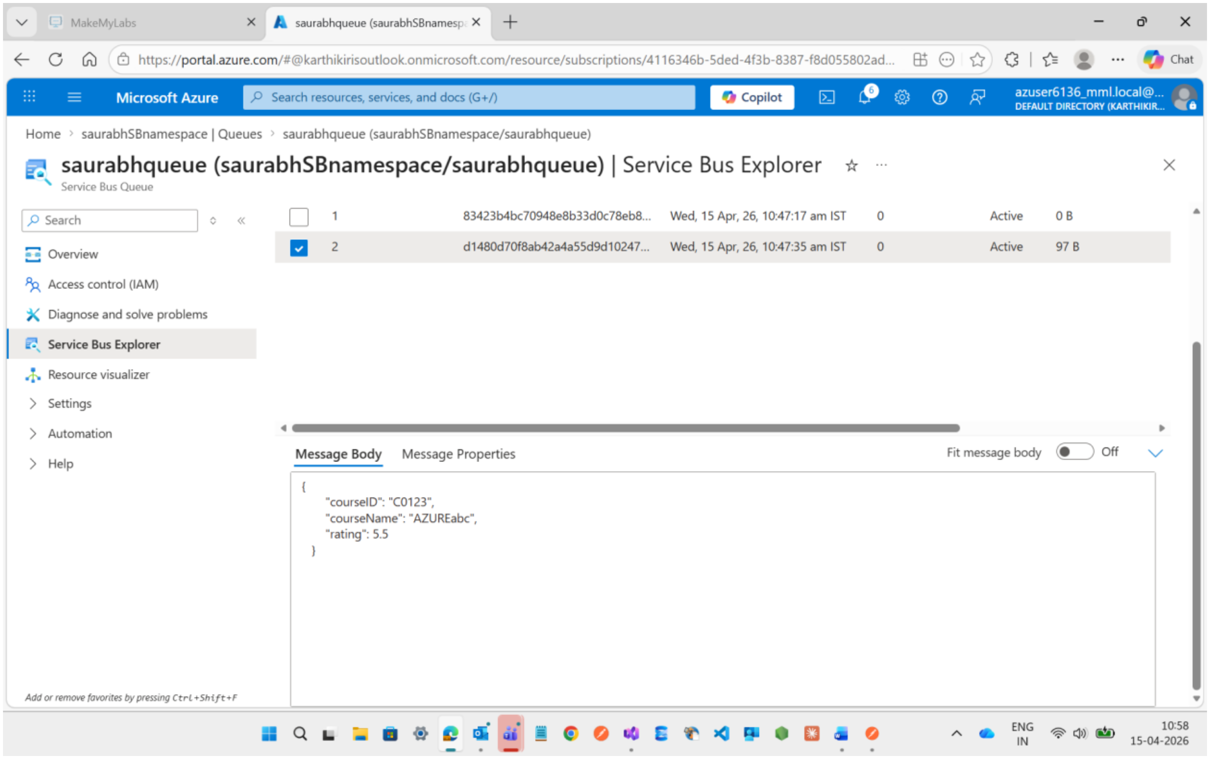
Now click to the Queue which I have created with the name saurabhQueue



Inside this there is one option with the name of Service Bus Explorer just open it

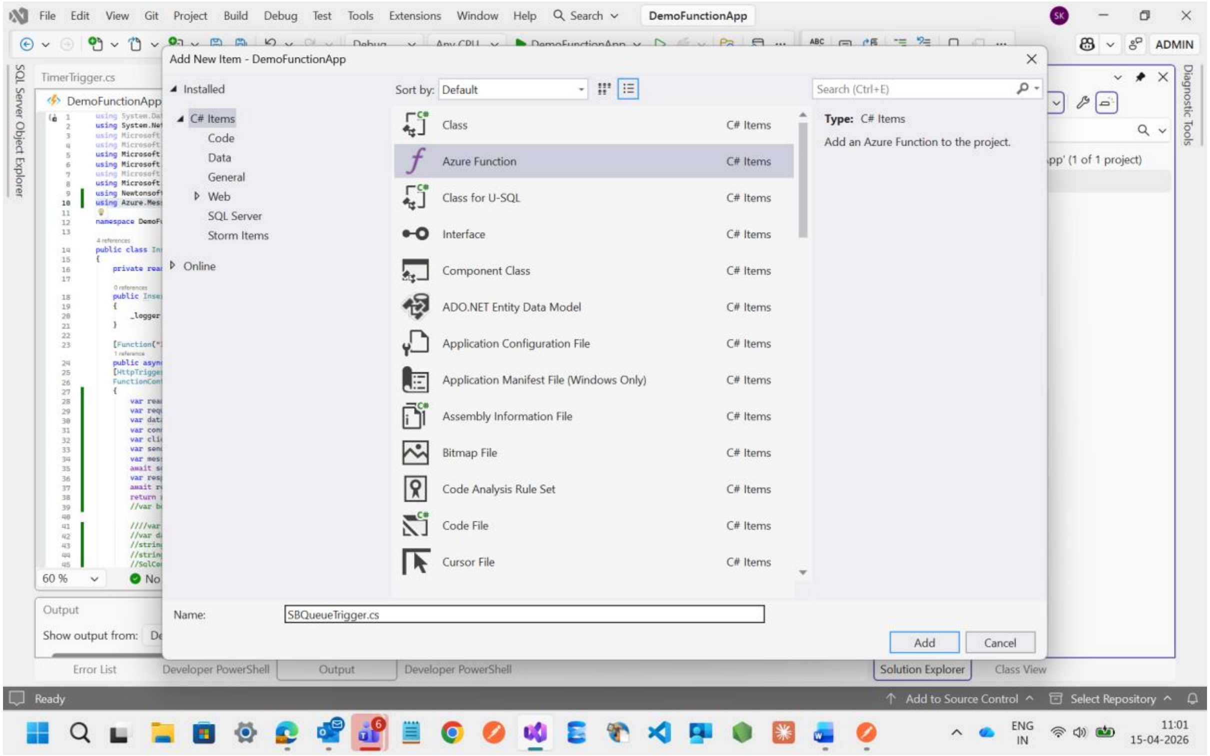


Now see at the above there is 2 Queue is available now I just click on the Peek from start option and see the result what's ever I have send through the Post Method

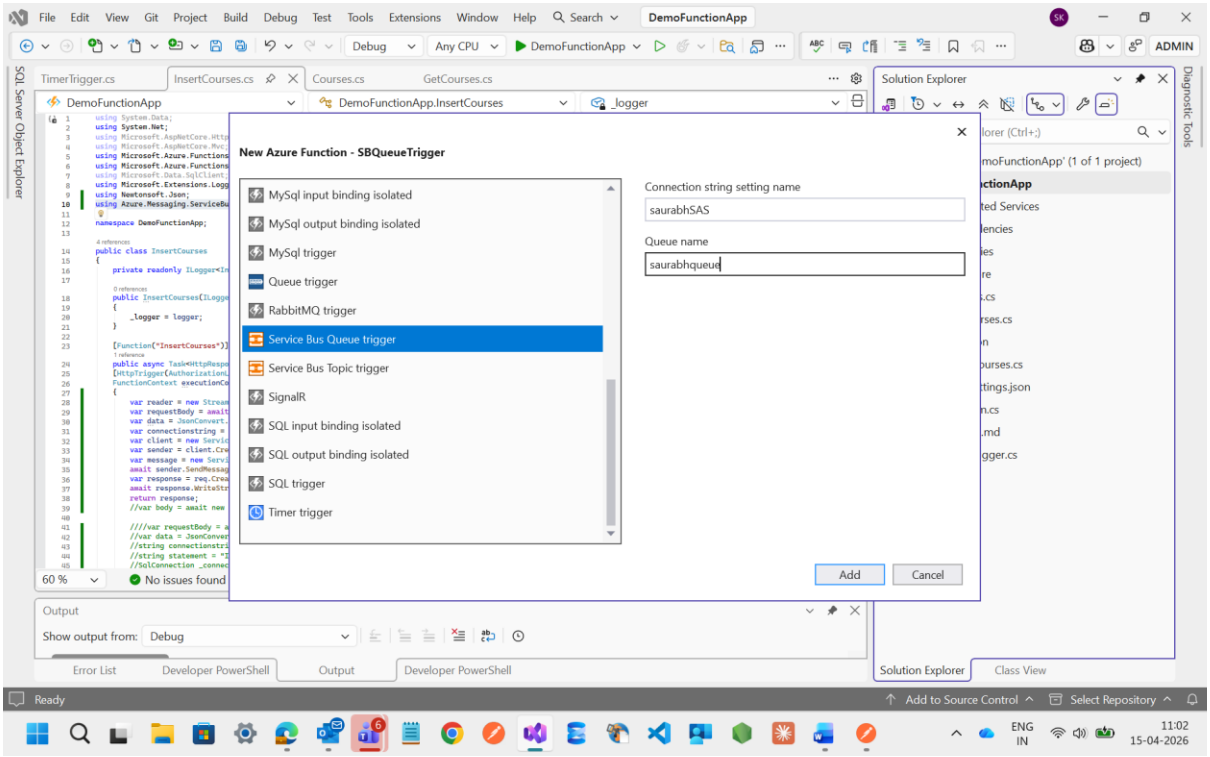


To doing this all only because of to subscribe it from here and insert inside the Database

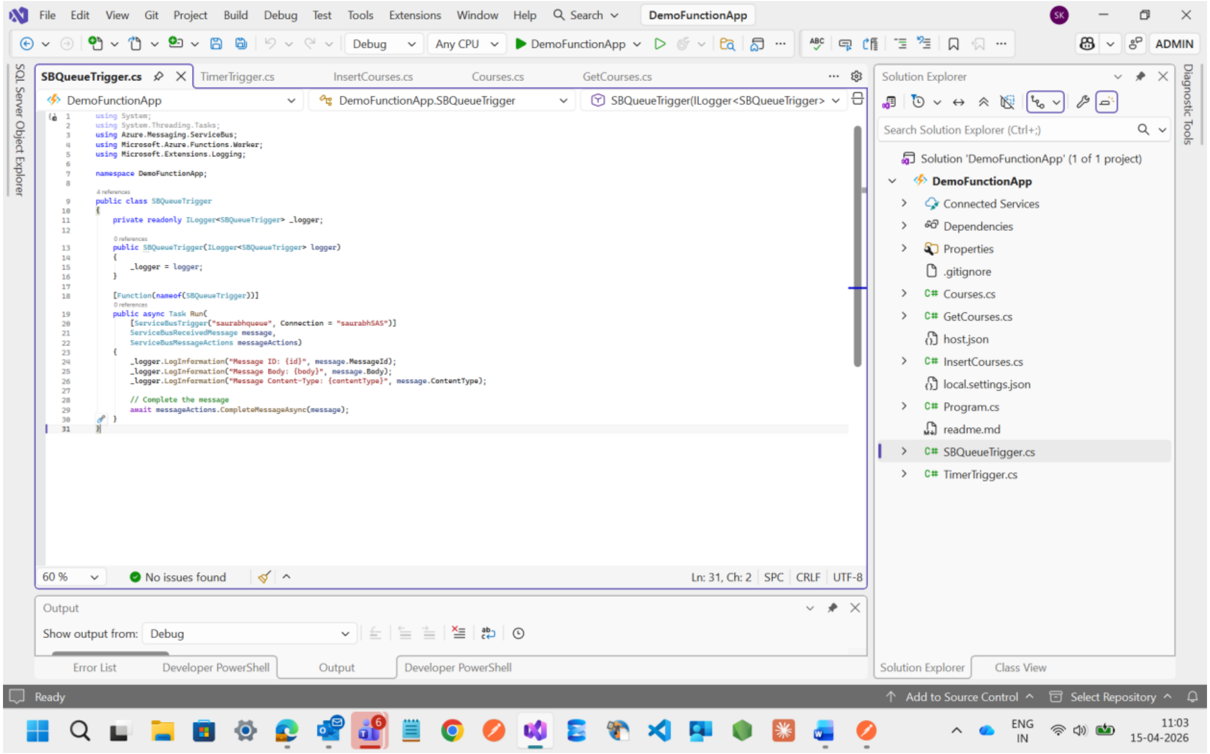
Then After I m going to create one Function inside the FunctionApp with the name of SBQueueTrigger



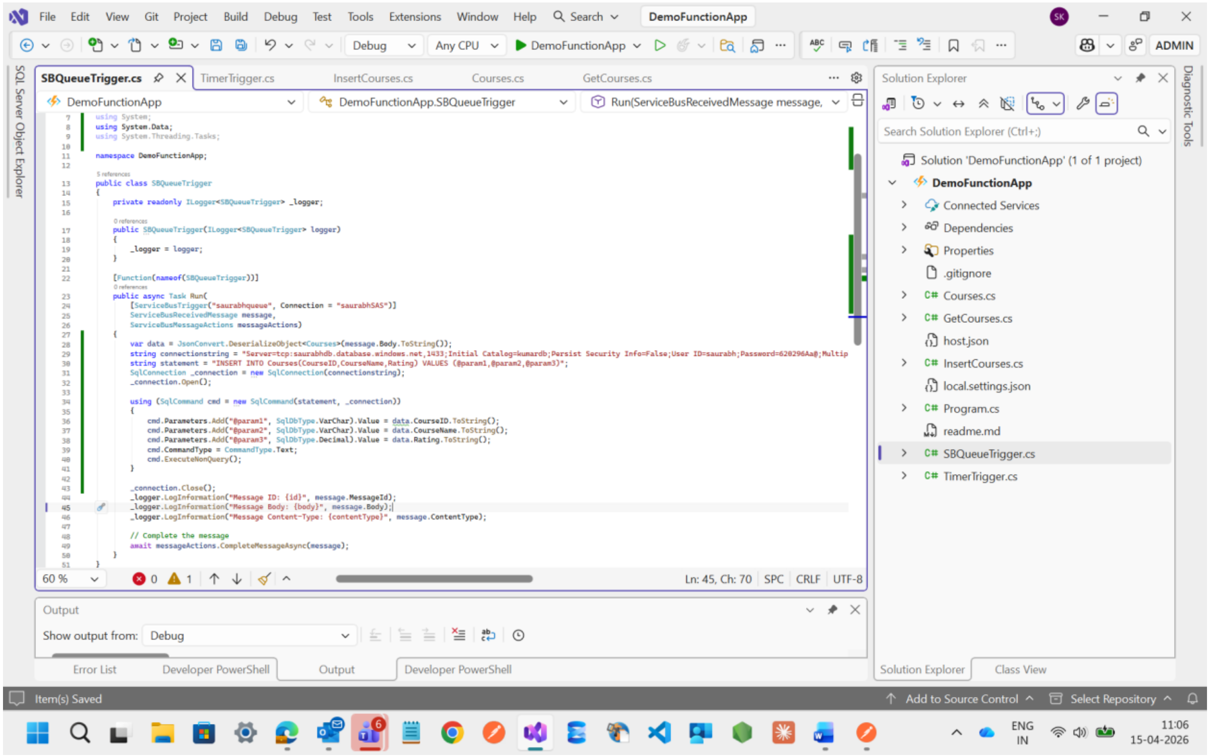
Here I select the Service Bus Queue Trigger and assign the connection string name and Queue name



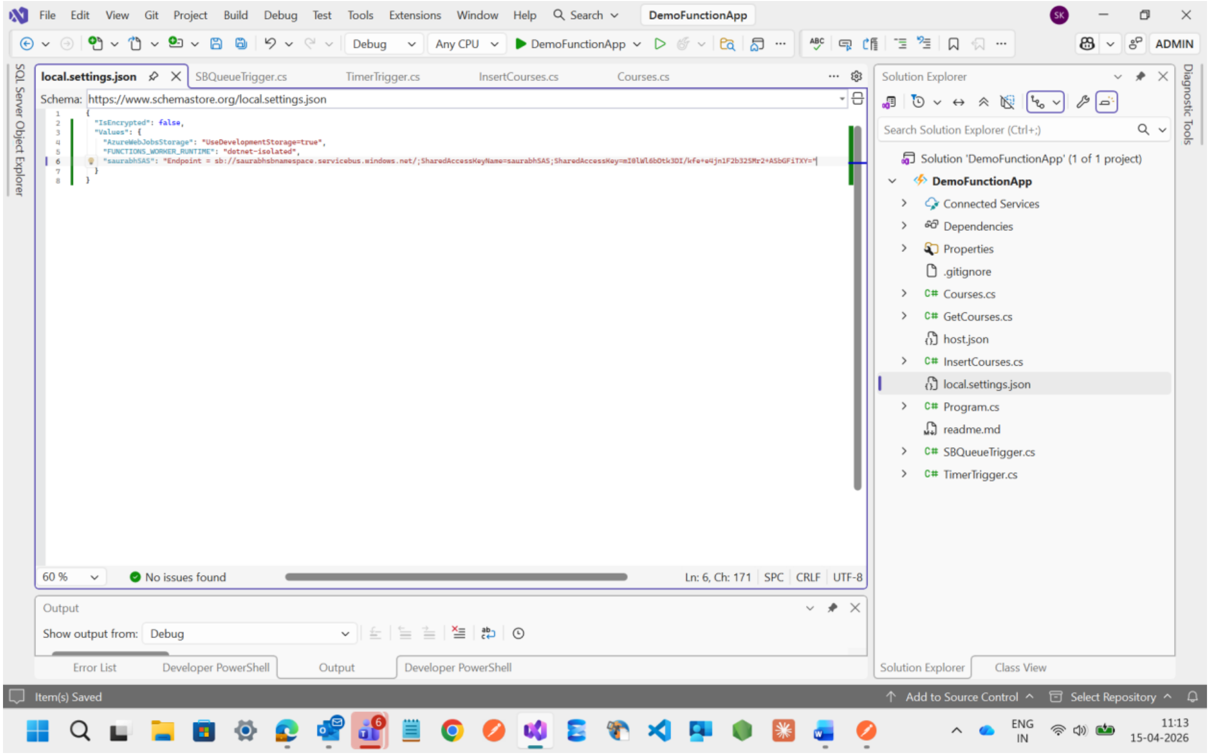
See I have created Service Bus Trigger Function file



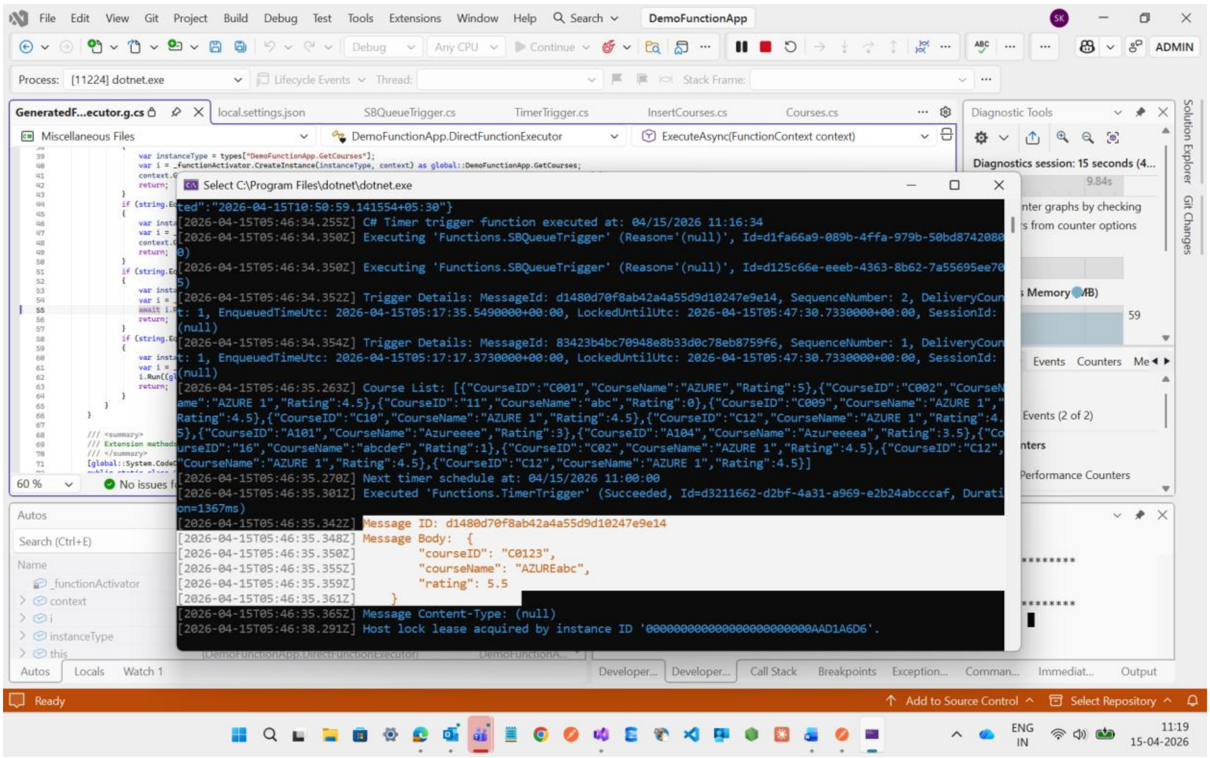
Then I will update the code by assigning the connection string



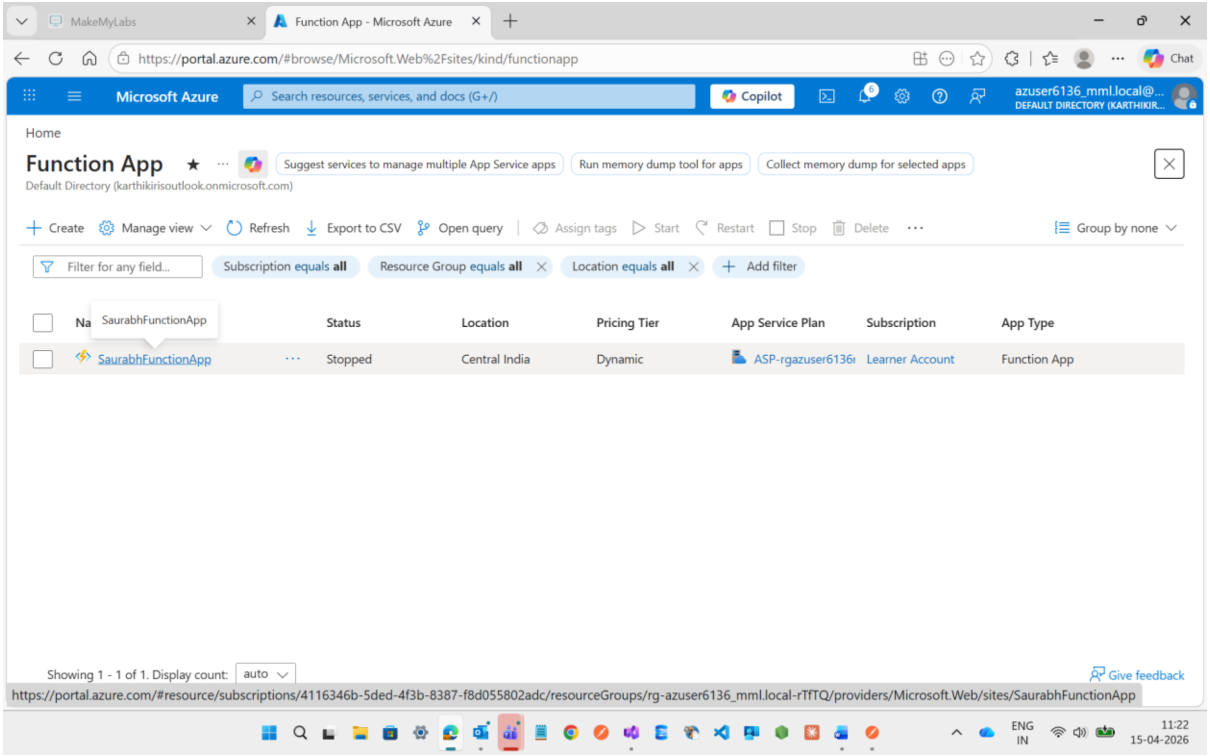
Then after this we need to assign the Connection string of Service bus namespace inside the local.settings.json



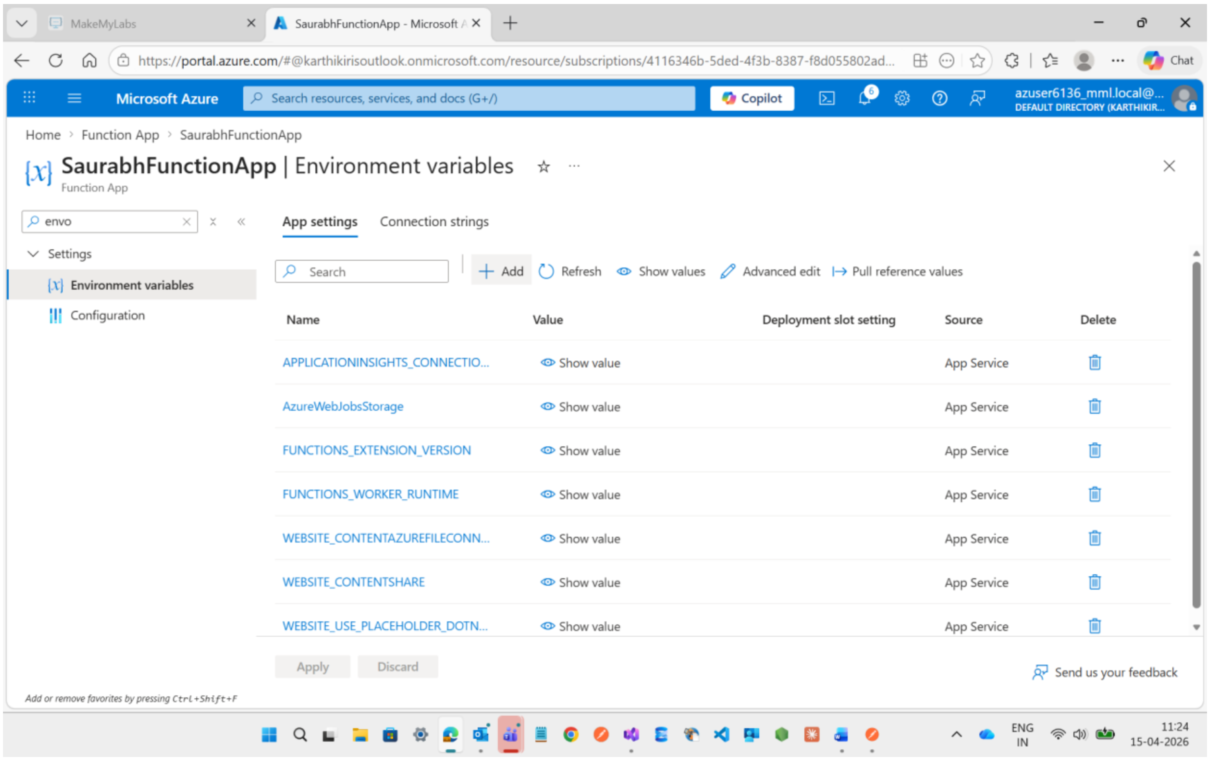
Then I will execute it and it will show the details what I was sended before through Post Method to the Queue



Then after this I will go to the Function App Portal to Azure

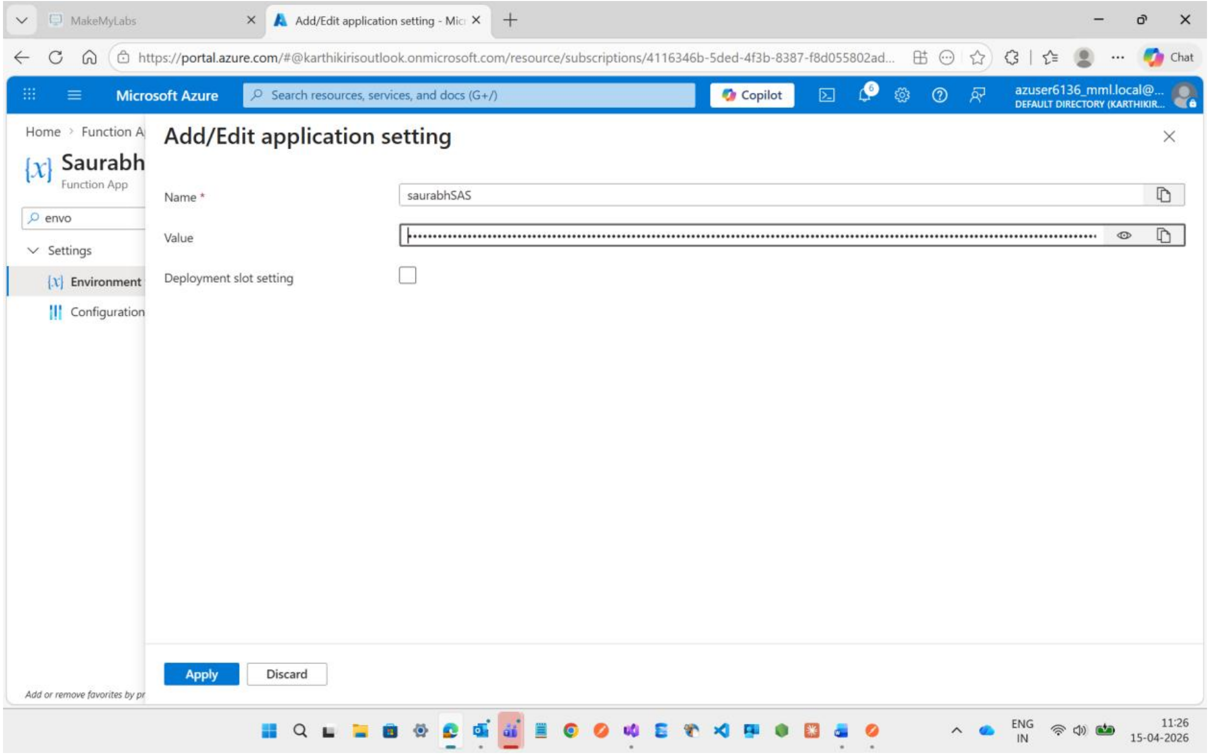


Then I will go to the Environment Variable to assign the Service Bus Connection string what I have assigned inside the local.settings.json to establish the connection while publishing the code to function app.

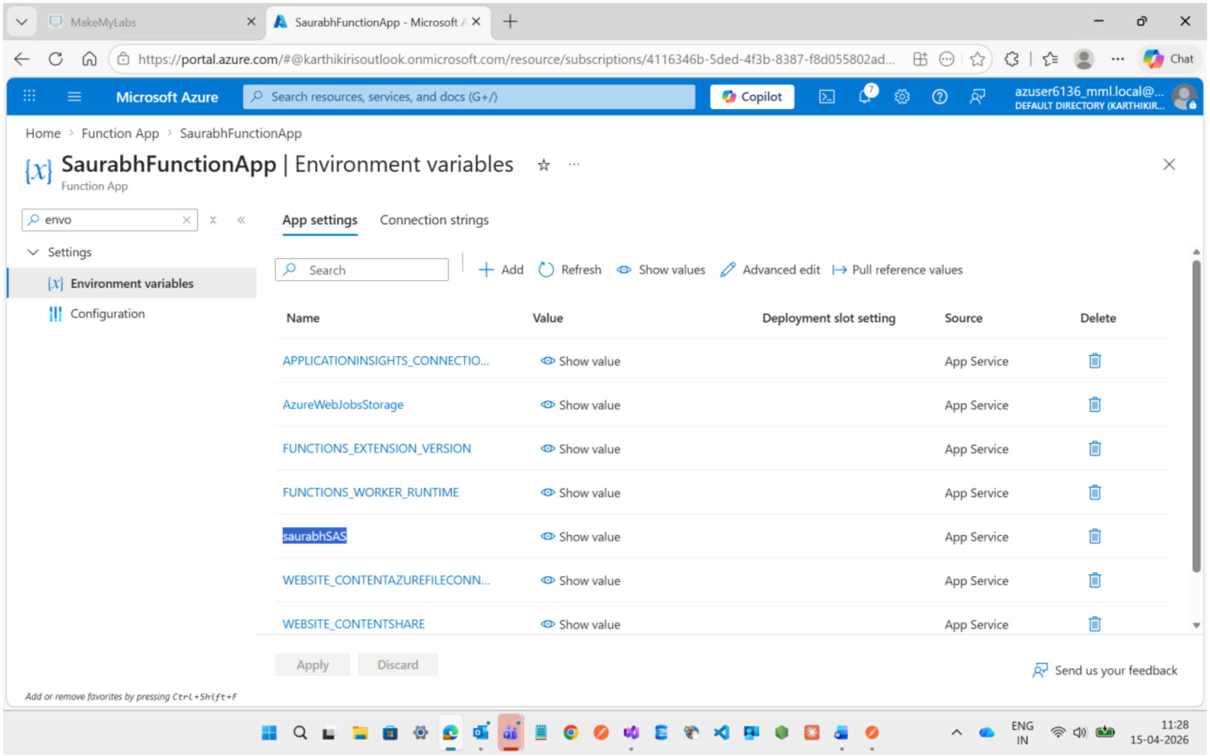


Here i add the name of connection string and the value of that connection string

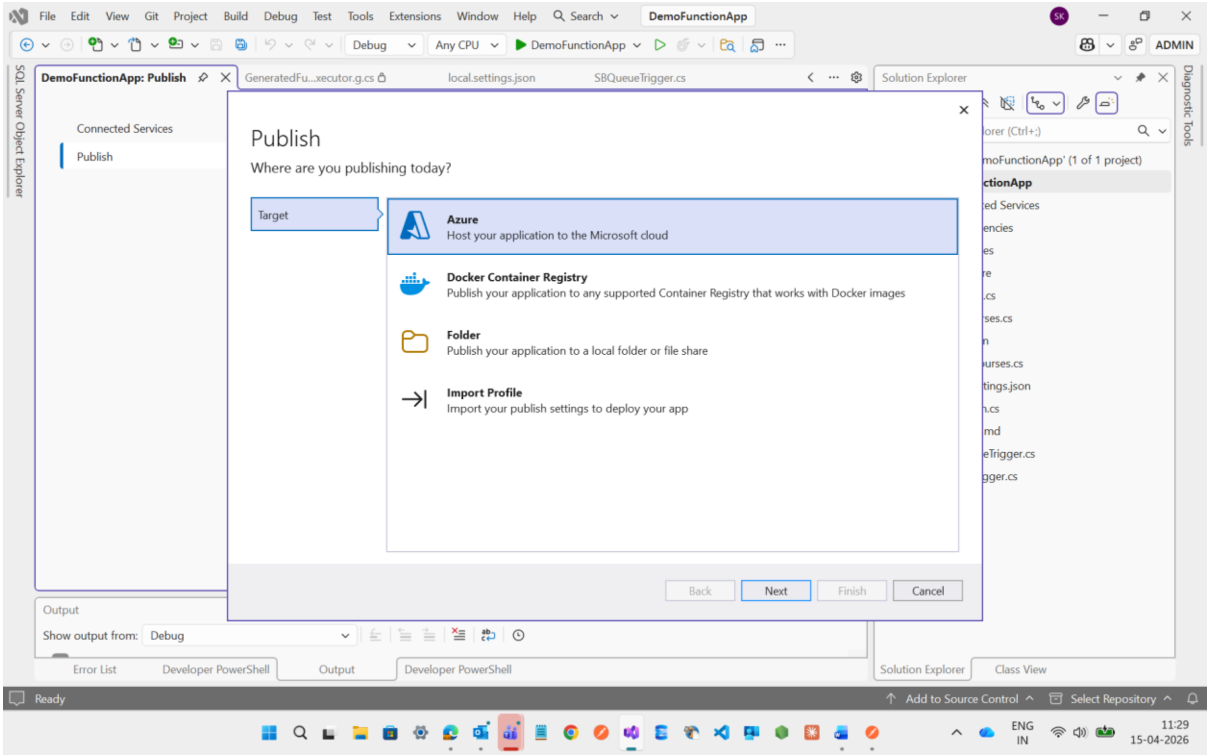
Endpoint =
sb://saurabhsbnamespace.servicebus.windows.net/;SharedAccessKeyName=saurabhSAS;SharedAccessKey=ml0lWl6bDtk3DI/kfe+e4jn1F2b32SMr2+ASbGFiTXY=

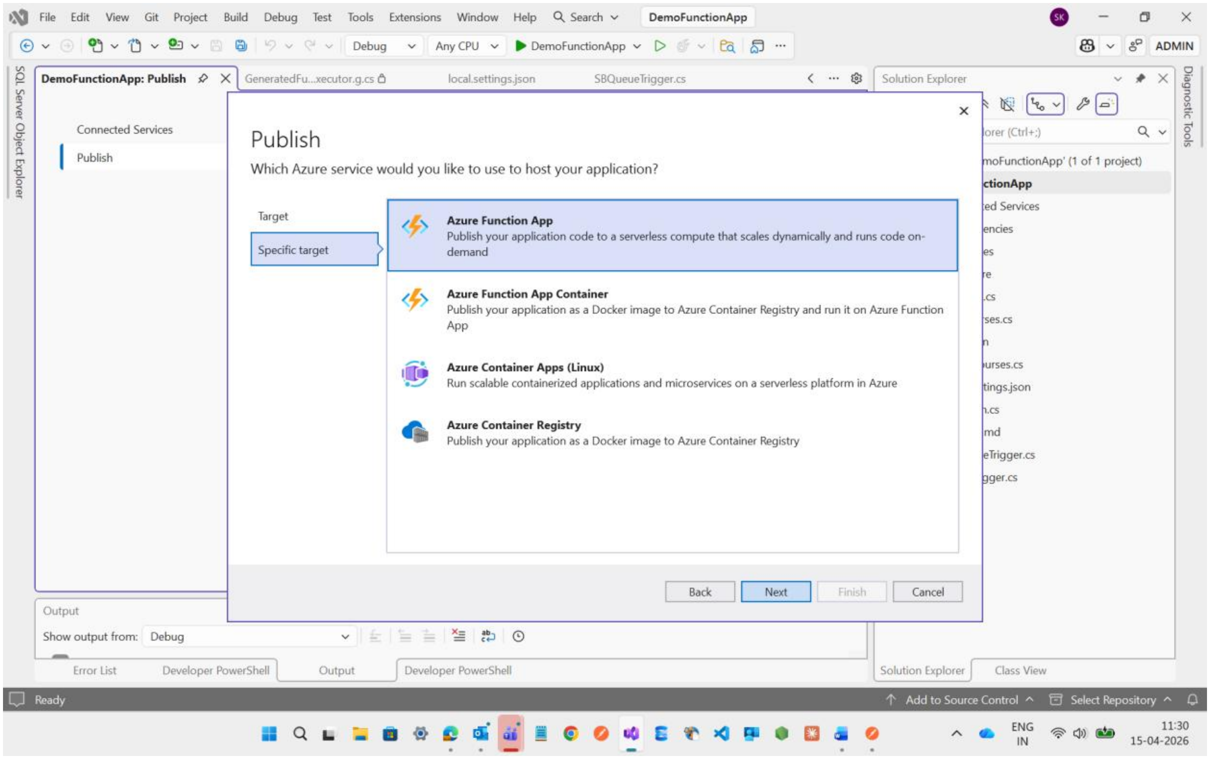


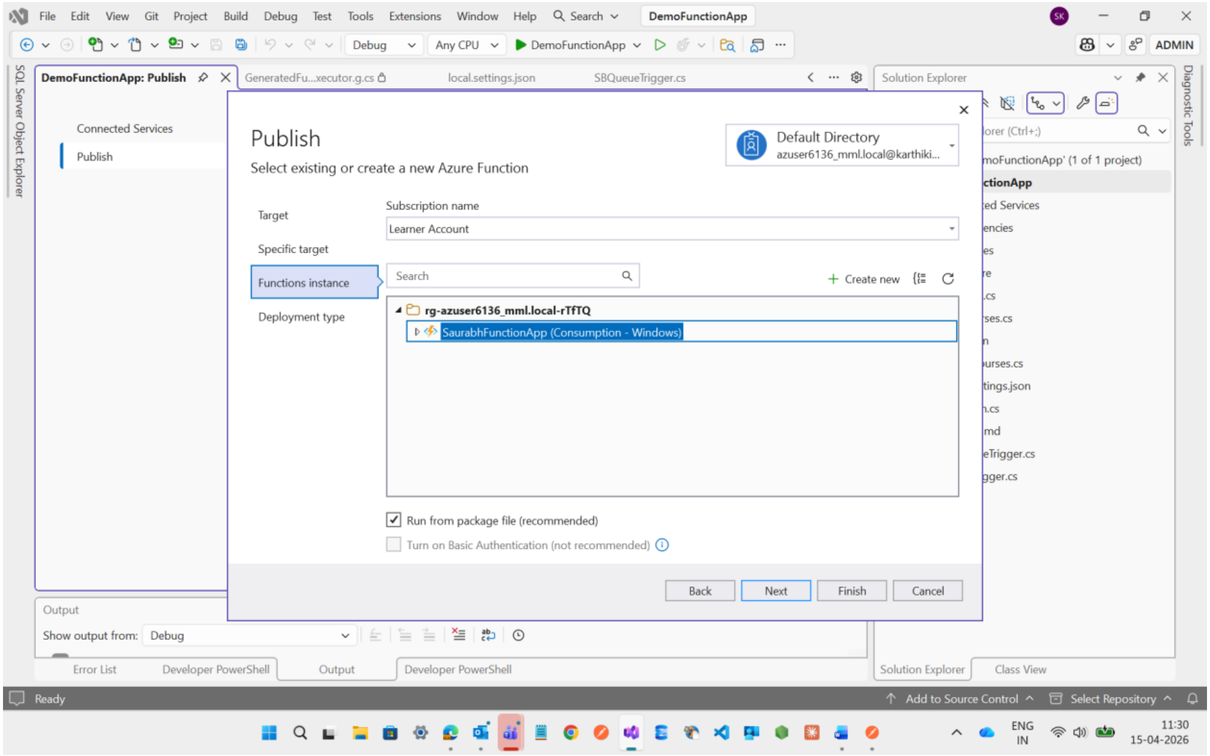
See here I have added the Environment Variable



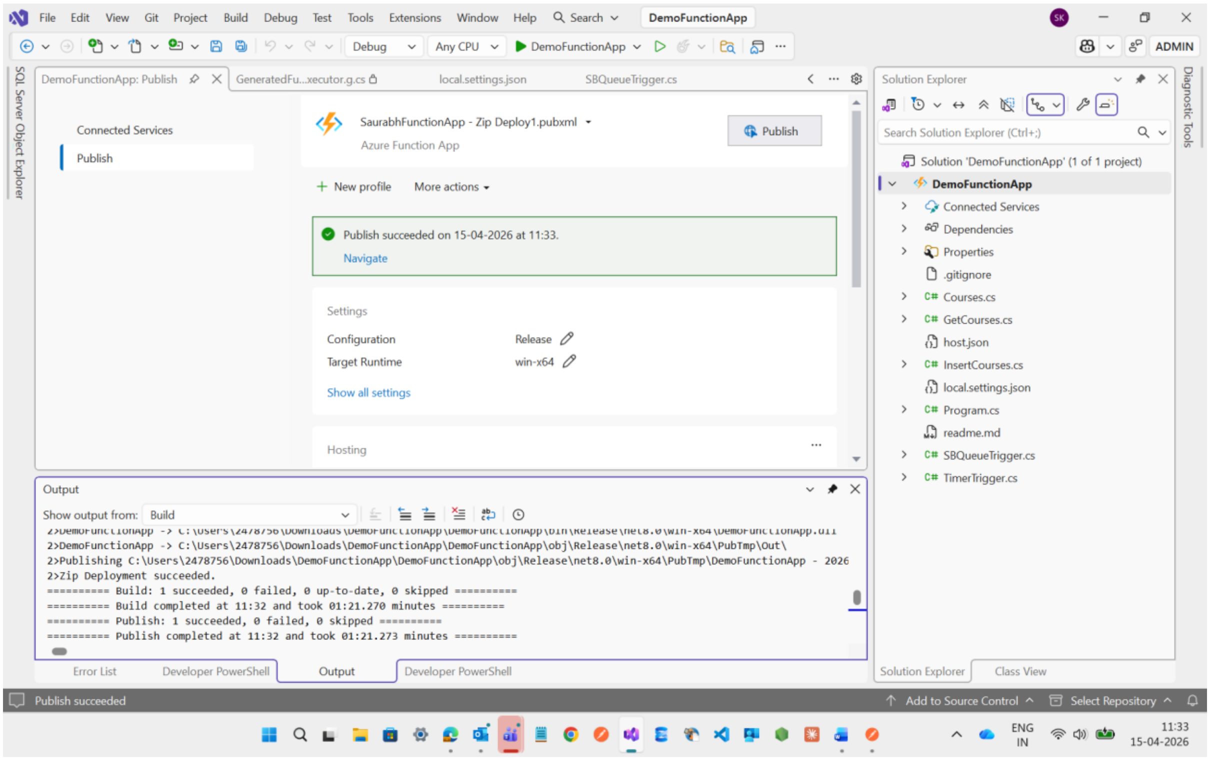
Now I m going to publish it through Visual studio


Publish
Where are you publishing today?
Target
Azure
Host your application to the Microsoft cloud
Docker Container Registry
Publish your application to any supported Container Registry that works with Docker images
Folder
Publish your application to a local folder or file share
Import Profile
Import your publish settings to deploy your app
Back
Next
Finish
Cancel

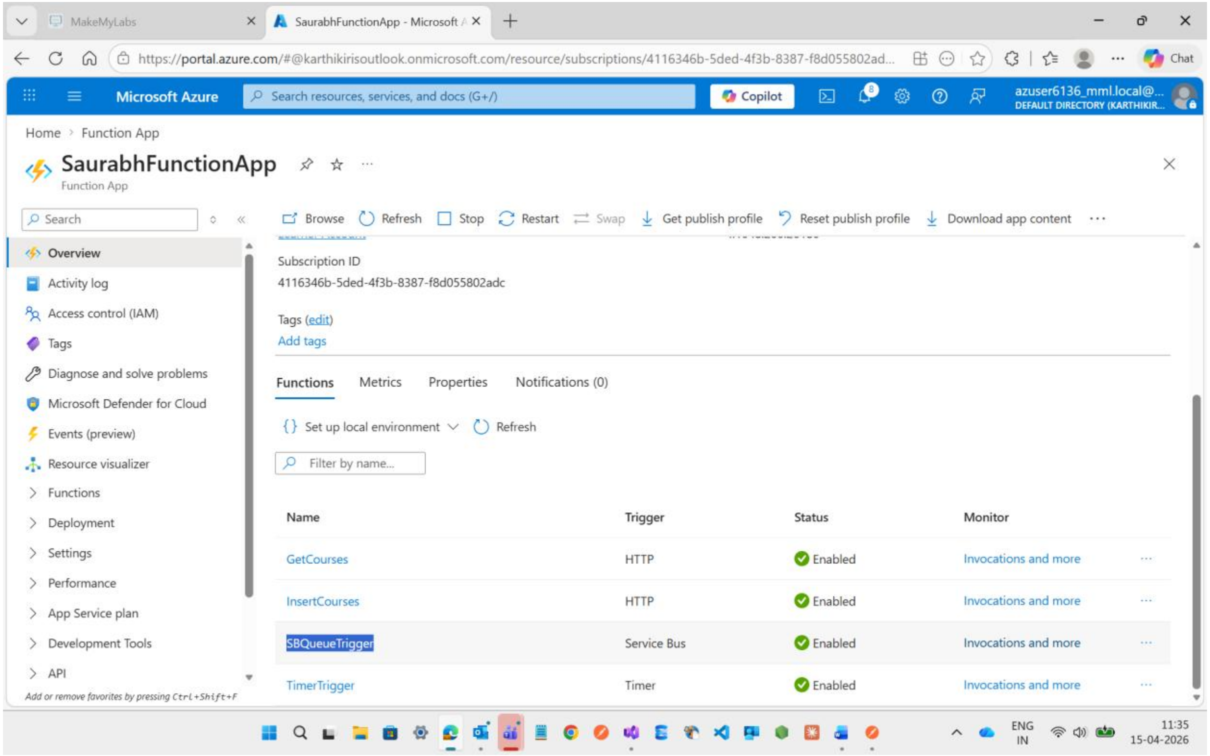

Publish
Which Azure service would you like to use to host your application?
Target
Specific target
Azure Function App
Publish your application code to a serverless compute that scales dynamically and runs code on-demand
Azure Function App Container
Publish your application as a Docker image to Azure Container Registry and run it on Azure Function App
Azure Container Apps (Linux)
Run scalable containerized applications and microservices on a serverless platform in Azure
Azure Container Registry
Publish your application as a Docker image to Azure Container Registry
Back
Next
Finish
Cancel



See below the publish is completed

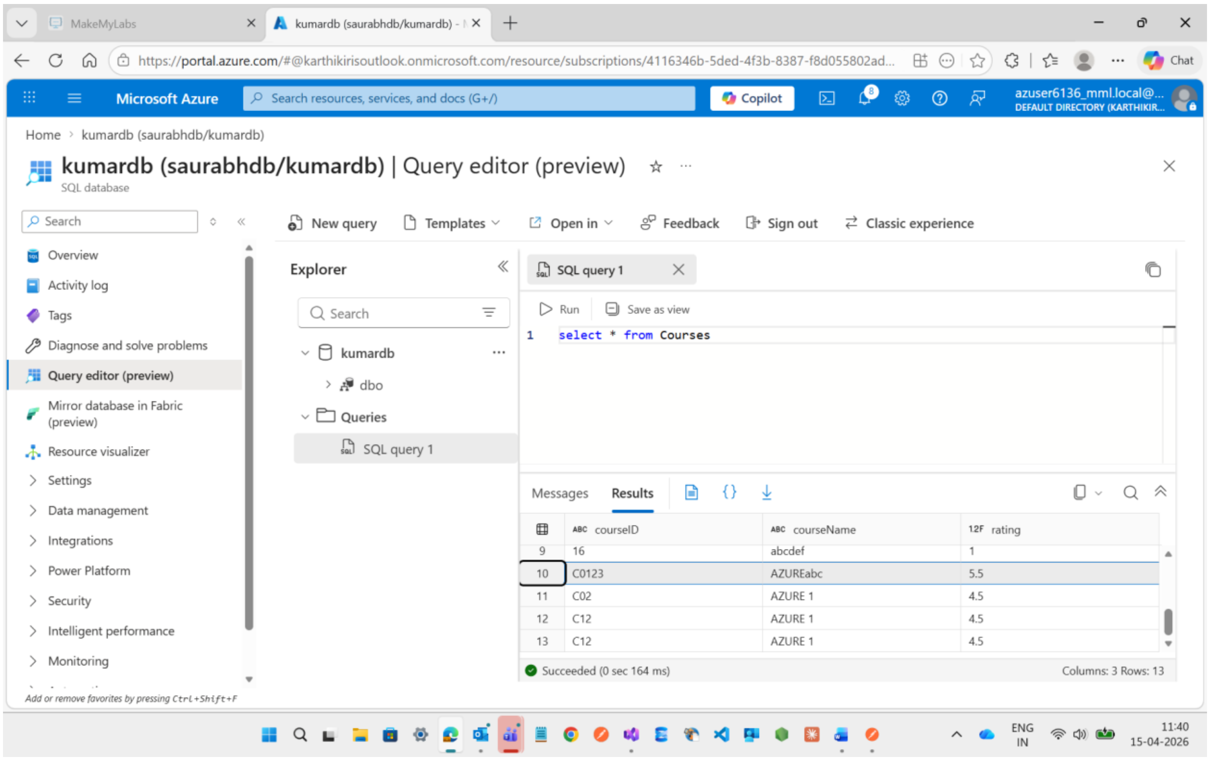


Then I will go to the Function App there the service bus trigger will appear

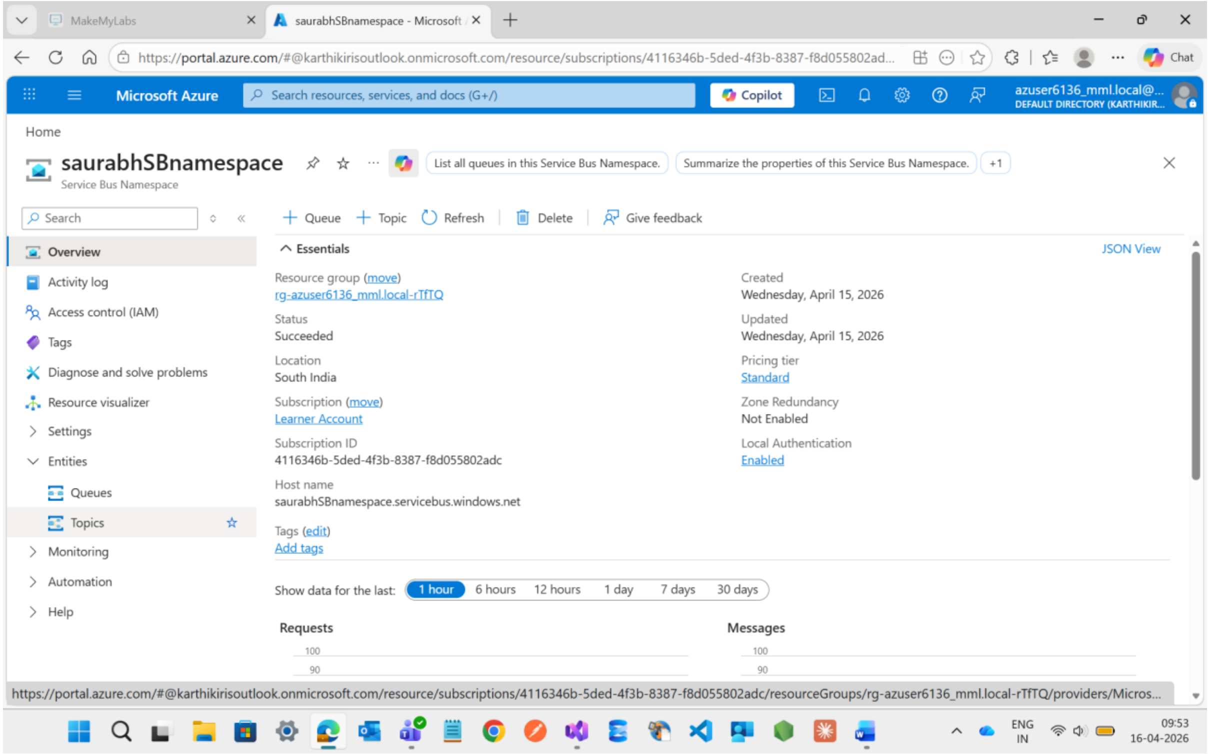


Then after we can only see the details which we provided through Post Method instead of Queue we can only see in our database

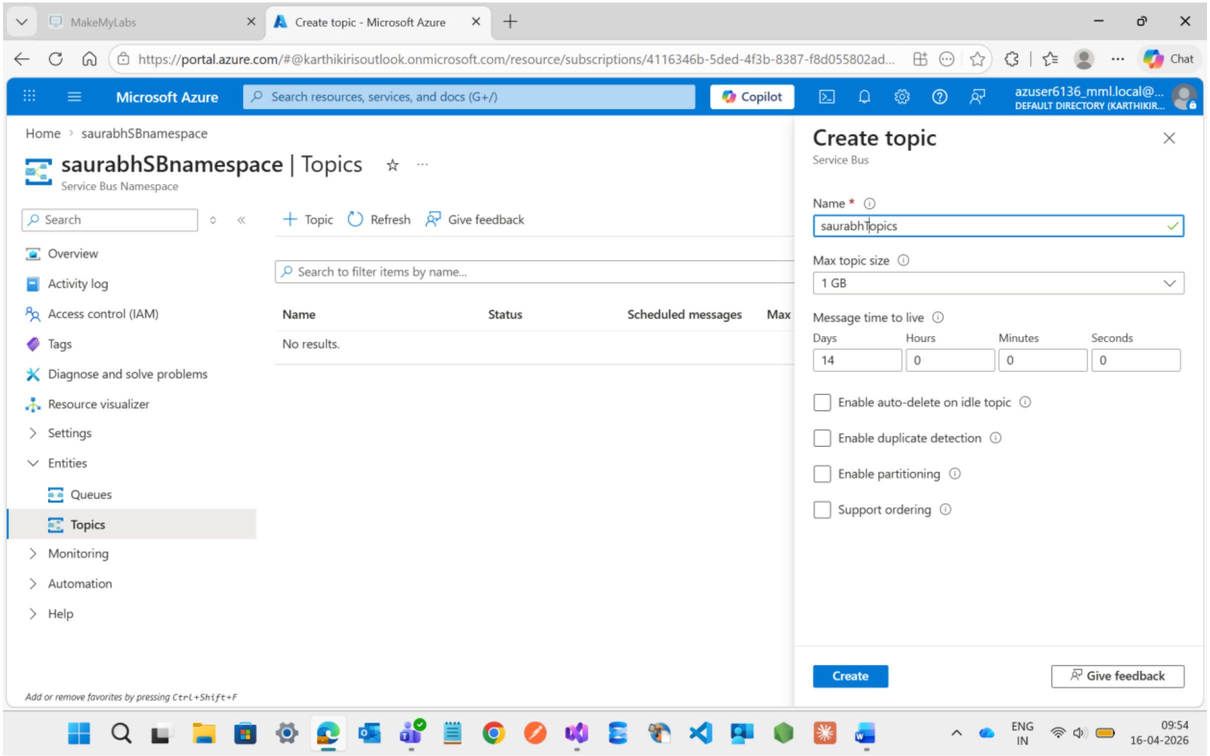
See below the details is clearly visible



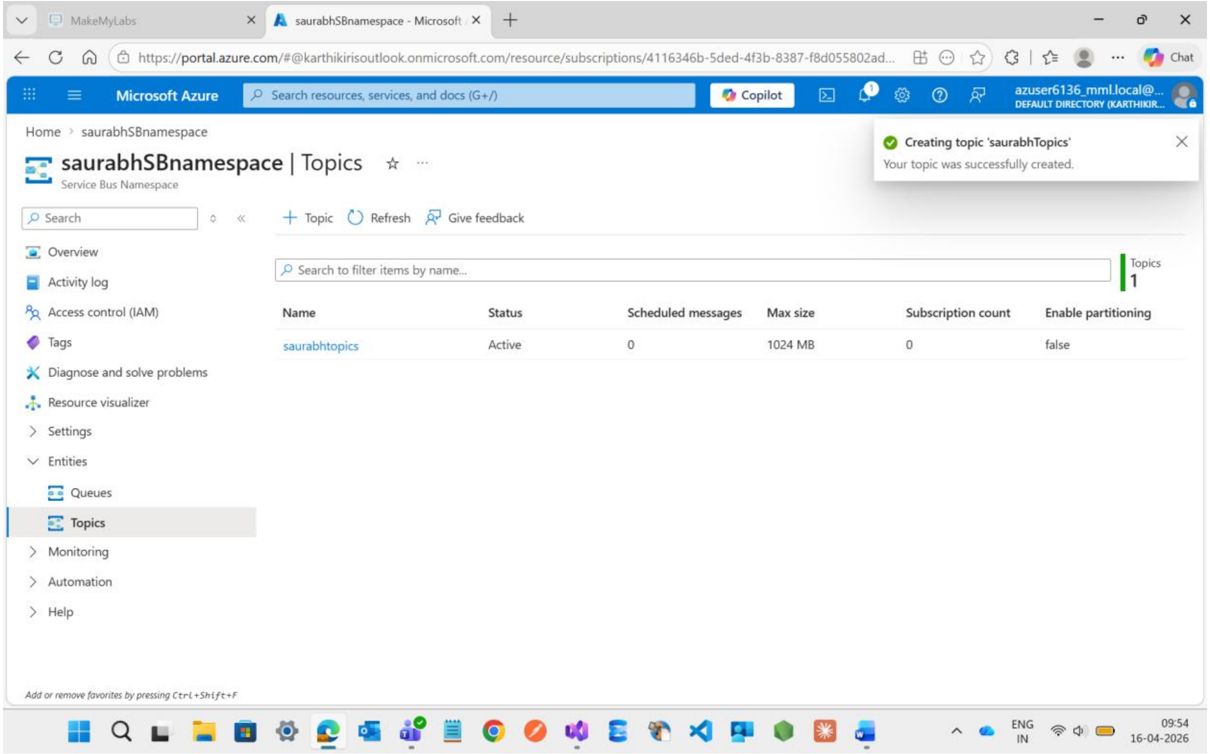
Now I m going to create a Topics



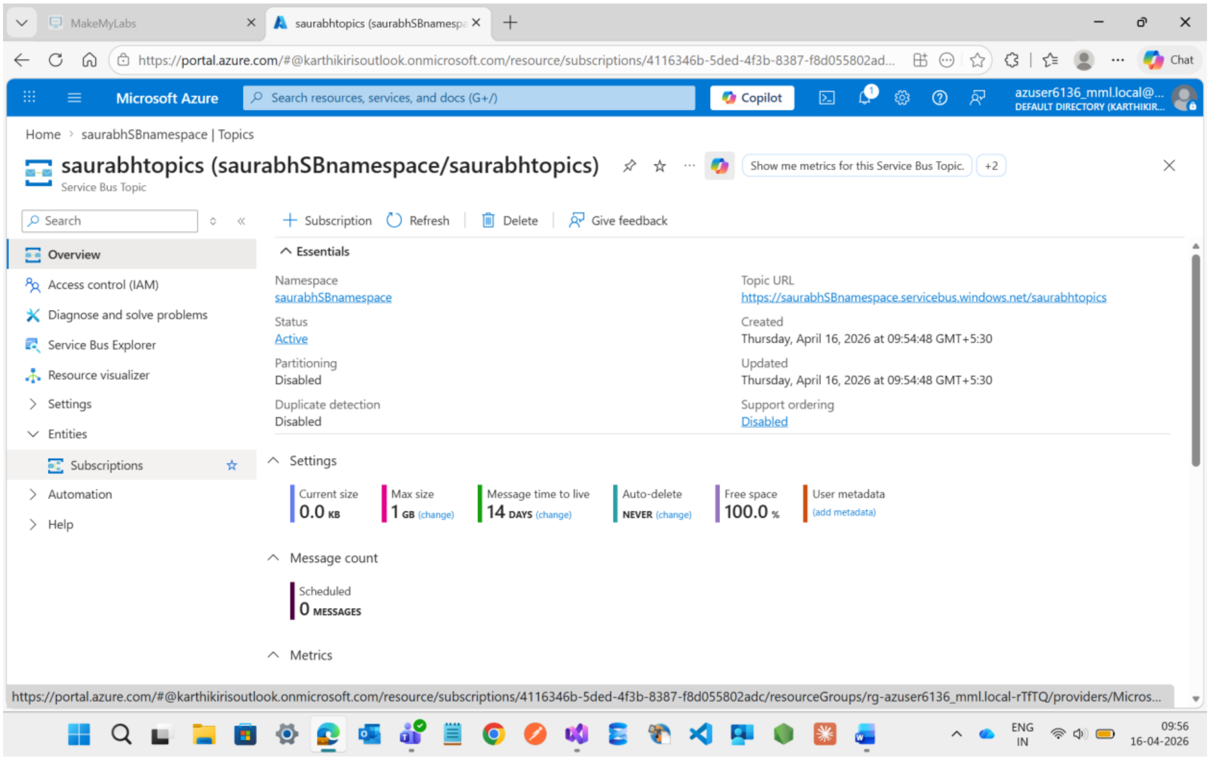
Here I m adding a Topics with the name of saurabhTopics



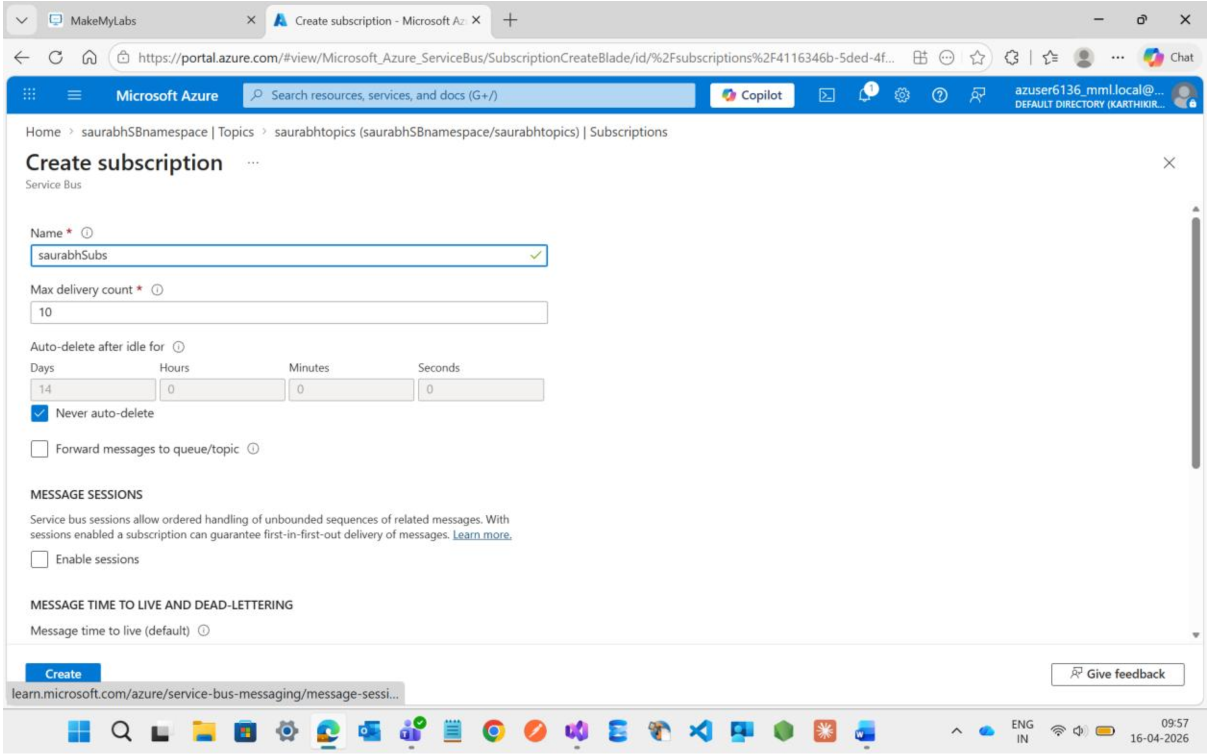
Successfully Created



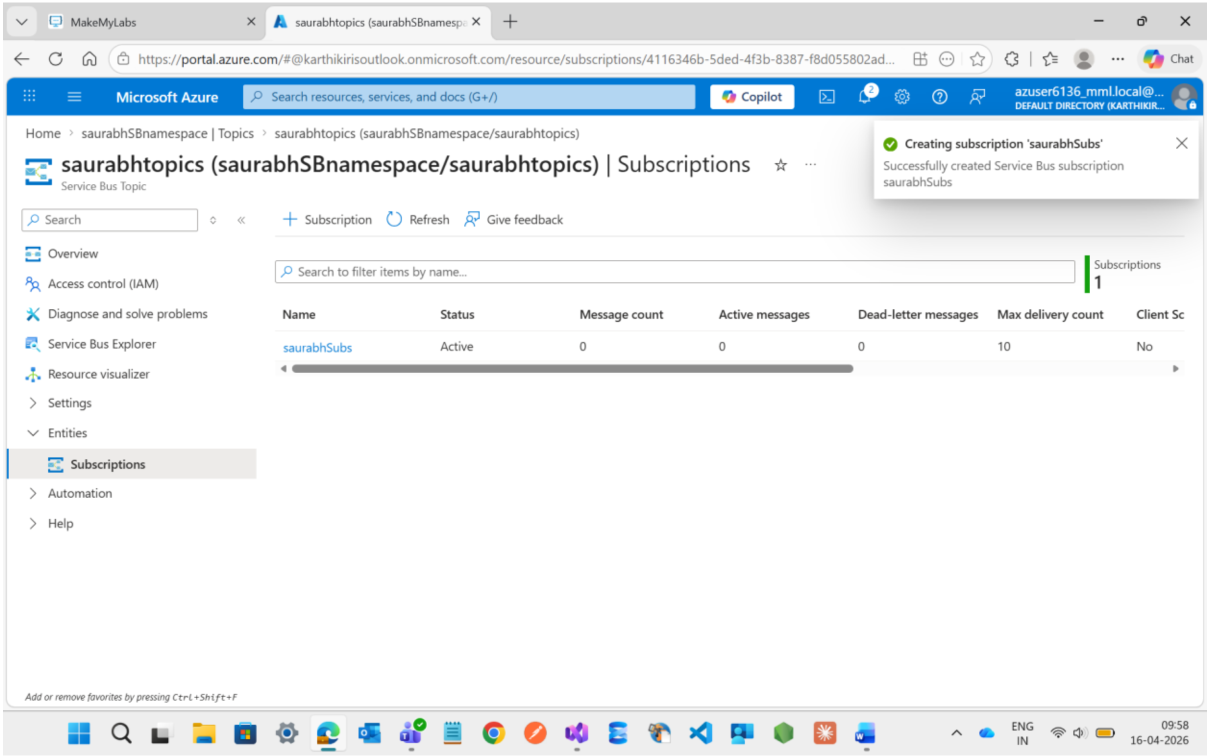
Now I m going inside the Topics and I m selecting the Subscriptions Option



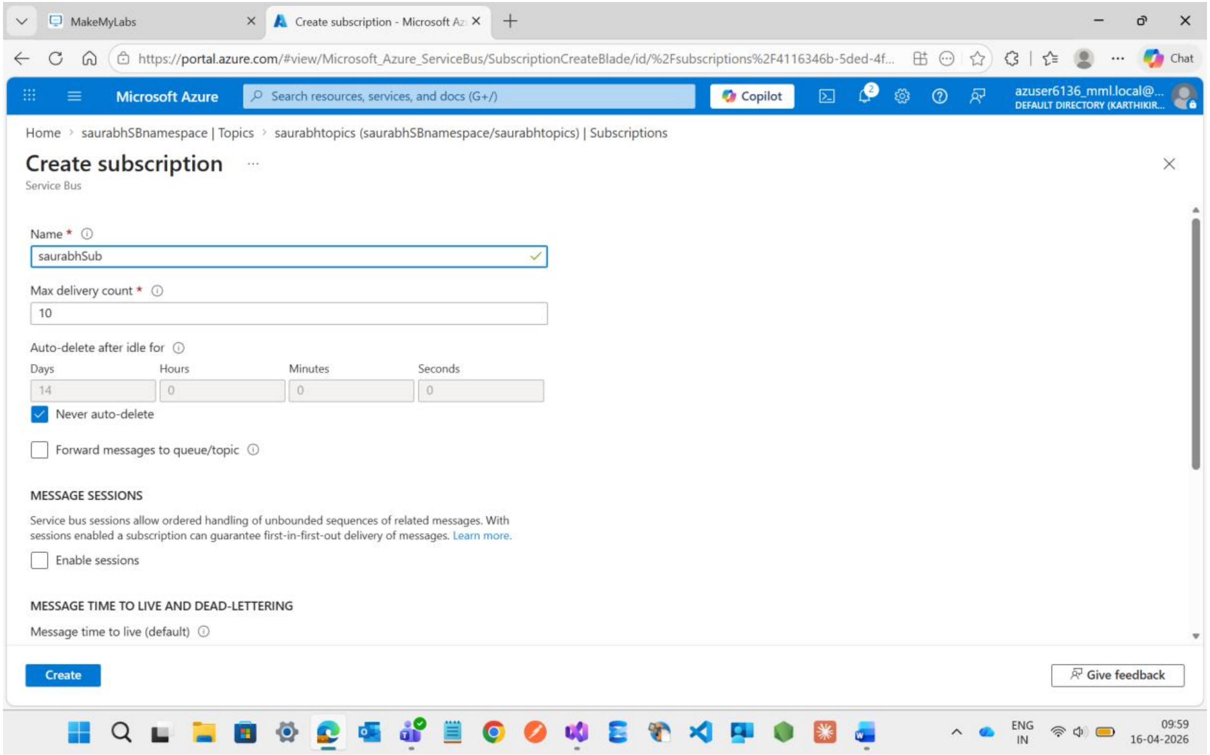
Now I m creating one Subscriptions with the name of saurabhSubs



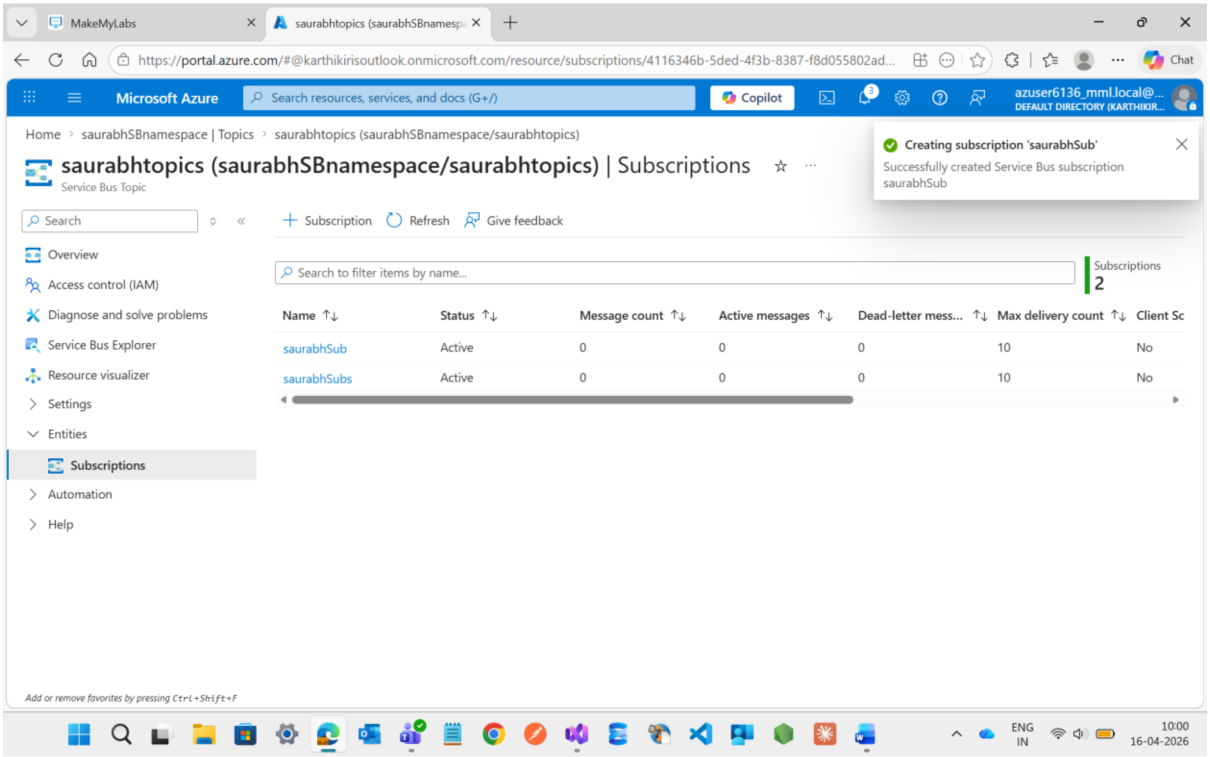
Successfully created



Again one more Subscriptions I m going to create with different name of saurabhSub



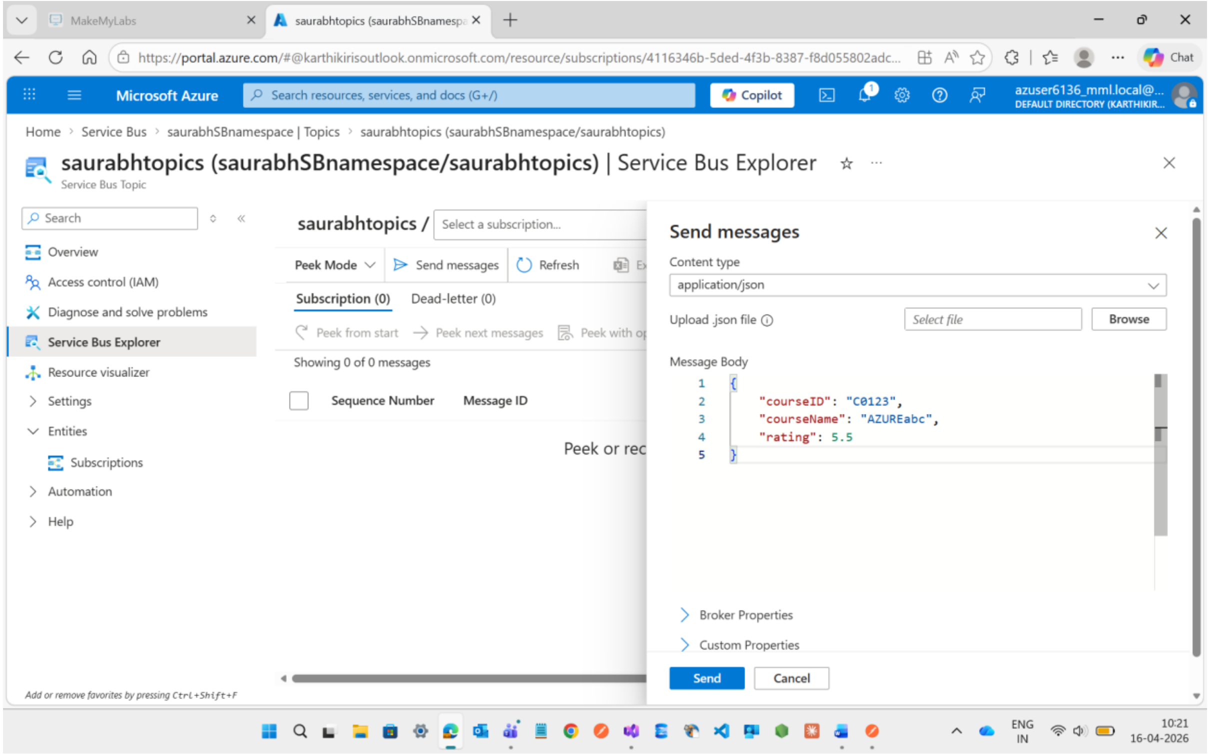
Successfully created



If any message will come it will available in both Topics.

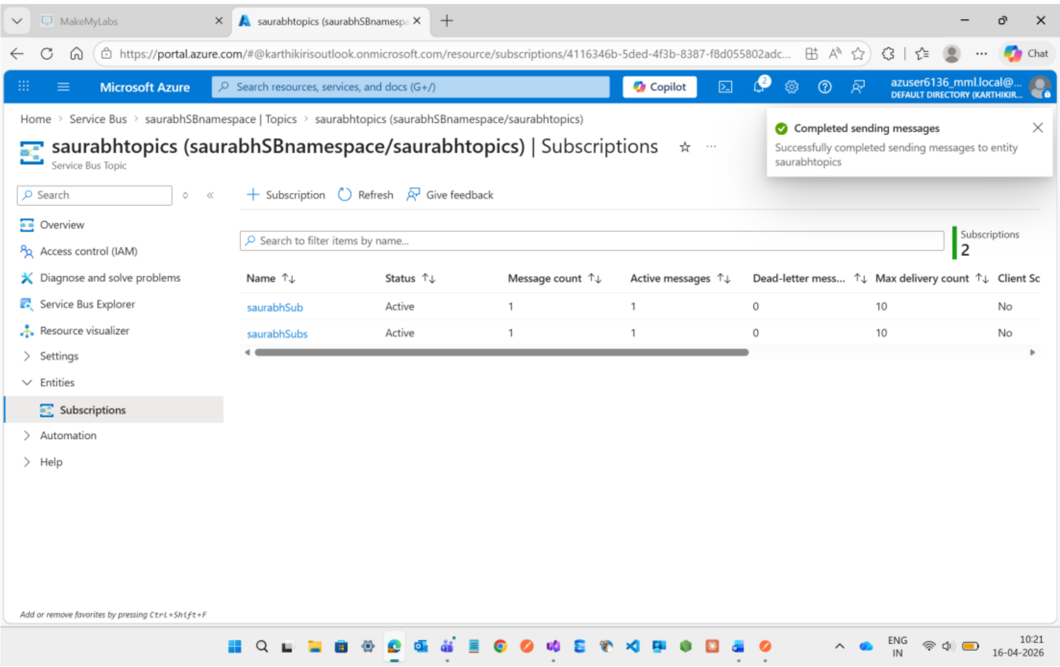
Now we can also send the message through the Service Bus Explorer

See inside the Topics there is Service Bus Explorer options and inside that there will be Send messages options is available we can write our message here it will directly goes to Subscriptions

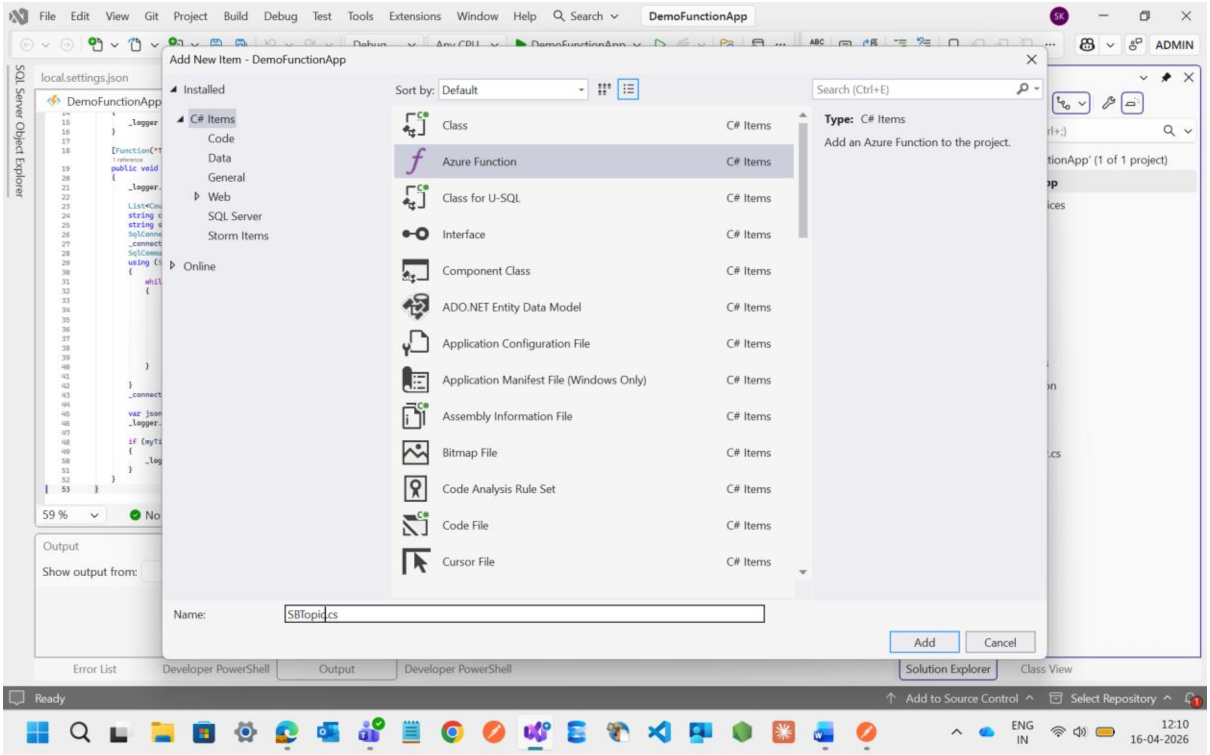


I m sending this message in application/json format

See it will showing 1 Message Count and 1 Active Messages is available

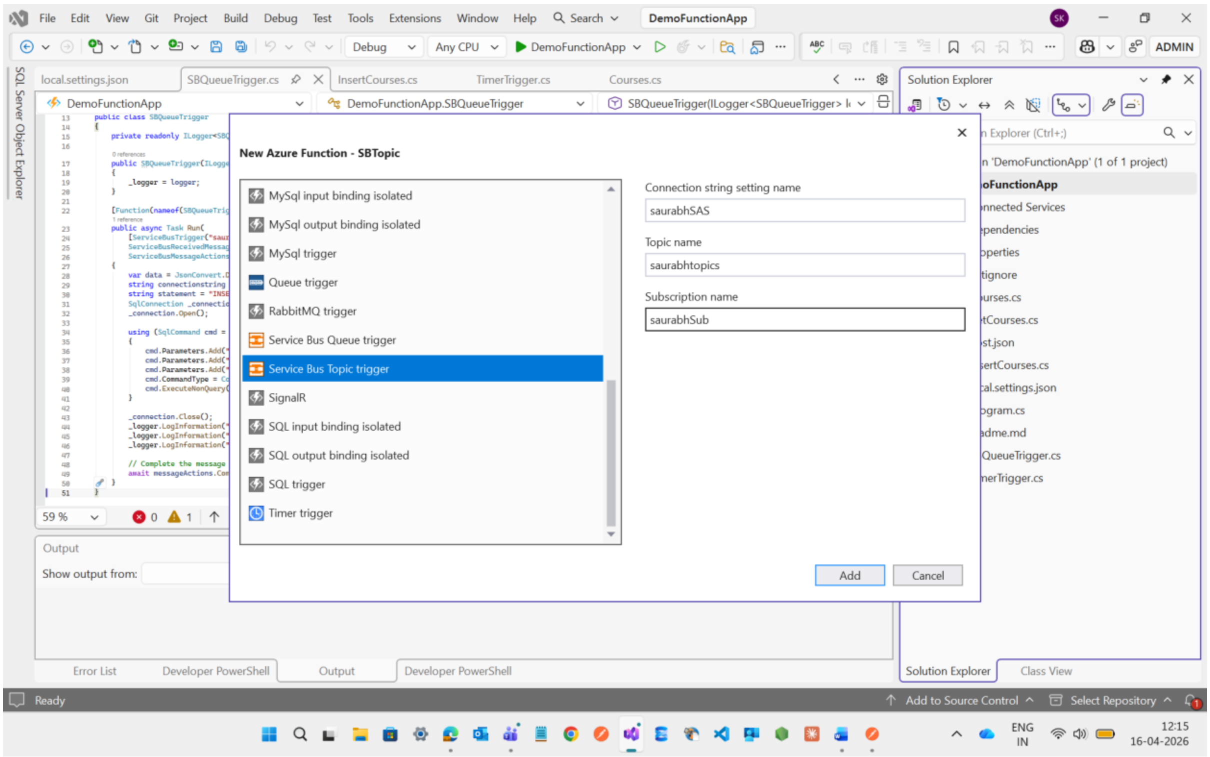


Now I m going to create one Function with the name of SBTopic

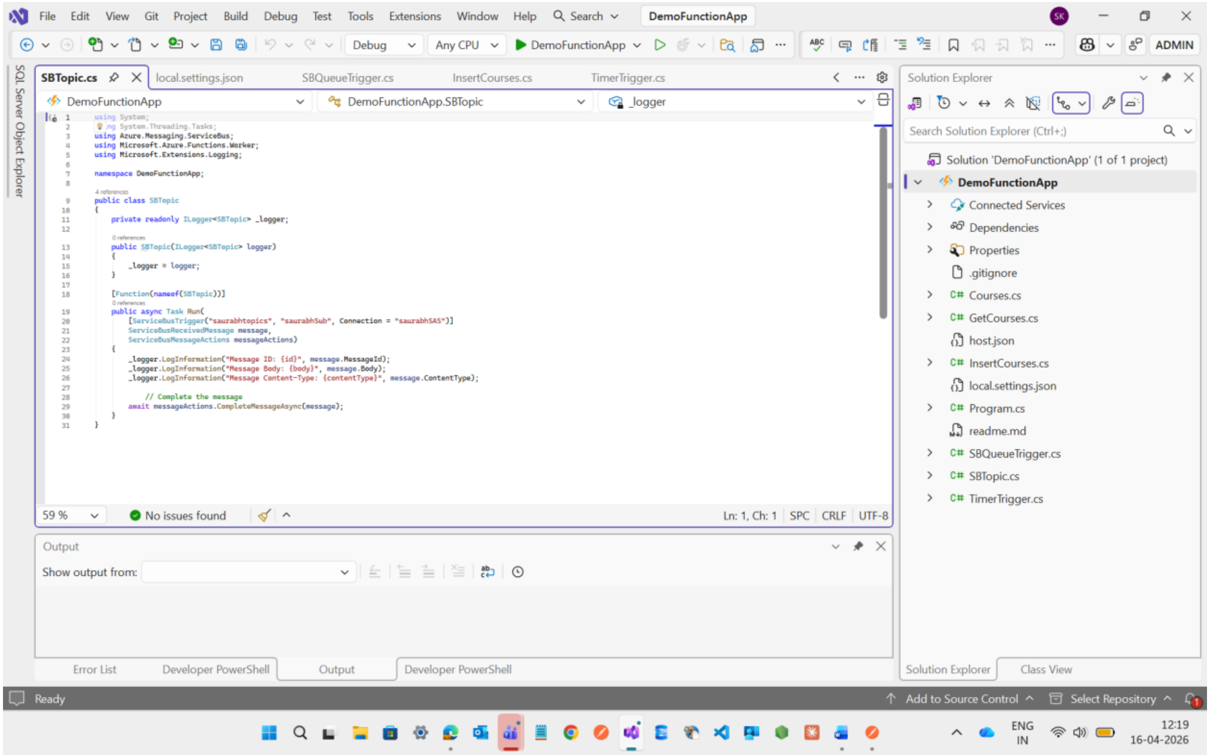


Here I have mentioned the Connection String, Topic name and Subscription name as well as these all are available on Service Bus Topics as well.

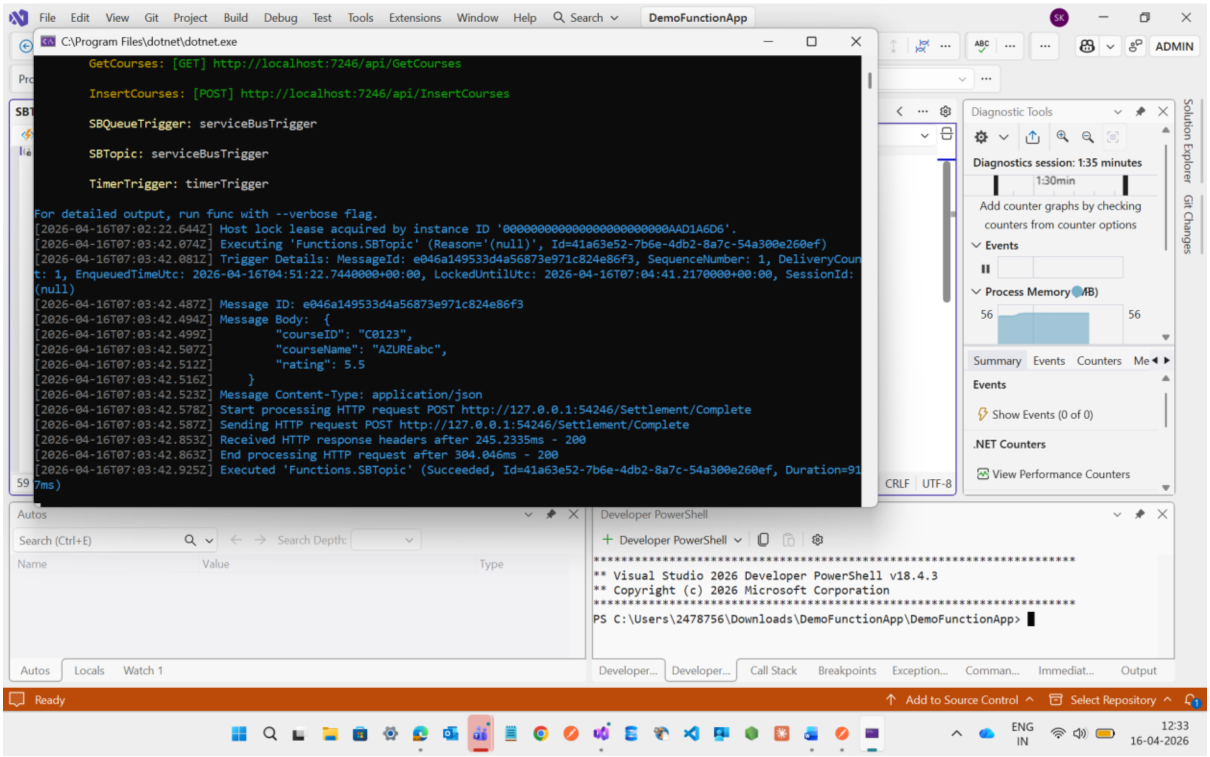
In place of subscription name I have created two subscription in Topics with different name but I choose one as per my choice



See Service Bus Topics File is created successfully



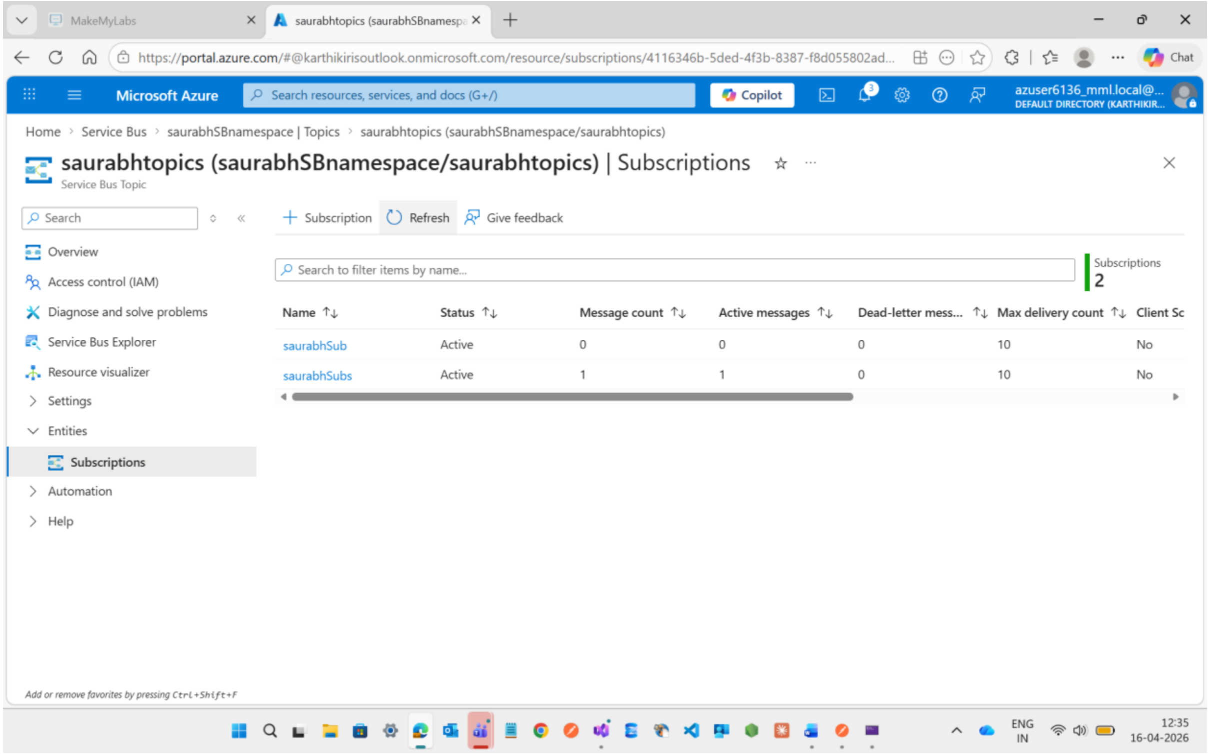
Now I will execute it to check the wheather the message from Subscription 1st saurabhSub is getting Message count 0 or not and 2nd Subscription which is saurabhSubs is still having the same Message Count 1 or not.



See on above execution part I will get the details which I had write inside the service explorer in Topics

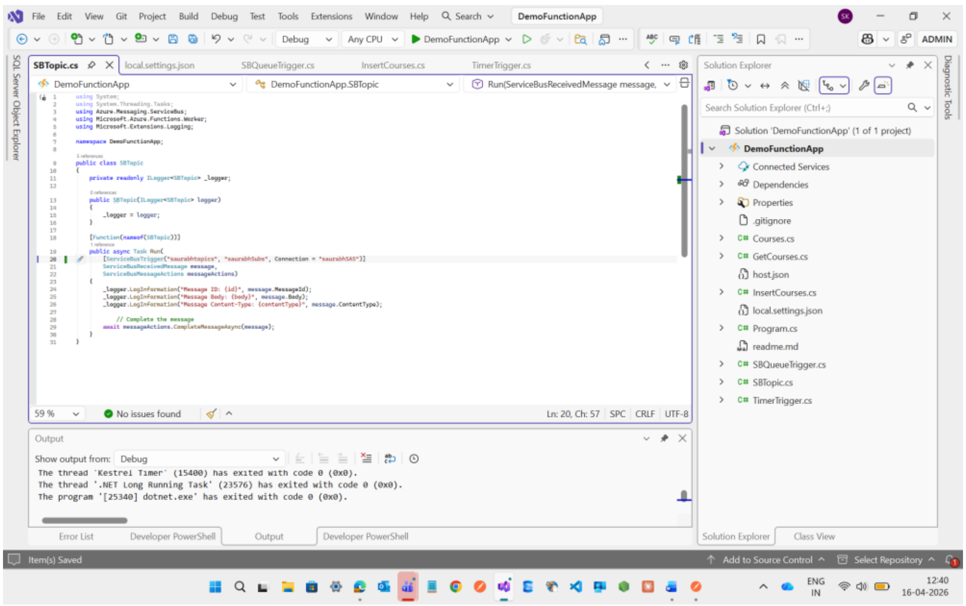
And in subscriptions 1st Message Count will become 0

And Subscriptions 2nd Message Count is still same

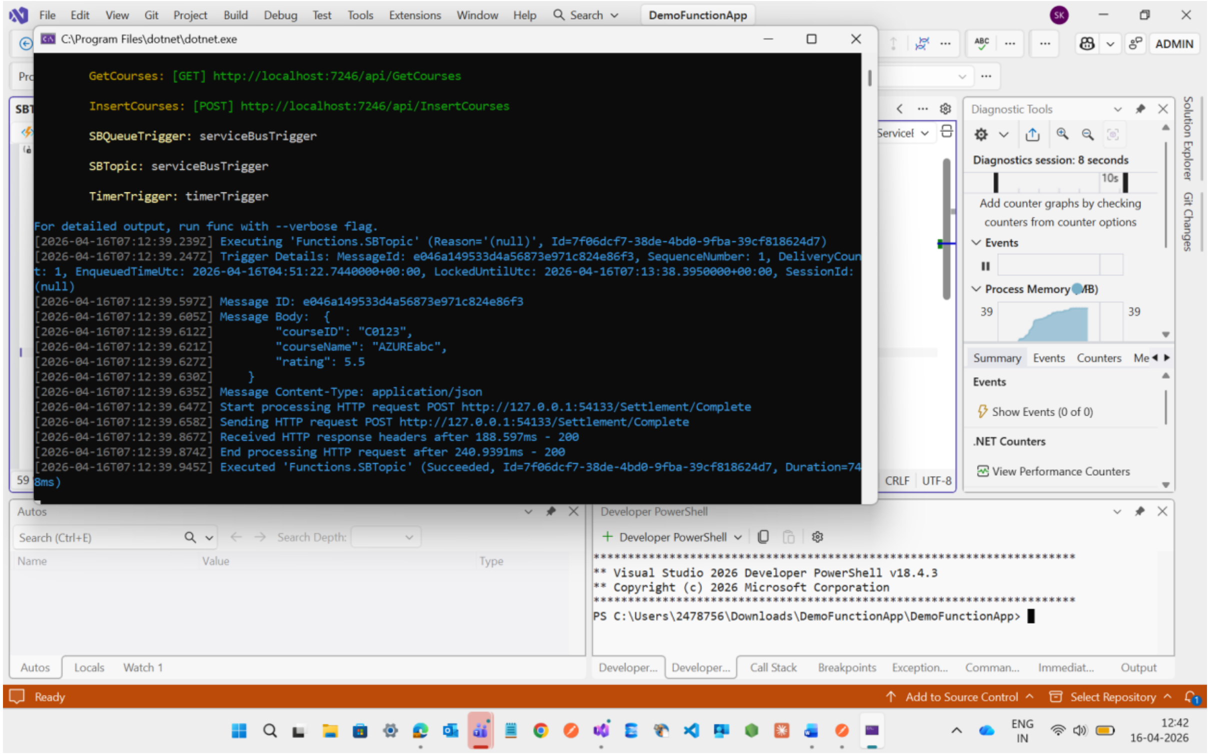


Then after this I will do the changes inside the code of Topic files.

Just change the subscription value like before it was saurabhSub now I will assigned it as saurabhSubs then both Subscription Message Count and Active message will become 0.

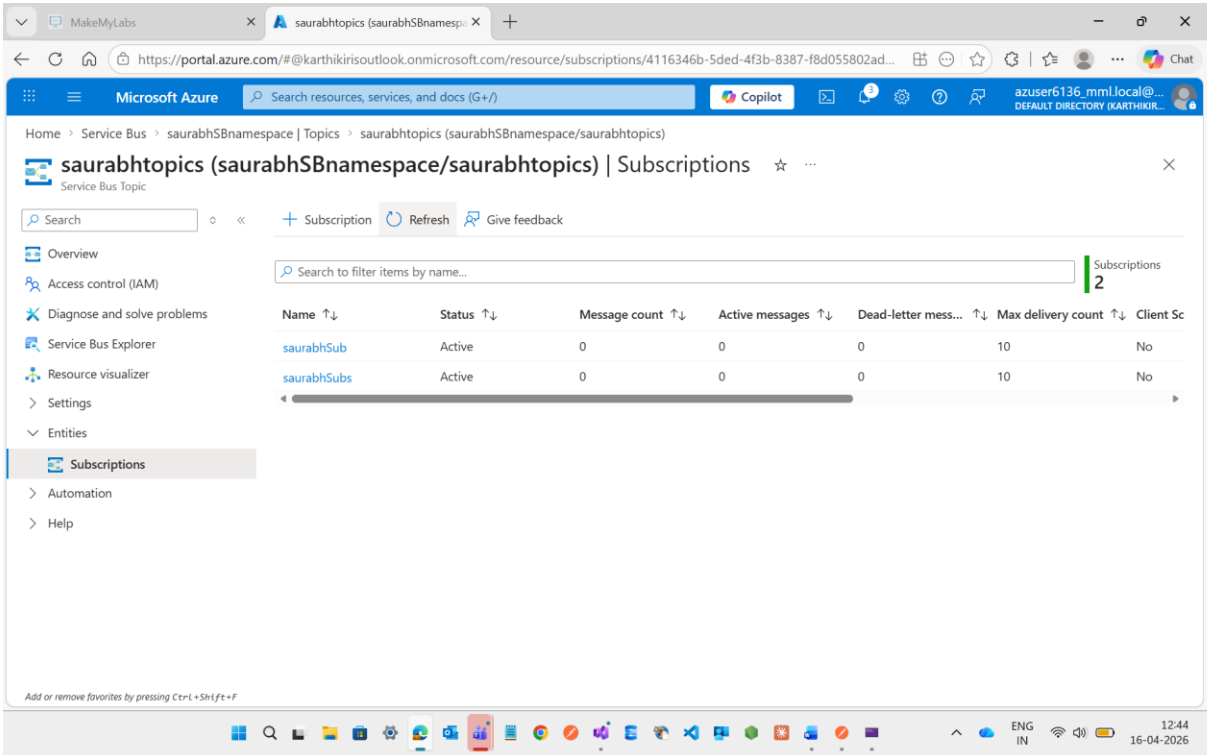


Now I will execute the code



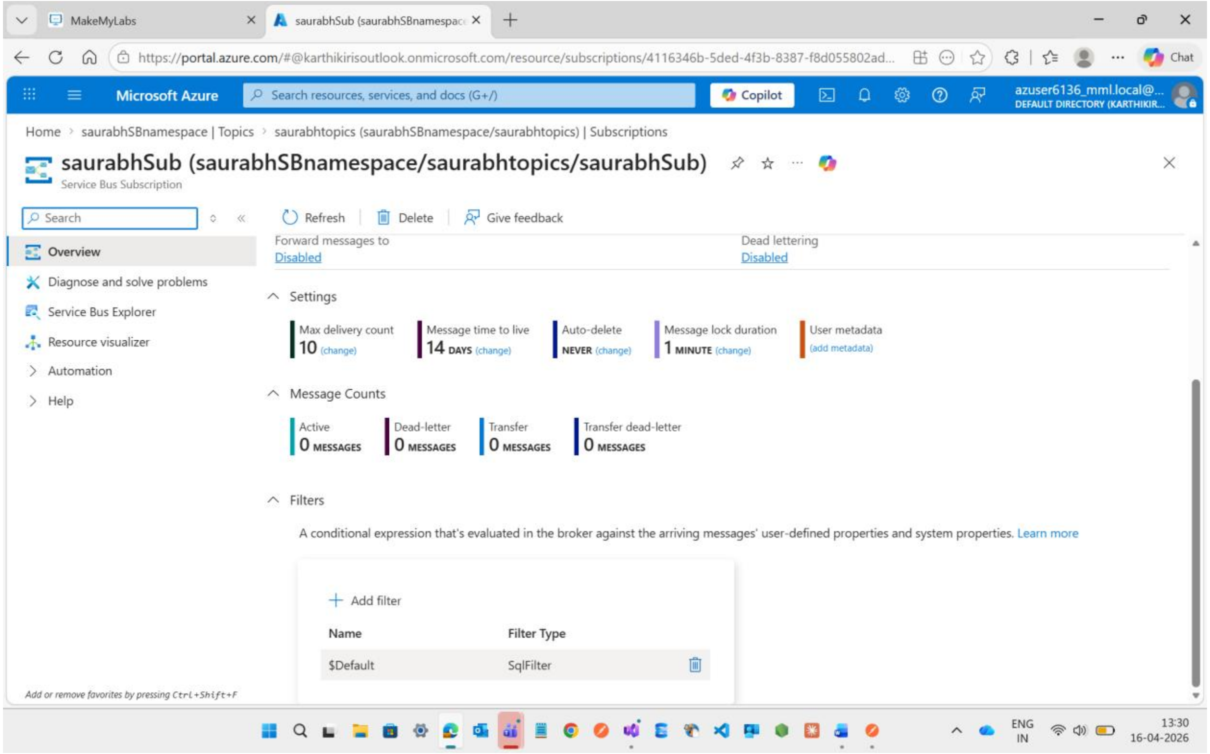
See the result Is showing the same as like what I have done it before

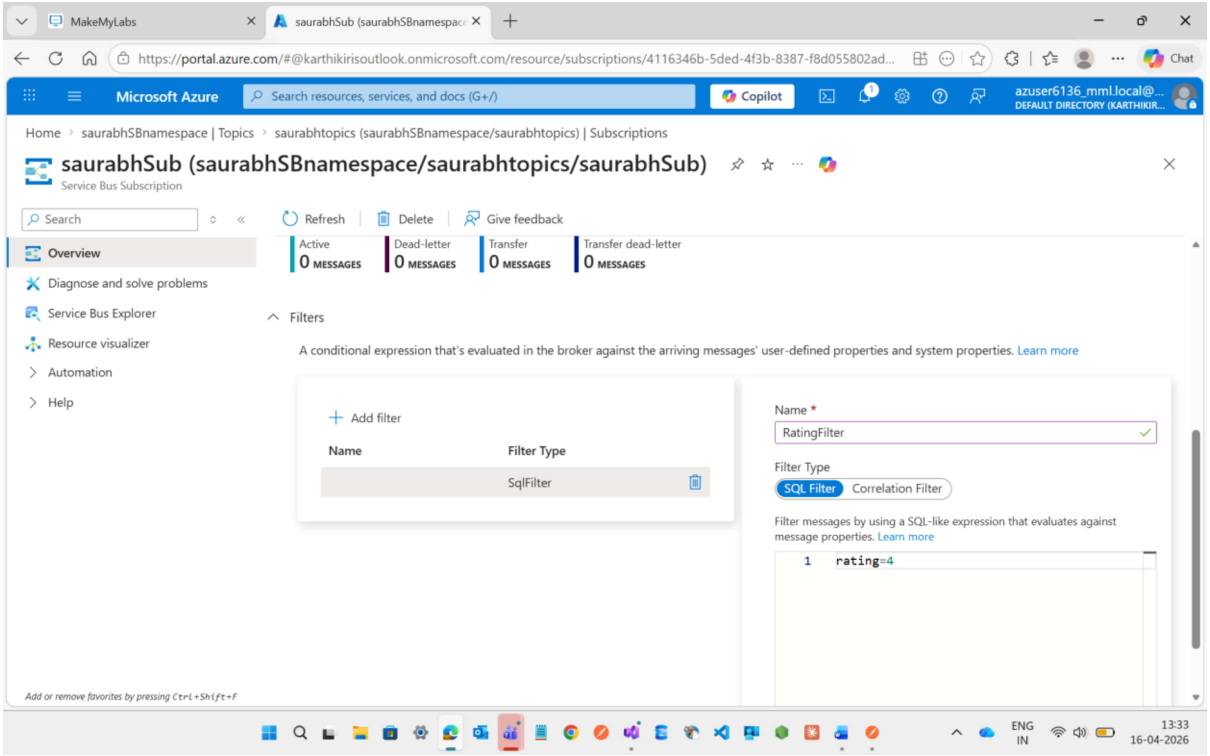
Now in Azure I will check the Topic Subscriptions saurabhSubs it will become 0.



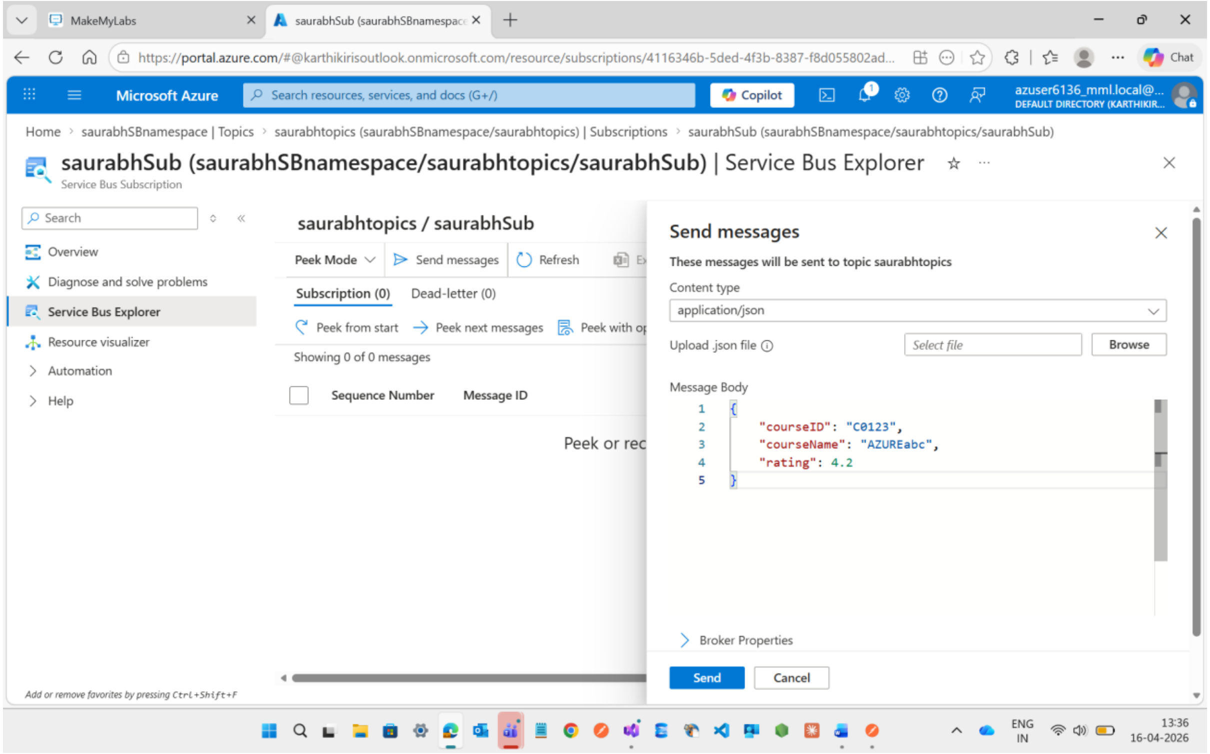
Then After doing this all implementation Topics Is having the Filtering Options to do the filter of message based on the requirement.

Now I m going inside the subscription 1st saurabhSub and Apply the filter it has already having default filter first I delete it and then make a new filter with the name of RatingFilter and assign a rating value.

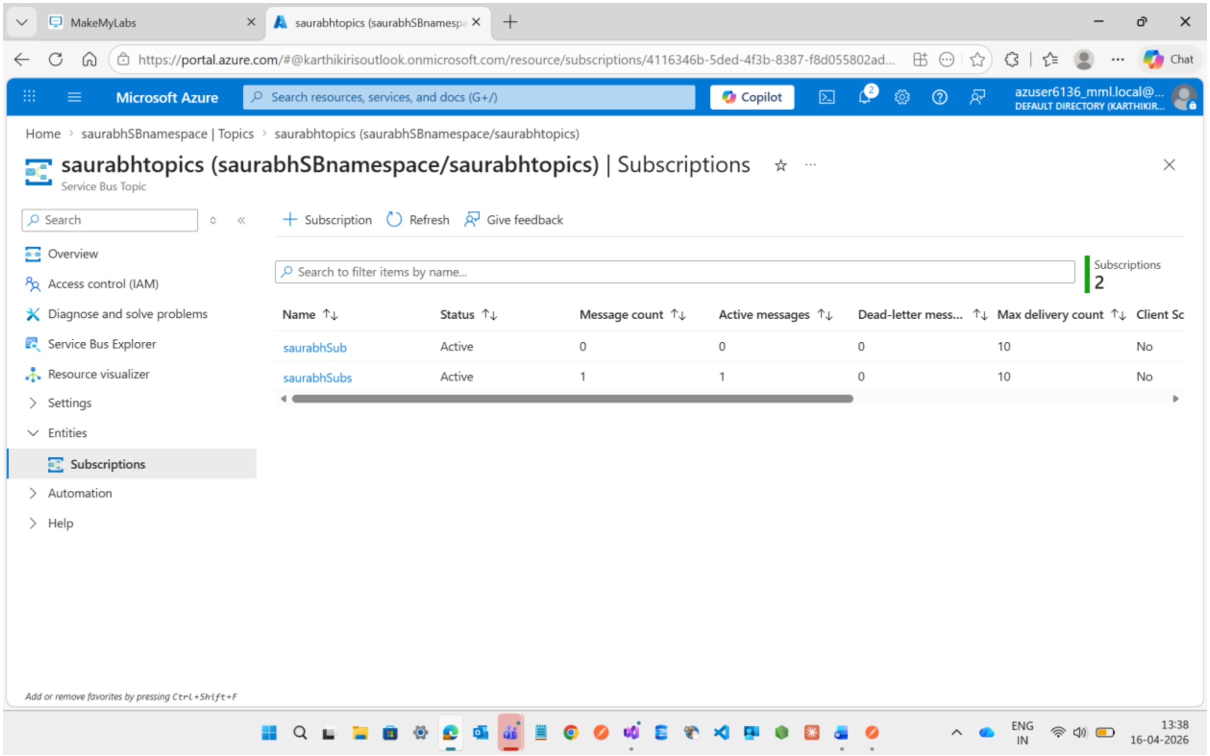




Then go to service explorer and write the details with rating value = 4 if it is less than and more than 4 then Subscription saurabhSub it will not accept it only saurabhSubs will accept it

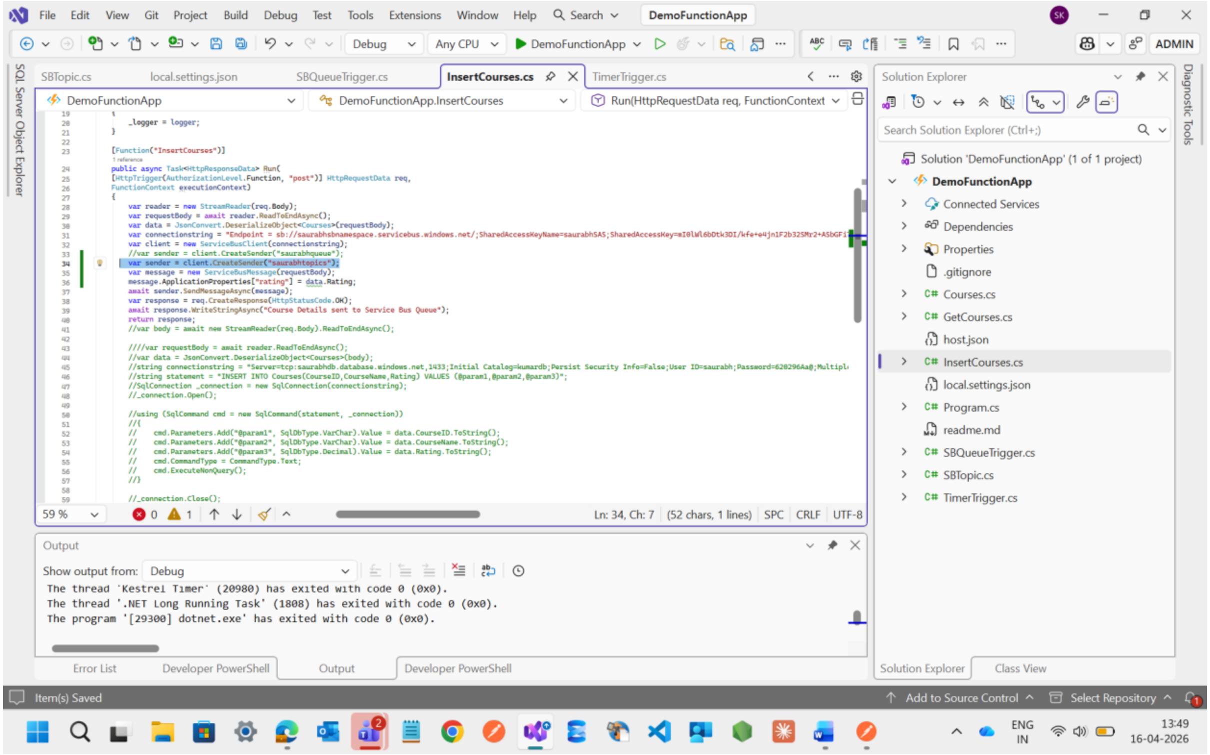


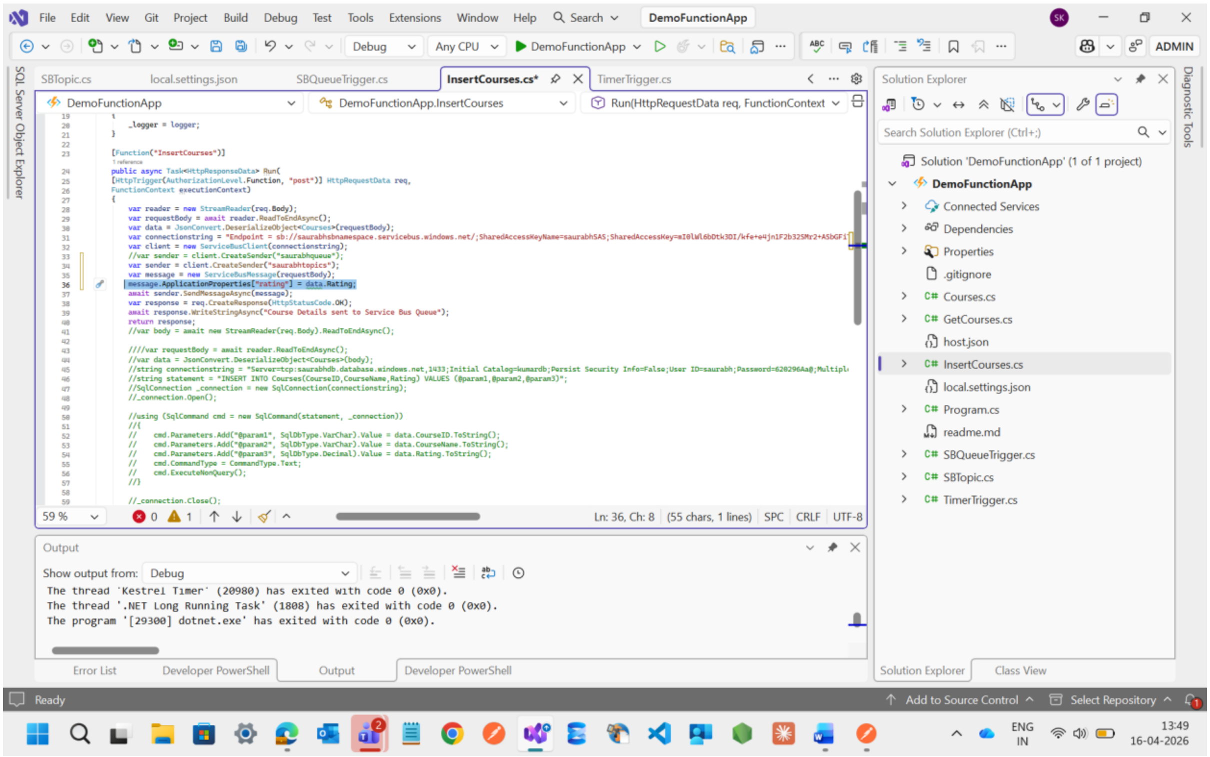
See below it will clearly show the message count of **saurabhSub is 0** and **saurabhSubs is 1**

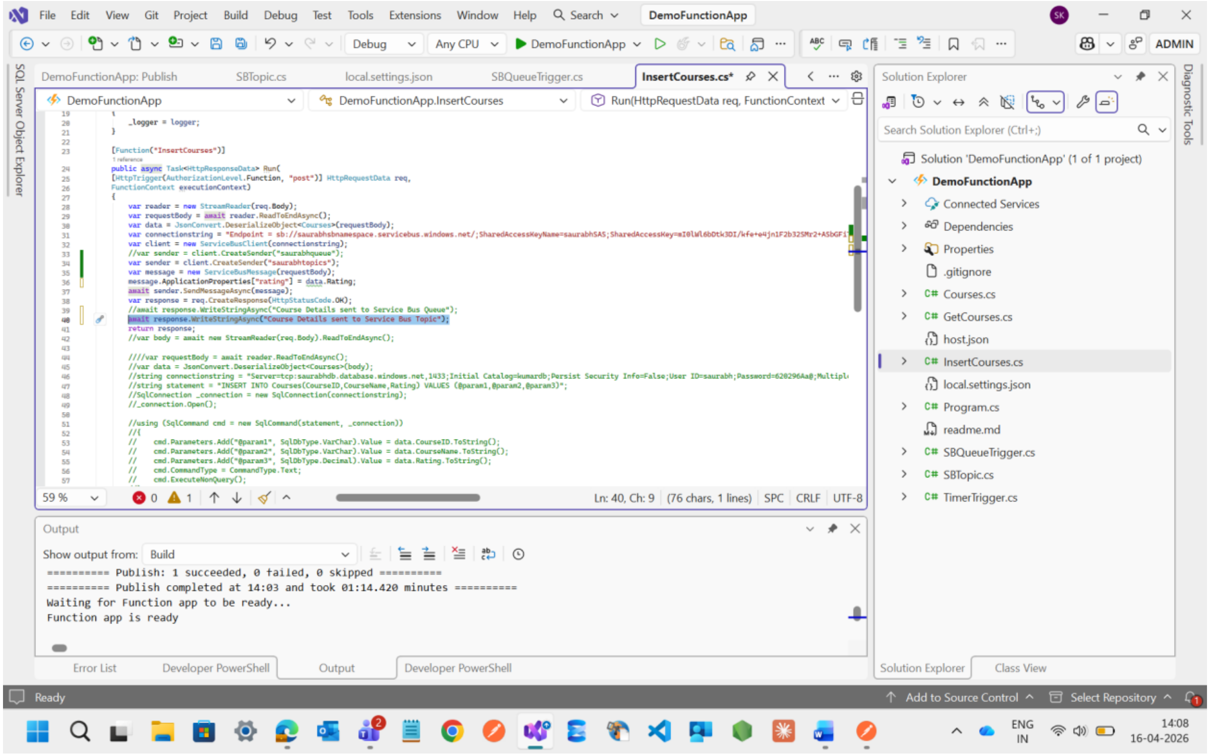


Now I m going to change the rating value is equal to 4 in this case Subscription saurabhSub become 1 and saurabhSubs become 2 in case of saurabhSubs it is not having a filter that's why it will accept all the request or messages.

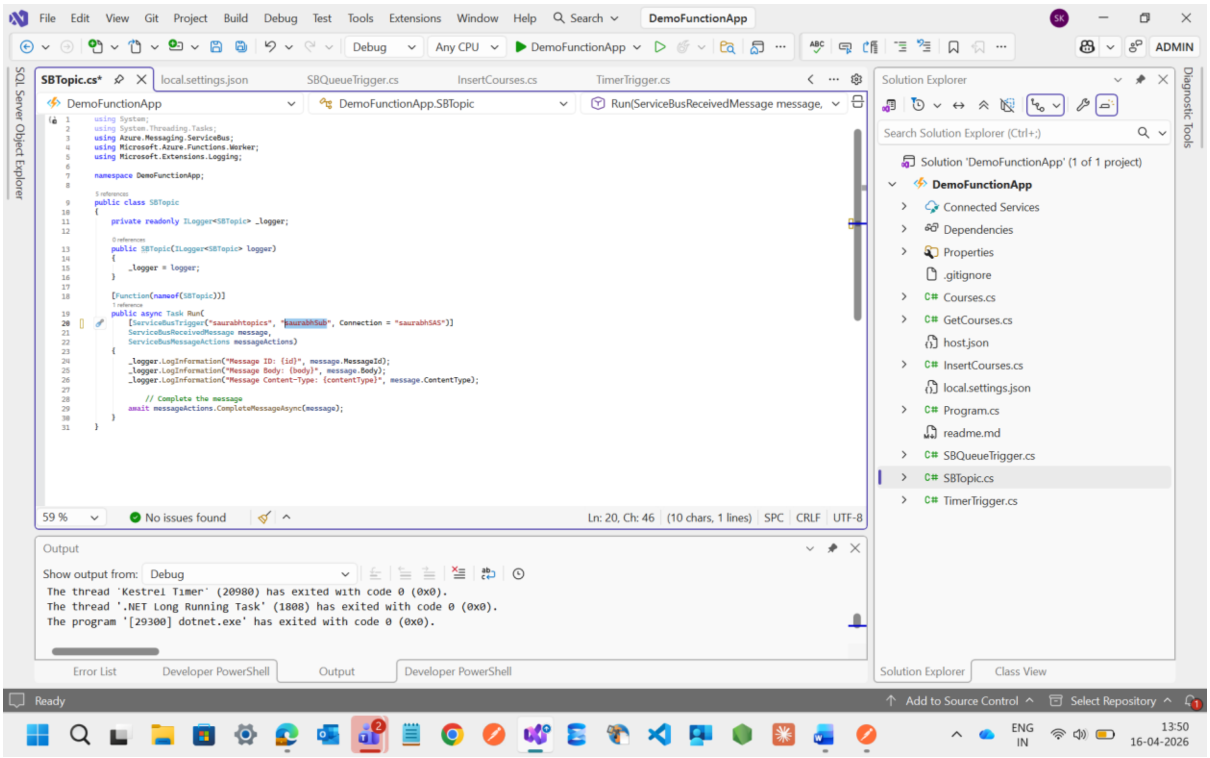
But before the above point to be work I need to do some changes inside insertCourses method instead of assign the saurabhqueue I just change to saurabhtopics and add one more line for rating filtering and also change the response Queue to Topic



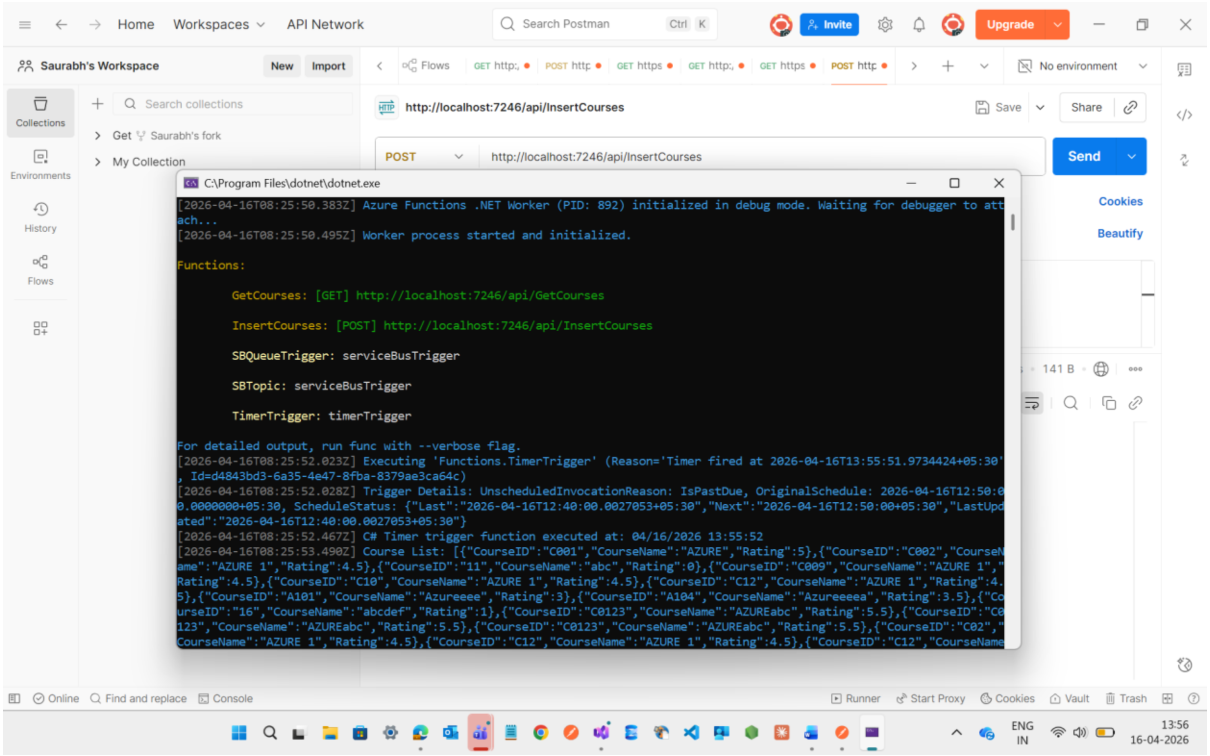




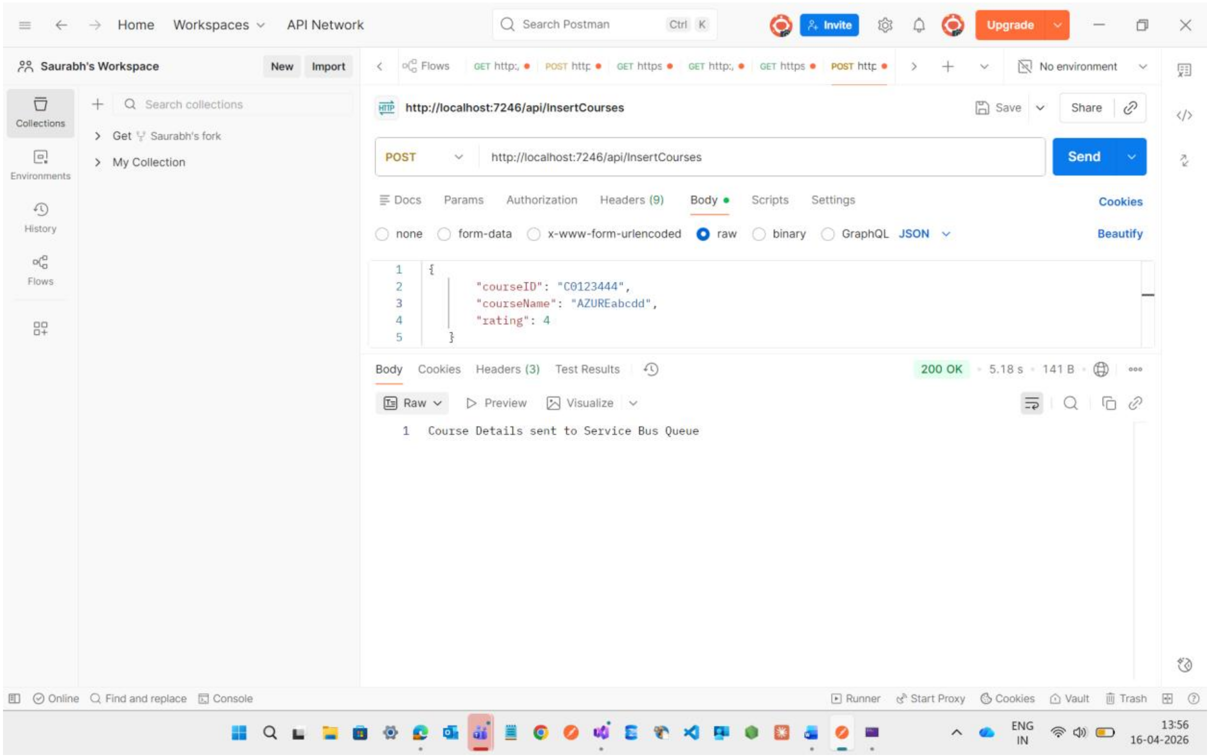
And also change the ServiceBusTrigger of Topics Subscriptions to saurabhSaus to saurabhSau then it will work



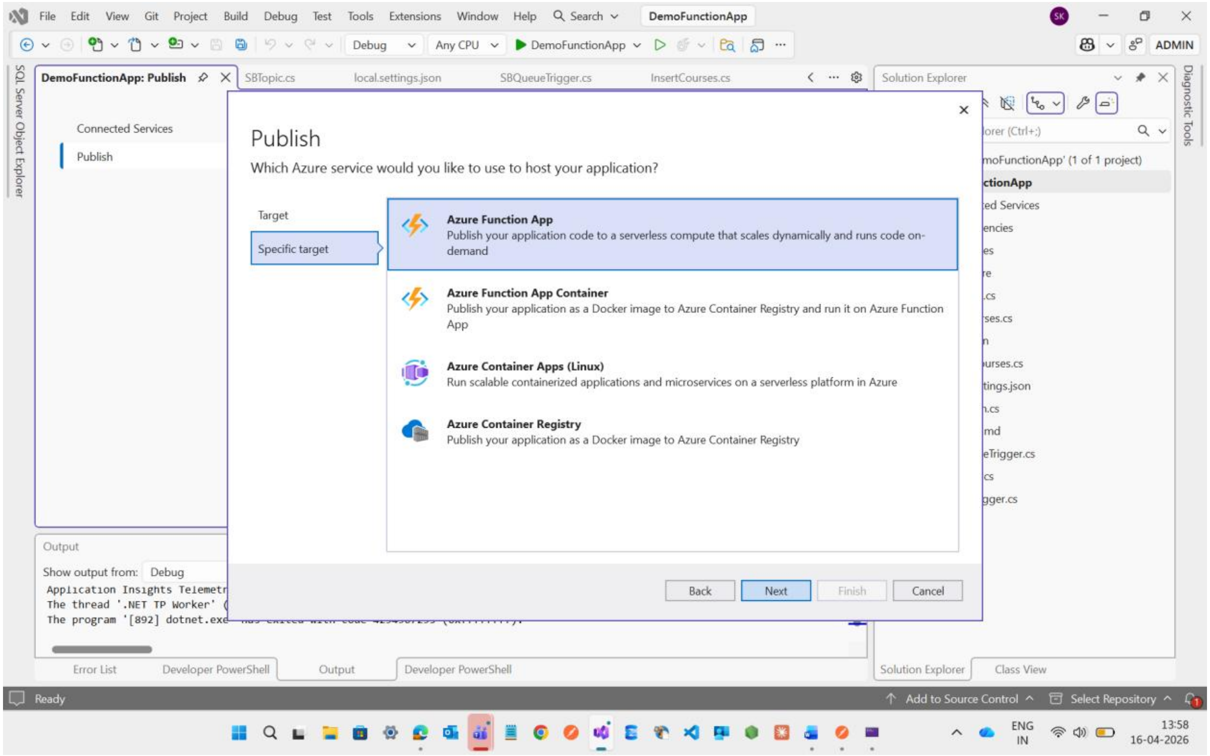
Now first I will try to check it locally wheather it is working or not for that I m going to execute it

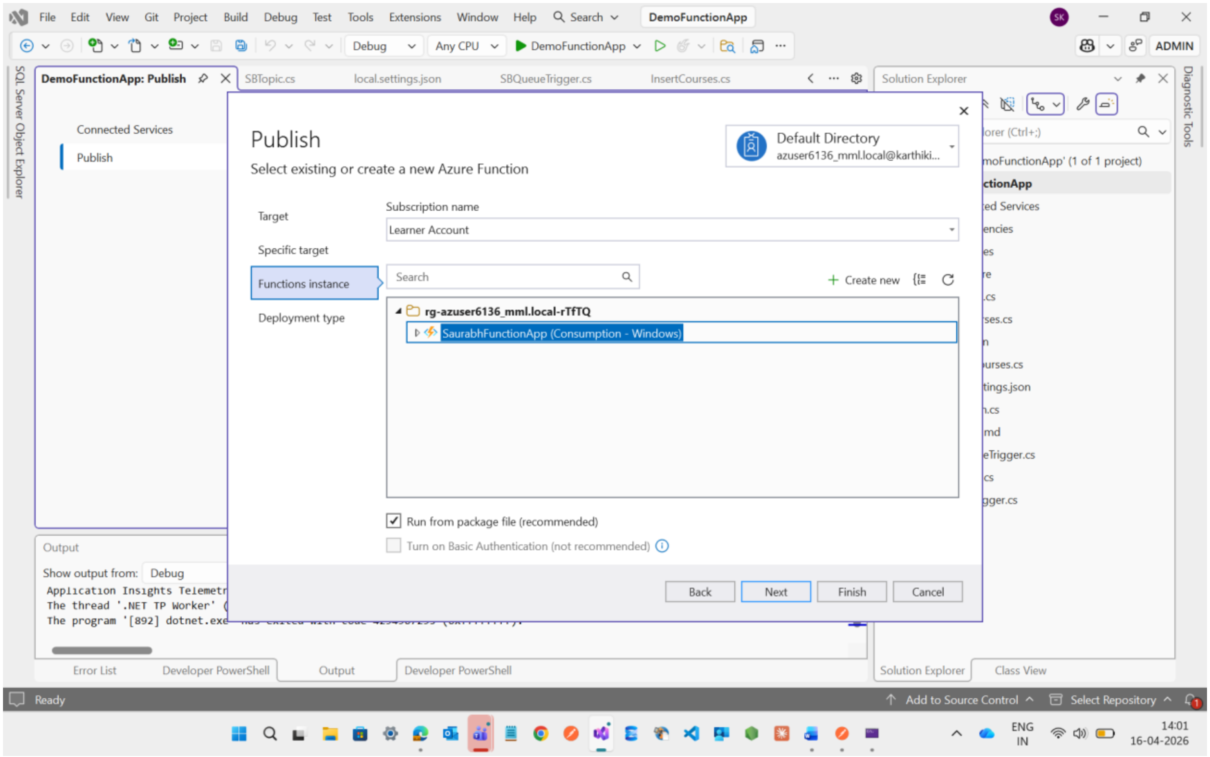


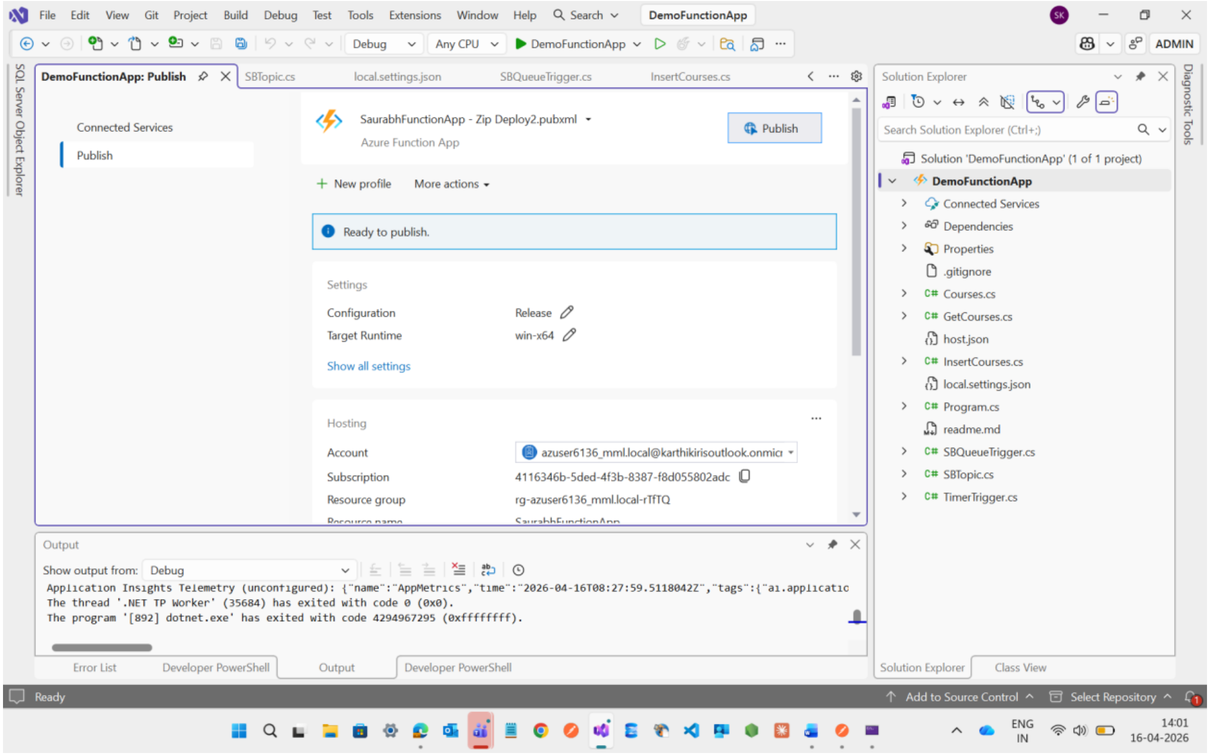
See below it will send the details to the queue



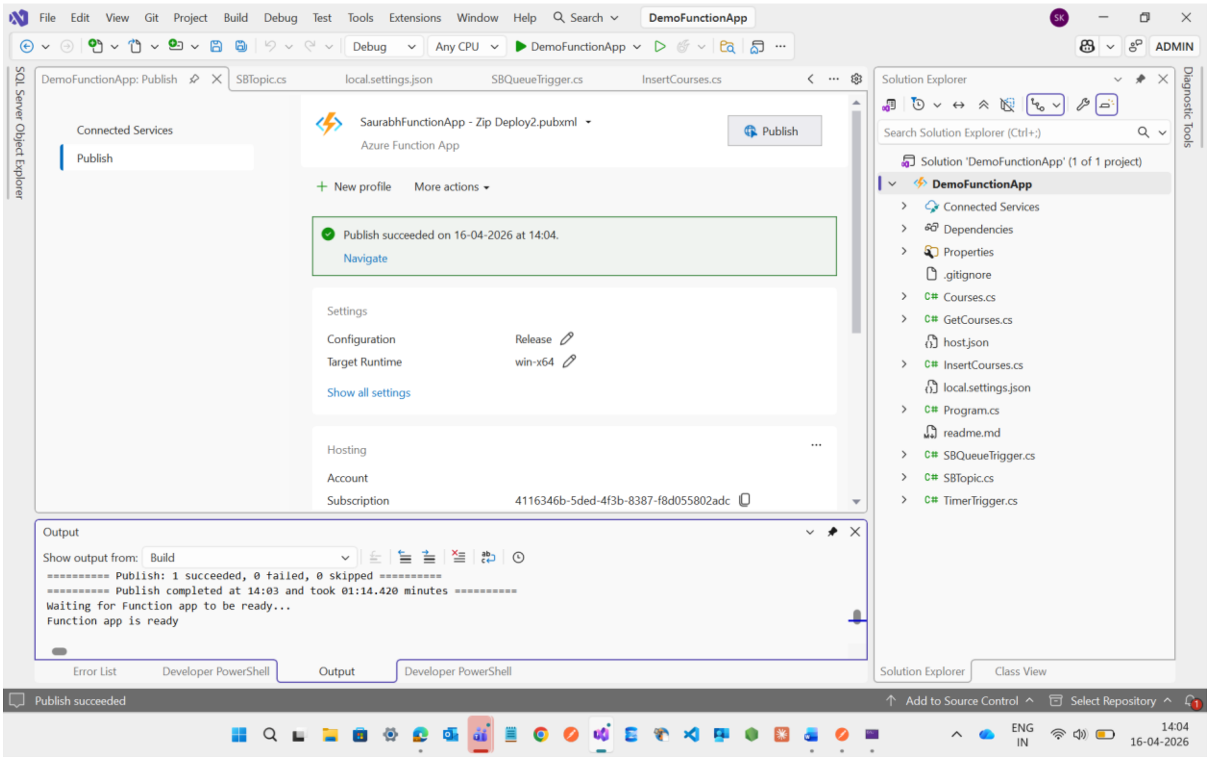
It is working Fine now I m going to Publish it to the Azure Function


DemoFunctionApp: Publish
Connected Services
Publish
Publish
Which Azure service would you like to use to host your application?
Target
Specific target
Azure Function App
Publish your application code to a serverless compute that scales dynamically and runs code on-demand
Azure Function App Container
Publish your application as a Docker image to Azure Container Registry and run it on Azure Function App
Azure Container Apps (Linux)
Run scalable containerized applications and microservices on a serverless platform in Azure
Azure Container Registry
Publish your application as a Docker image to Azure Container Registry
Back
Next
Finish
Cancel
Solution Explorer
Output
Ready
Add to Source Control
Select Repository
13:58
16-04-2026

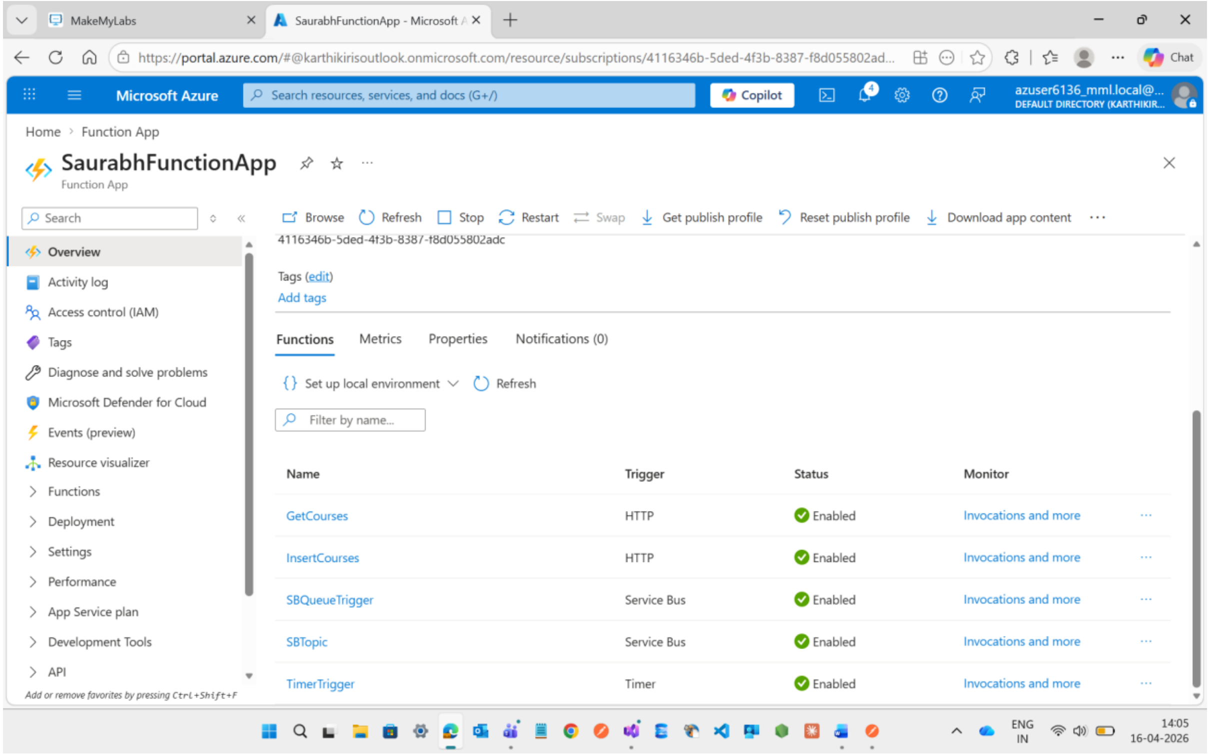

DemoFunctionApp: Publish
Connected Services
Publish
Publish
Select existing or create a new Azure Function
Default Directory
azuser6136_mml.local@karthiki...
Target
Specific target
Functions instance
Deployment type
Subscription name
Learner Account
Search
Create new
rg-azuser6136_mml.local-rTfTQ
SaurabhFunctionApp (Consumption - Windows)
Run from package file (recommended)
Turn on Basic Authentication (not recommended)
Back
Next
Finish
Cancel
Solution Explorer
Output
Ready
Add to Source Control
Select Repository
14:01
16-04-2026



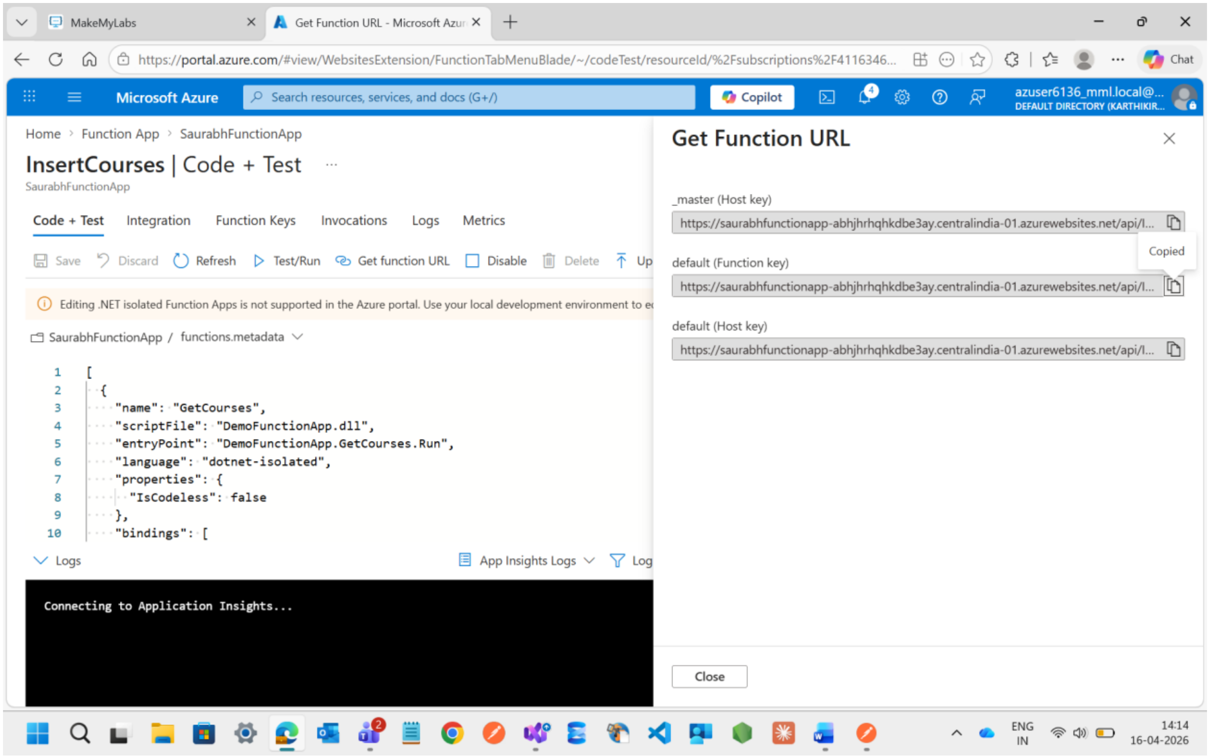
Now Publish Succeeded



See Topic will added here after publishing

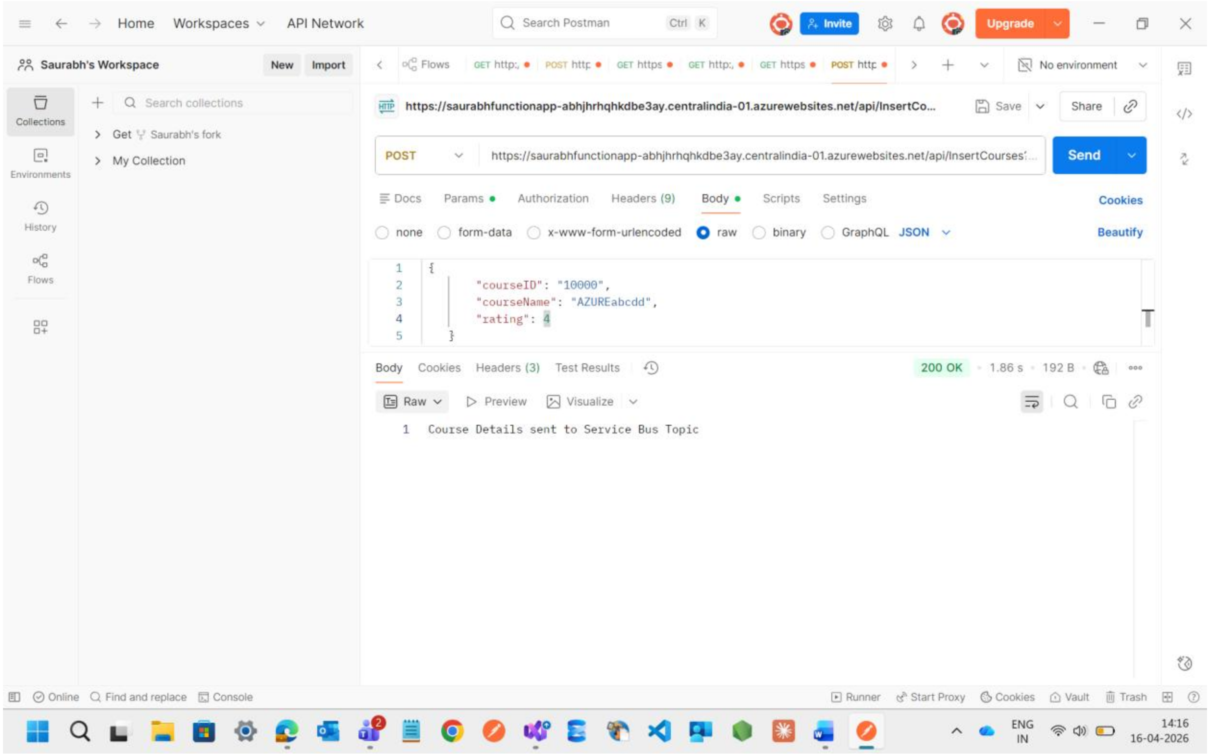


Now I m going to function app and try with InsertMethod wheather the Topic will accept the message with the rating 4 or not



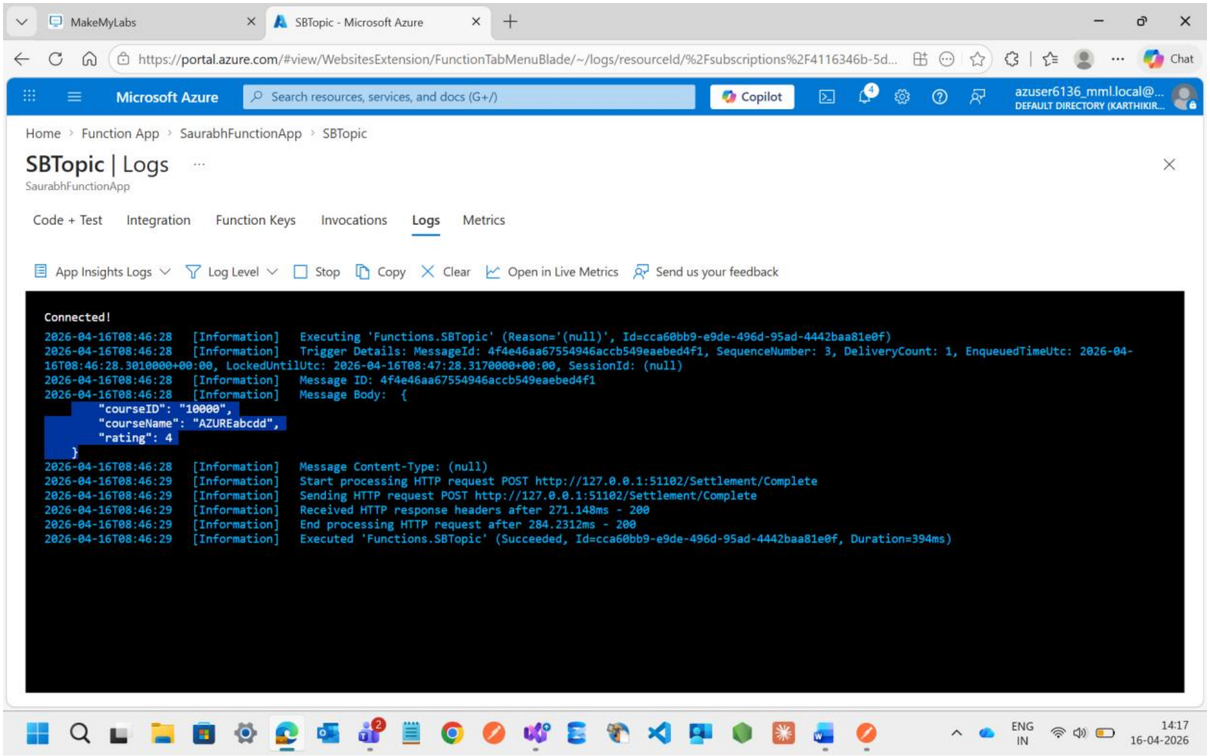
Then after this I will go to the Topic and try to check the logs when I try to hit the InsertMethod url through Postman it will work or not

See through postman message will work and it goes to topics



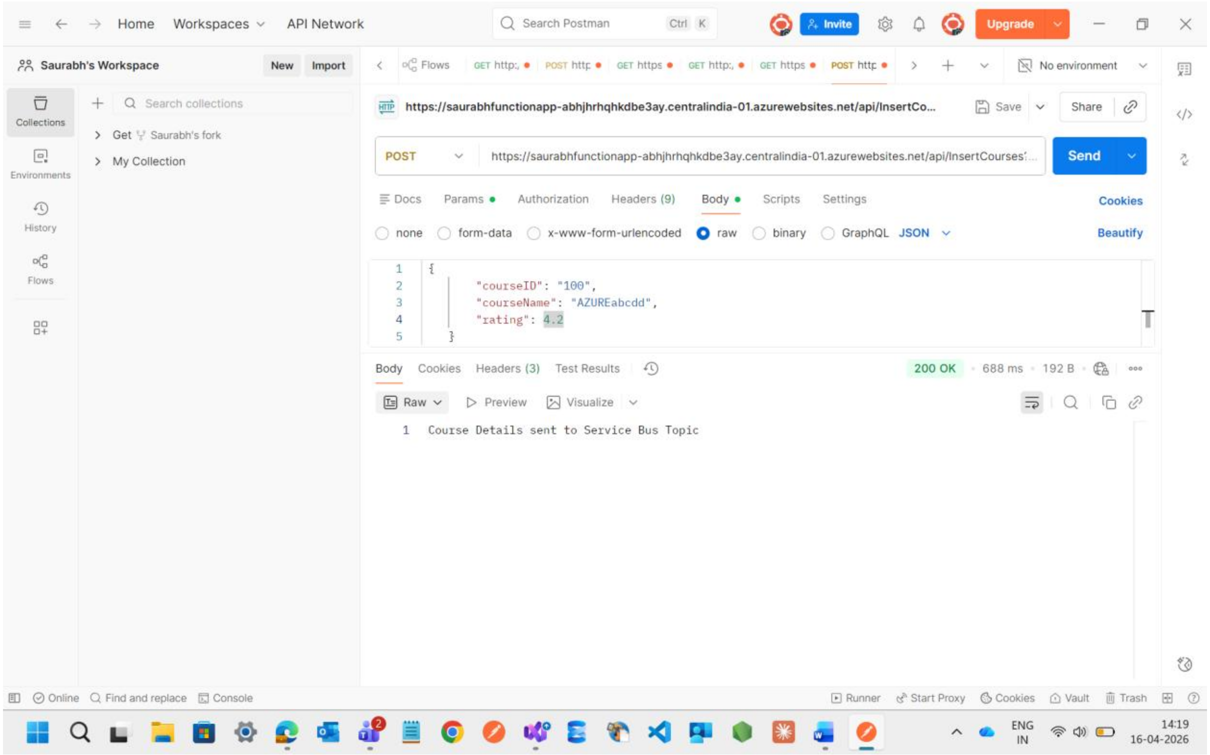
now I m going to check Inside the Topic Logs.

See here it will work fine



Now when I try to assign the rating more than 4 it will not store the log inside the Topic Logs

See in the postman message will clearly delivery but it will not store in Topic Logs



See here Rating 4.2 is not visible due to rating filtering which I was assigned inside the Subscriptions saurabhSau inside the Topics